

Week 2: Visual Encodings

CMSC471: Introduction to Information Visualization

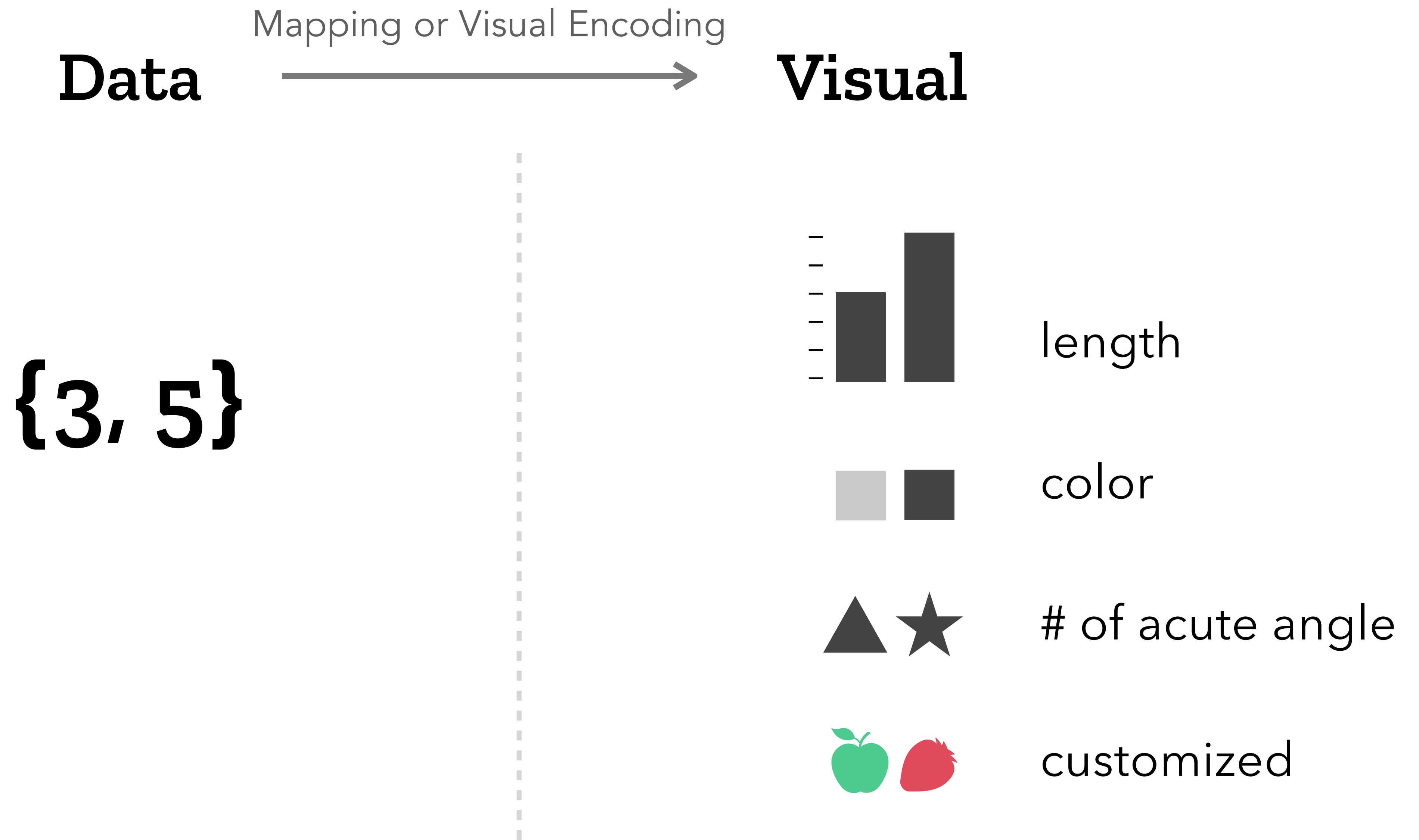
Fumeng Yang, Assistant Professor

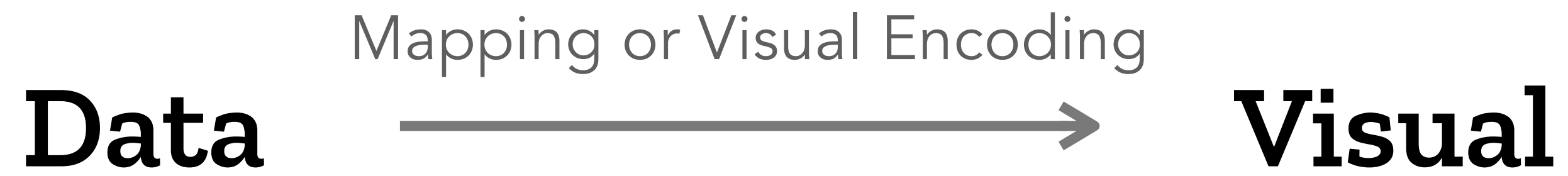
February 9, 2026

University of Maryland, College Park

<https://fumeng-yang.github.io/CMSC471>

What is a data visualization?





Dataset types

tabular, network, spatial, etc.

Variable (attribute) types

nominal, ordinal, etc.

Visual channels

length, color, opacity, etc.

Graphical marks

elements: point, line, area, etc.

Dataset types

Tabular

A collection of records
with named attributes.

[illegible]

Dataset types

Tabular

A collection of records with named attributes.

Networks

Nodes and links can also have attributes



Dataset types

Tabular

A collection of records with named attributes.

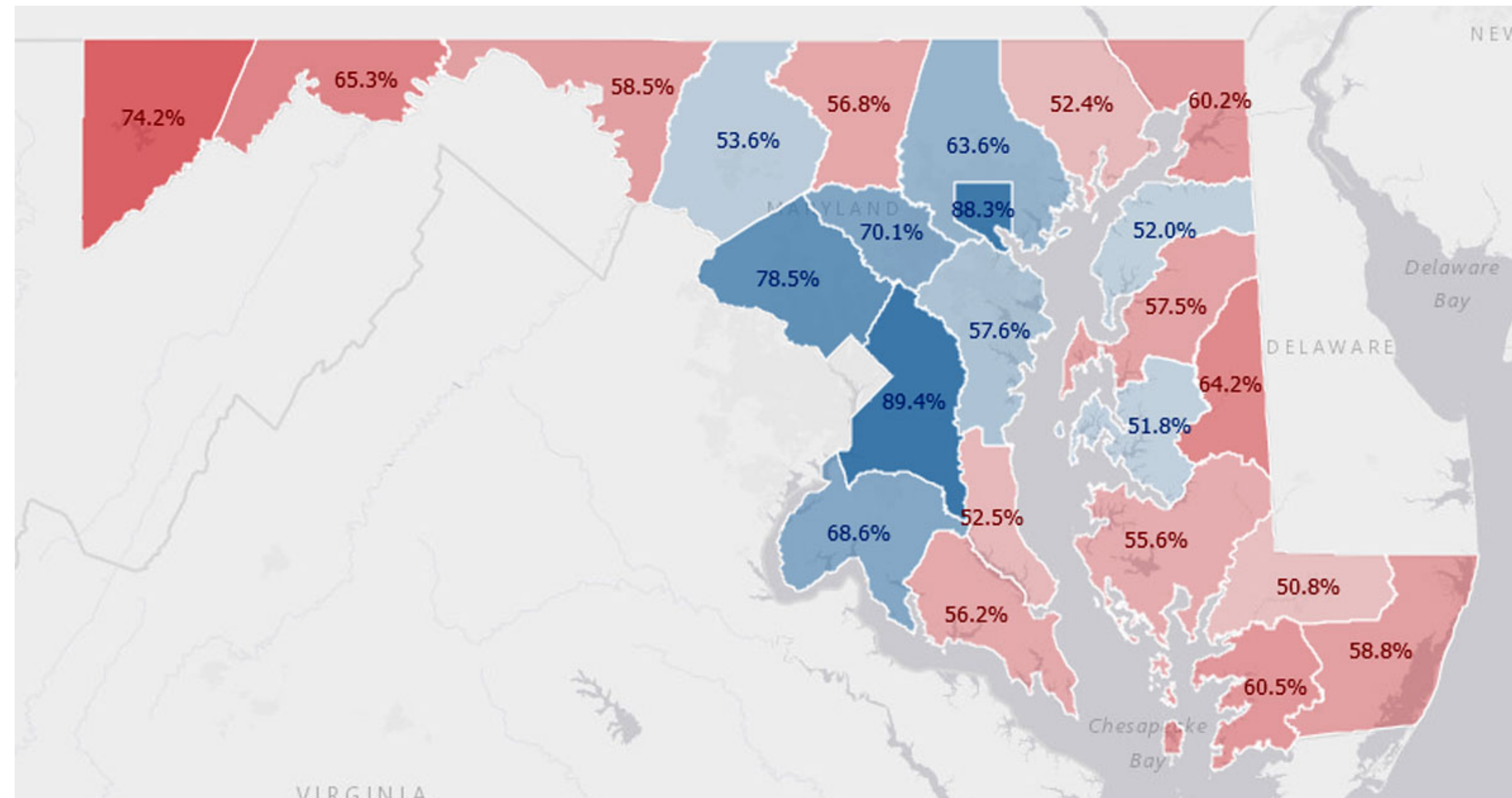
Networks

Nodes and links can also have attributes

Spatial

Continuous "fields" vs discrete "positions"

2022 Maryland Gubernatorial Election Results by County



<https://www.engagetu.com/2022/12/21/maryland-2022-gubernatorial-election-results-by-county/>

Variable (attribute) types

Nominal/Categorical

=, ≠

Labels or categories. E.g., Fruits: apples, bananas, cantaloupes, ...

Ordinal

=, ≠, <, >

Ordered. E.g., Quality of meat: Grade A, AA, AAA

Quantitative (Interval)

=, ≠, <, >, -

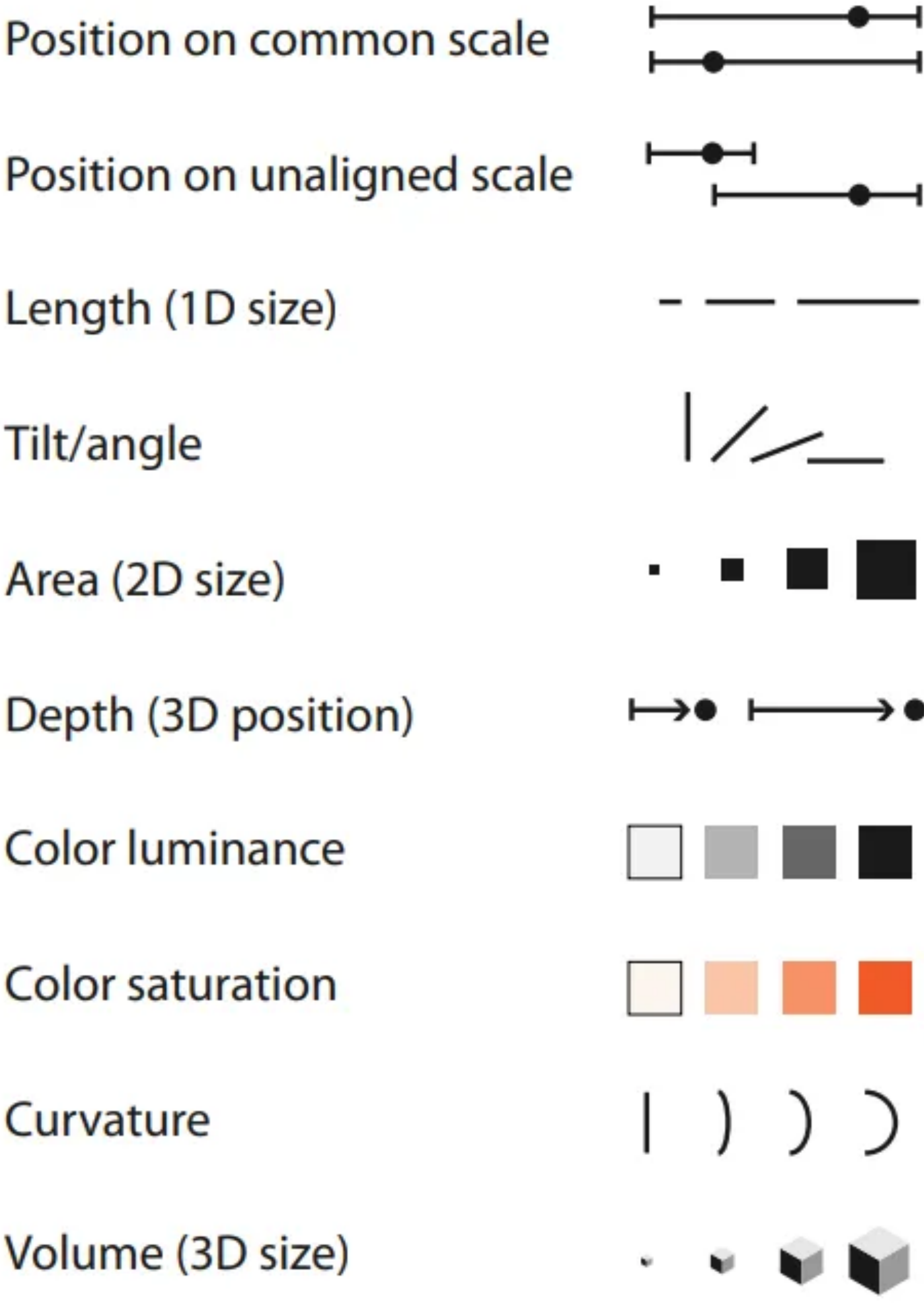
Interval (zero can be arbitrarily located). E.g., Dates: Jan 19, 2018; Location: (Lat 42.36, -71.09).

Quantitative (Ratio)

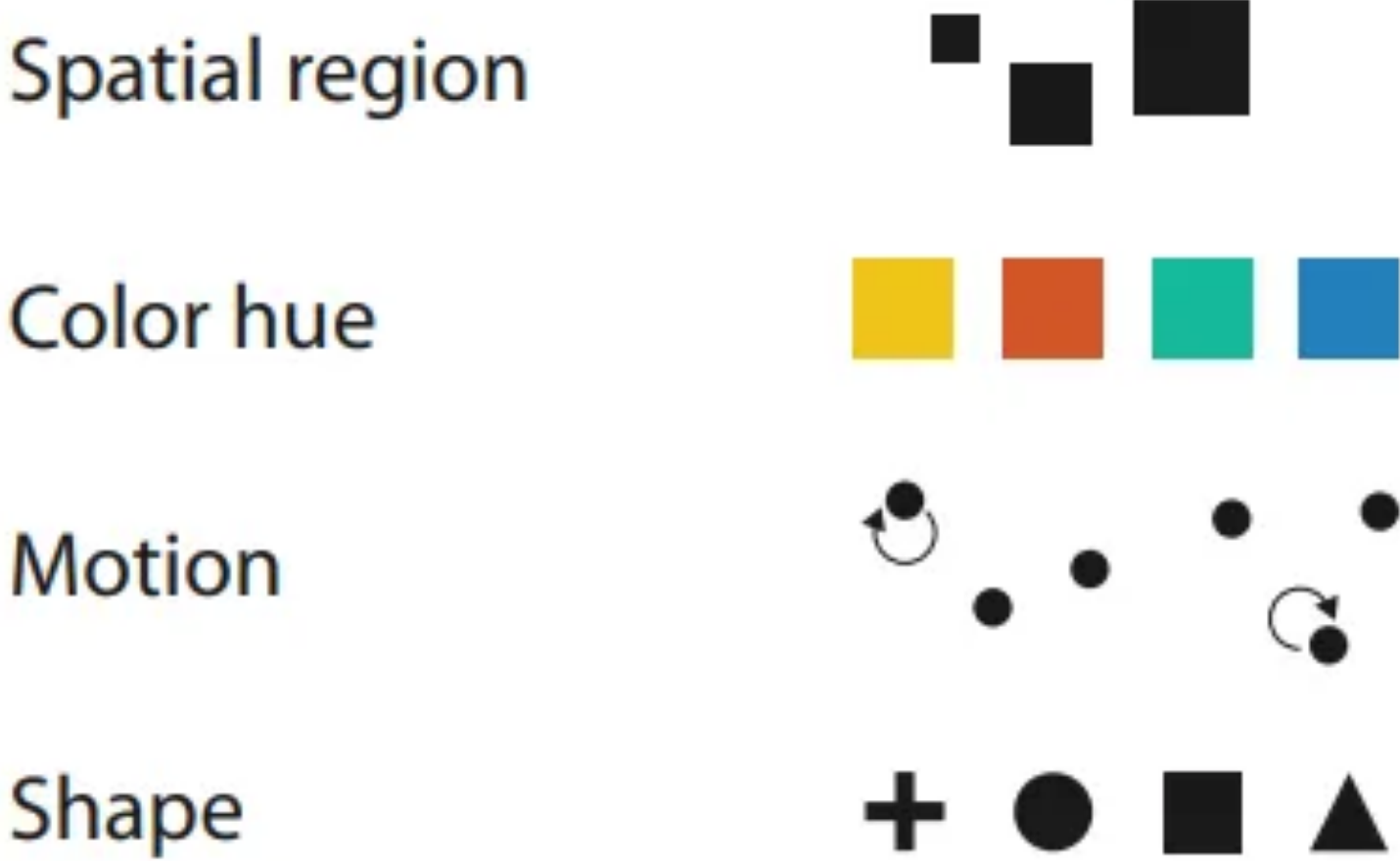
=, ≠, <, >, -, %

Fixed zero. E.g., Physical measurement: length, mass, temperature, counts and amounts.

➔ **Magnitude Channels: Ordered Attributes**

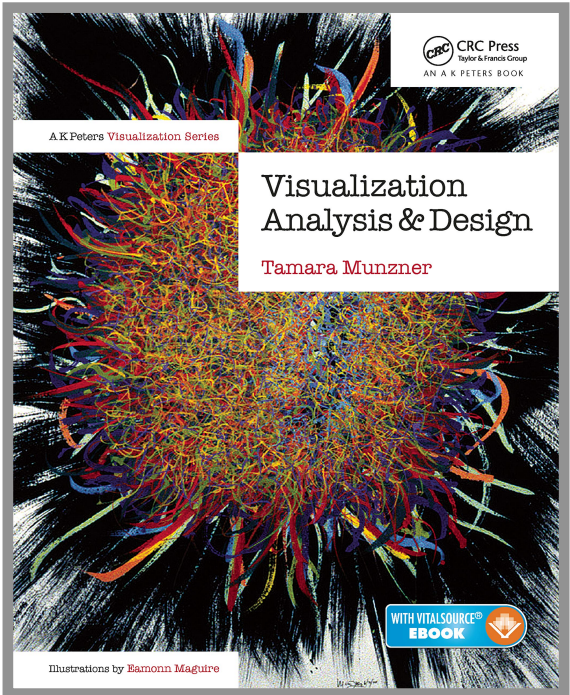


➔ **Identity Channels: Categorical Attributes**

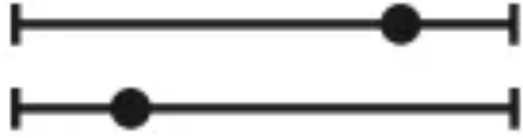


Most
Effectiveness
Least

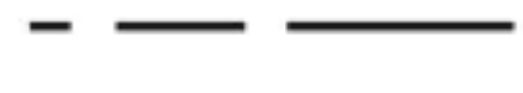
Same
Same

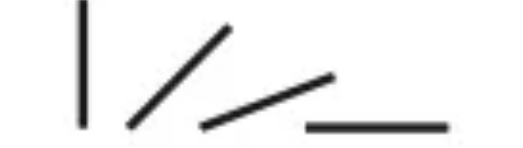


➔ **Magnitude Channels: Ordered Attributes**

Position on common scale 

Position on unaligned scale 

Length (1D size) 

Tilt/angle 

Area (2D size) 

Depth (3D position) 

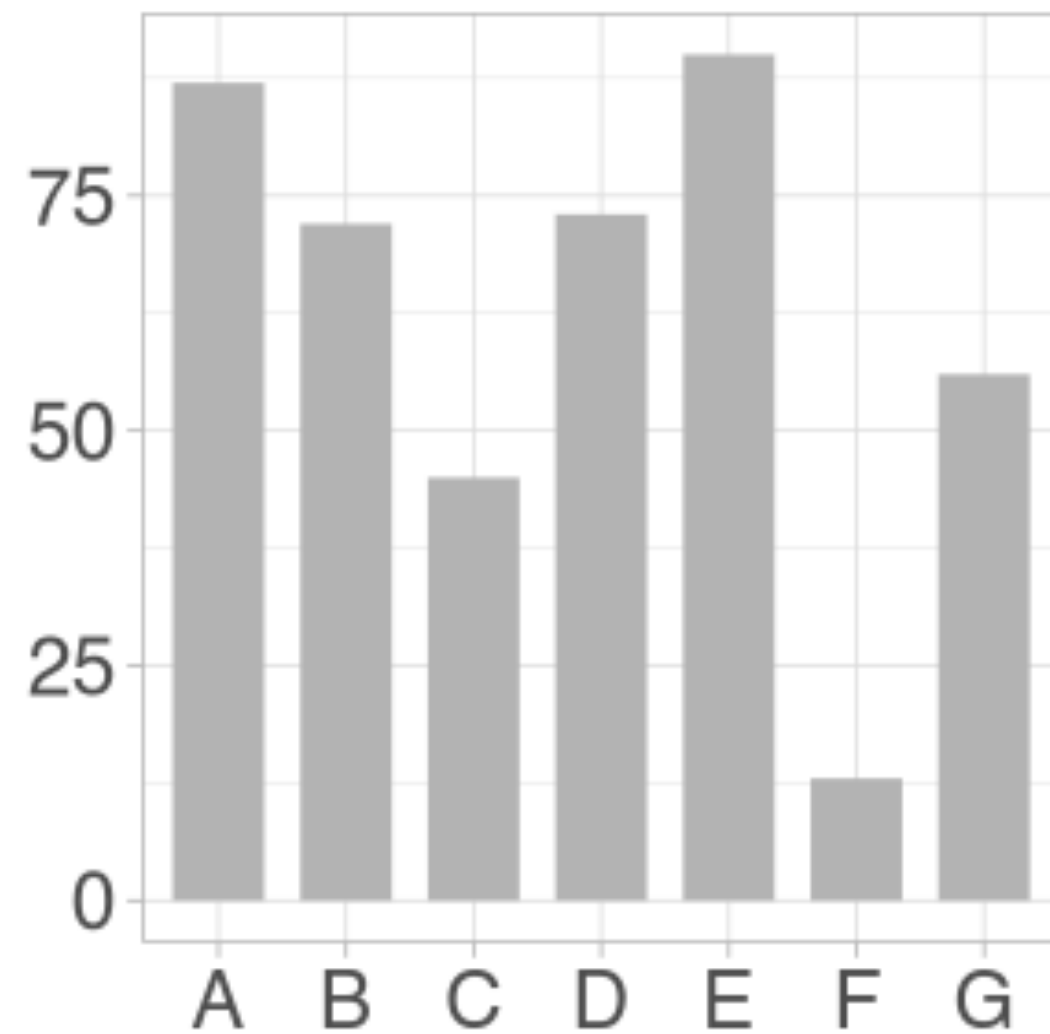
Color luminance 

Color saturation 

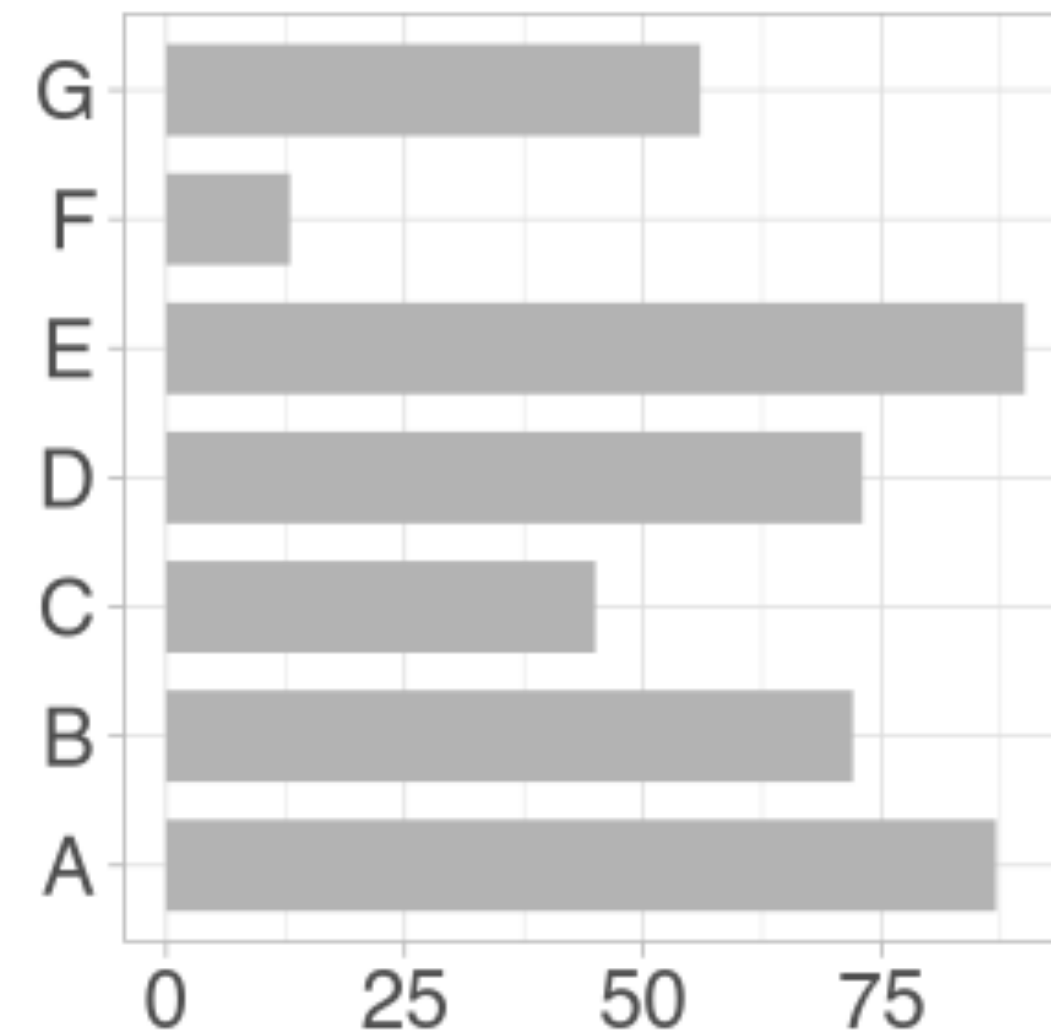
Curvature 

Volume (3D size) 

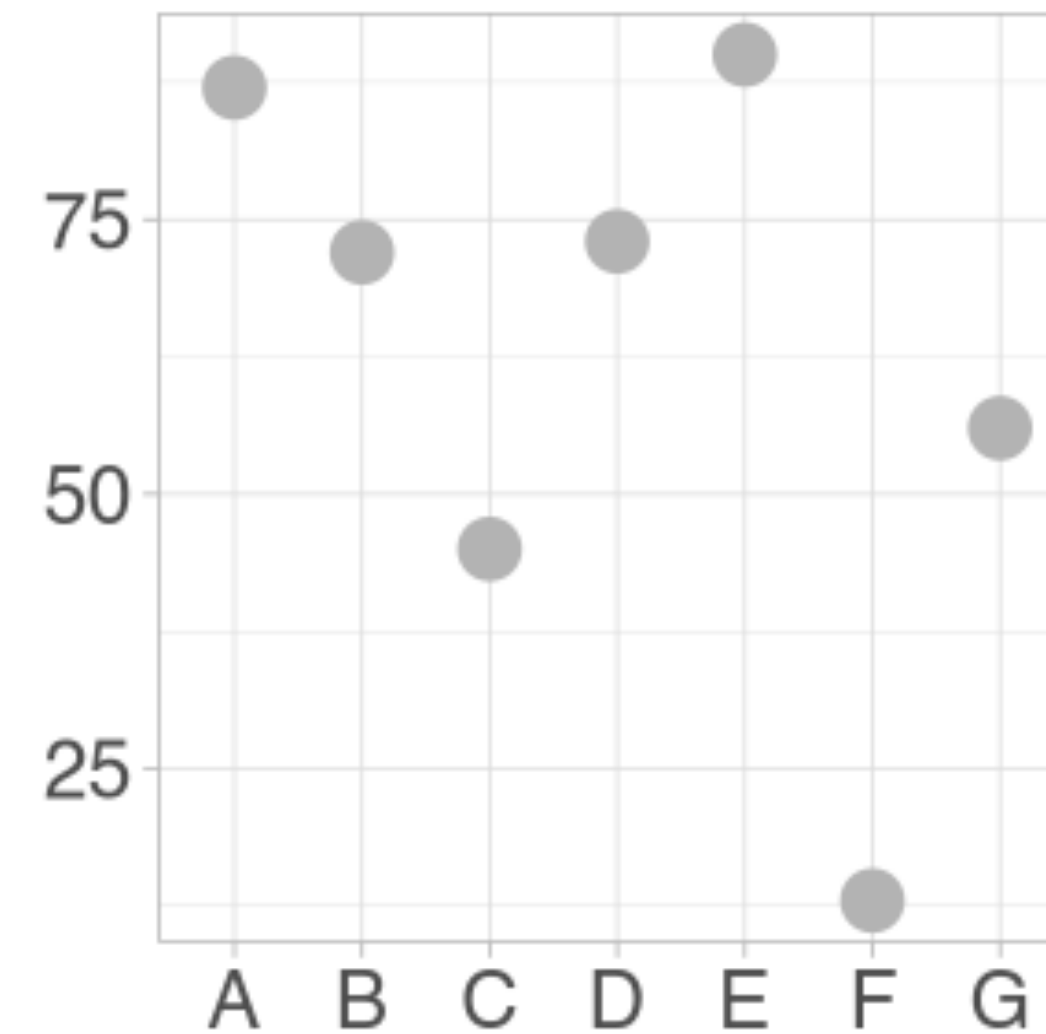
Visual encoding: 1 ordinal, 1 quantitative



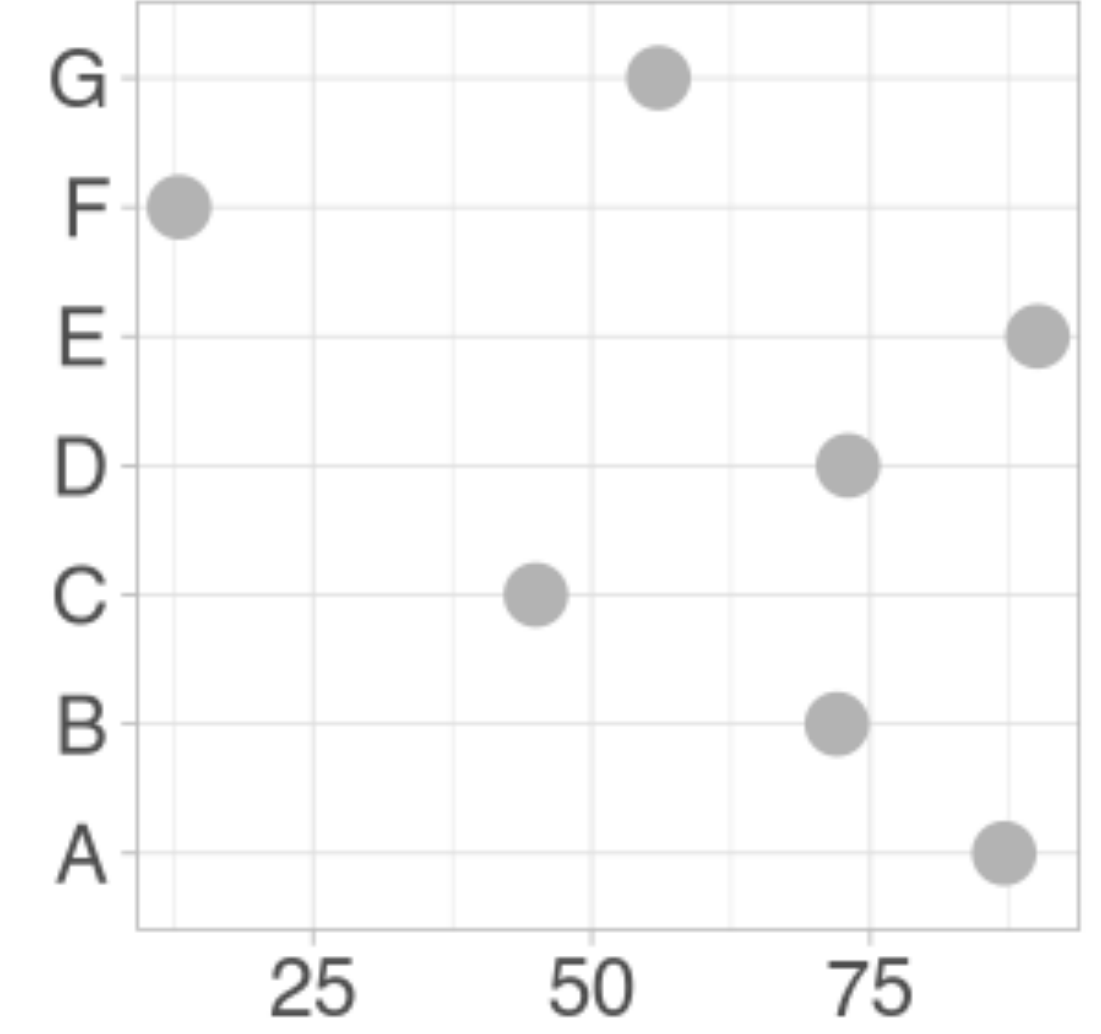
mark: bar ■
 $d_{\text{ordinal}} \rightarrow x$
 $d_{\text{quantitative}} \rightarrow y$



mark: bar ■
 $d_{\text{ordinal}} \rightarrow y$
 $d_{\text{quantitative}} \rightarrow x$

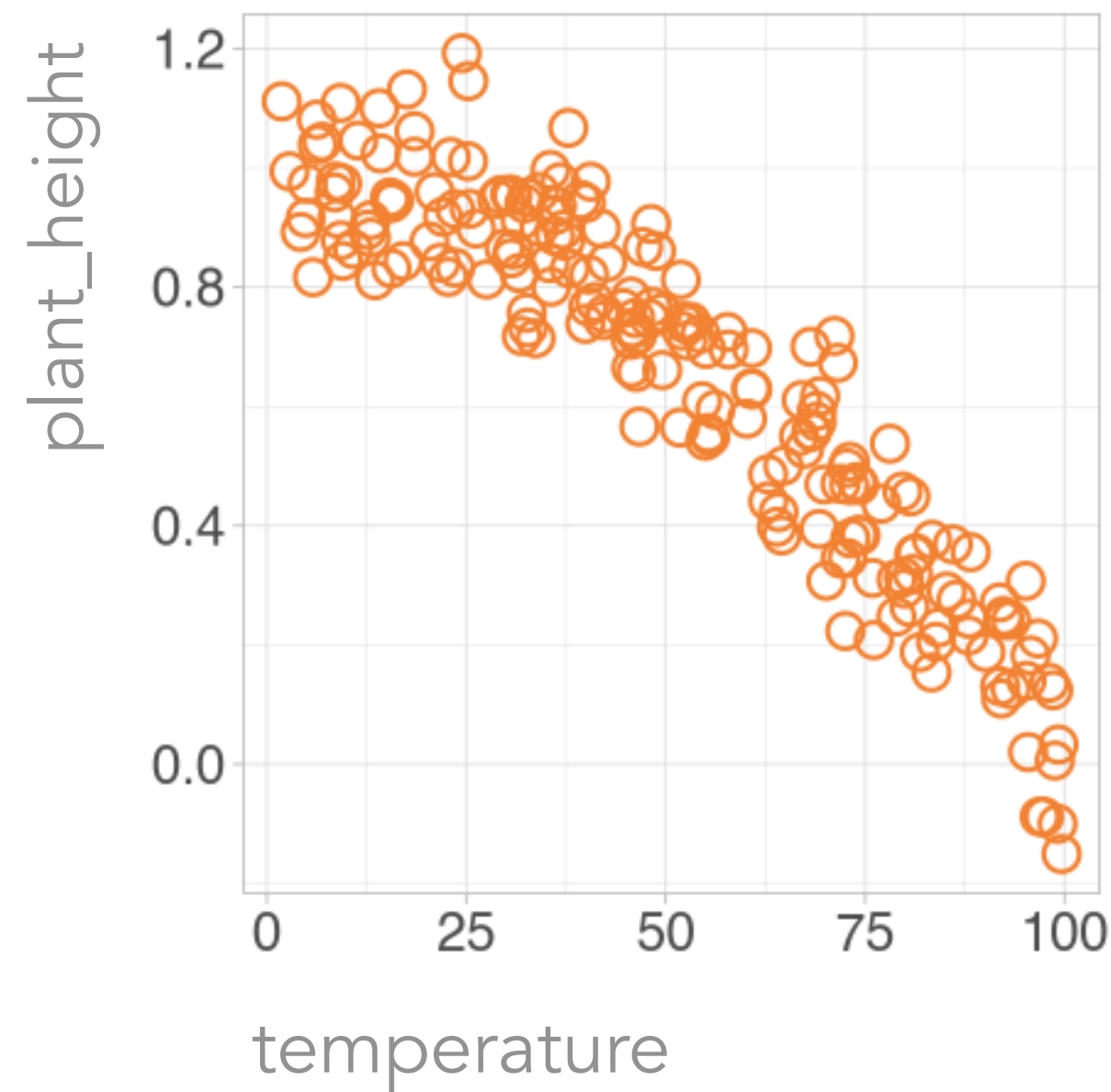


mark: point ●
 $d_{\text{ordinal}} \rightarrow x$
 $d_{\text{quantitative}} \rightarrow y$



mark: point ●
 $d_{\text{quantitative}} \rightarrow x$
 $d_{\text{ordinal}} \rightarrow y$

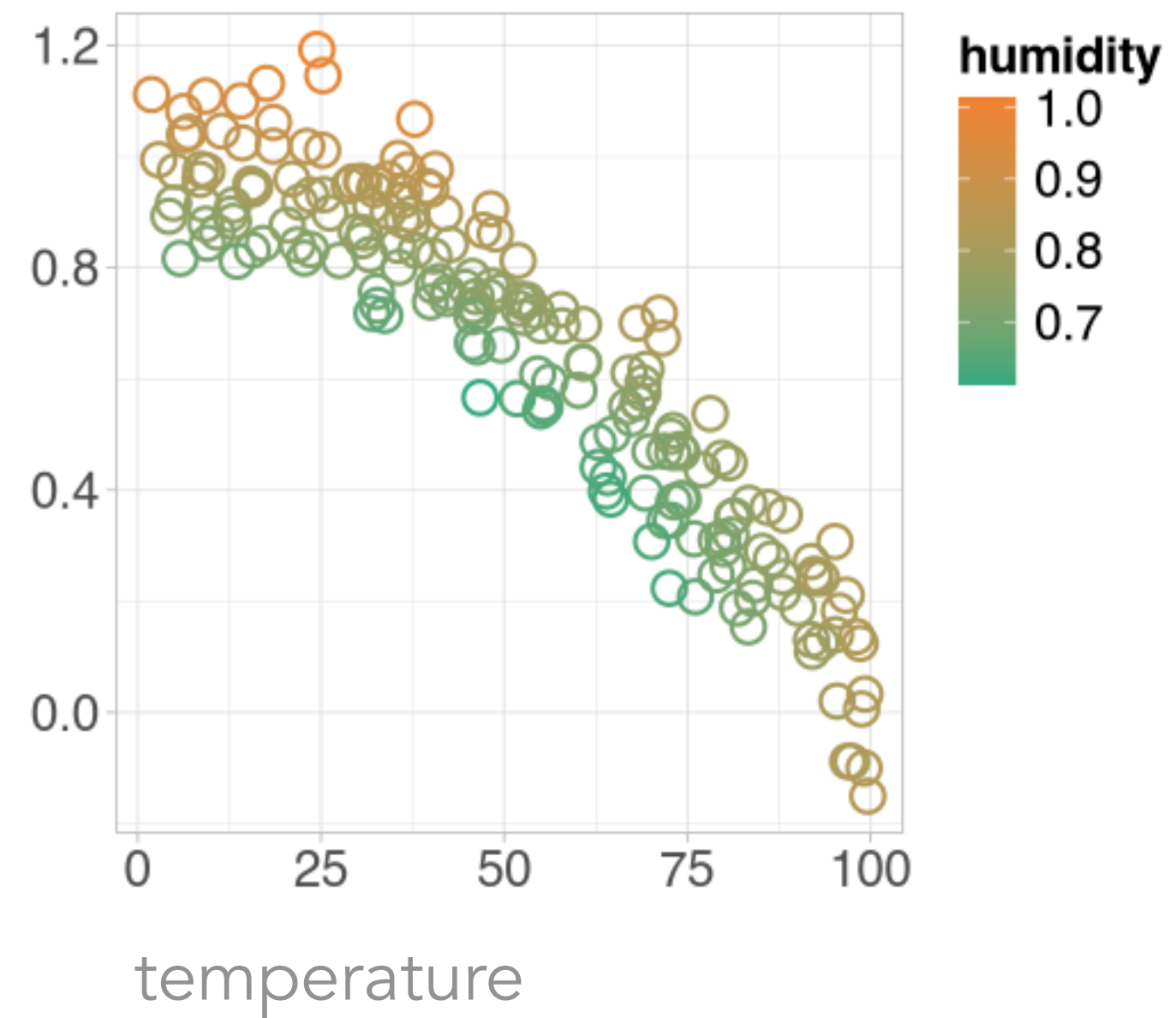
Visual encoding



mark: point

$d_{\text{quantitative}} \rightarrow x$

$d_{\text{quantitative}} \rightarrow y$

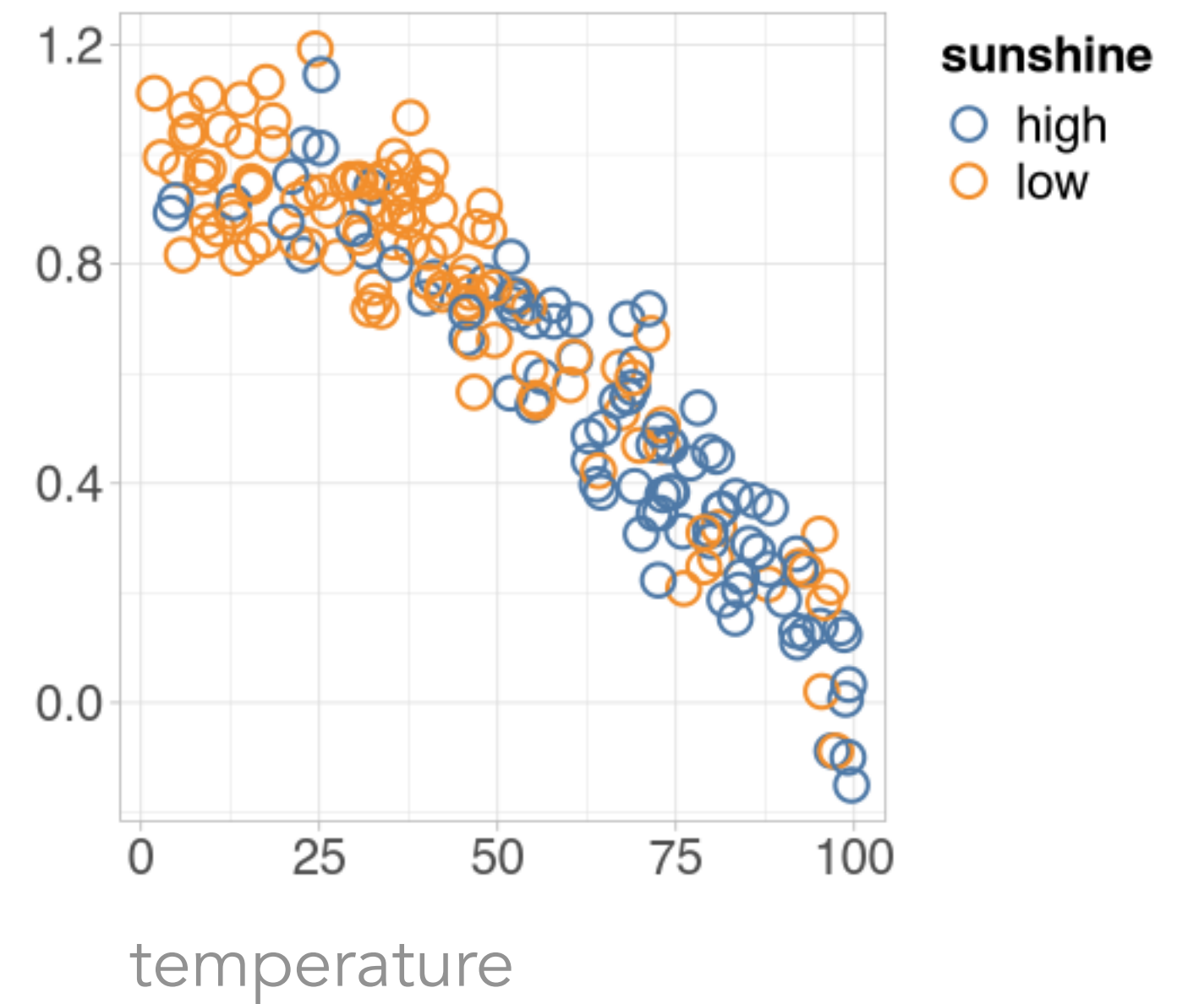


mark: point

$d_{\text{quantitative}} \rightarrow x$

$d_{\text{quantitative}} \rightarrow y$

$d_{\text{quantitative}} \rightarrow \text{color}$



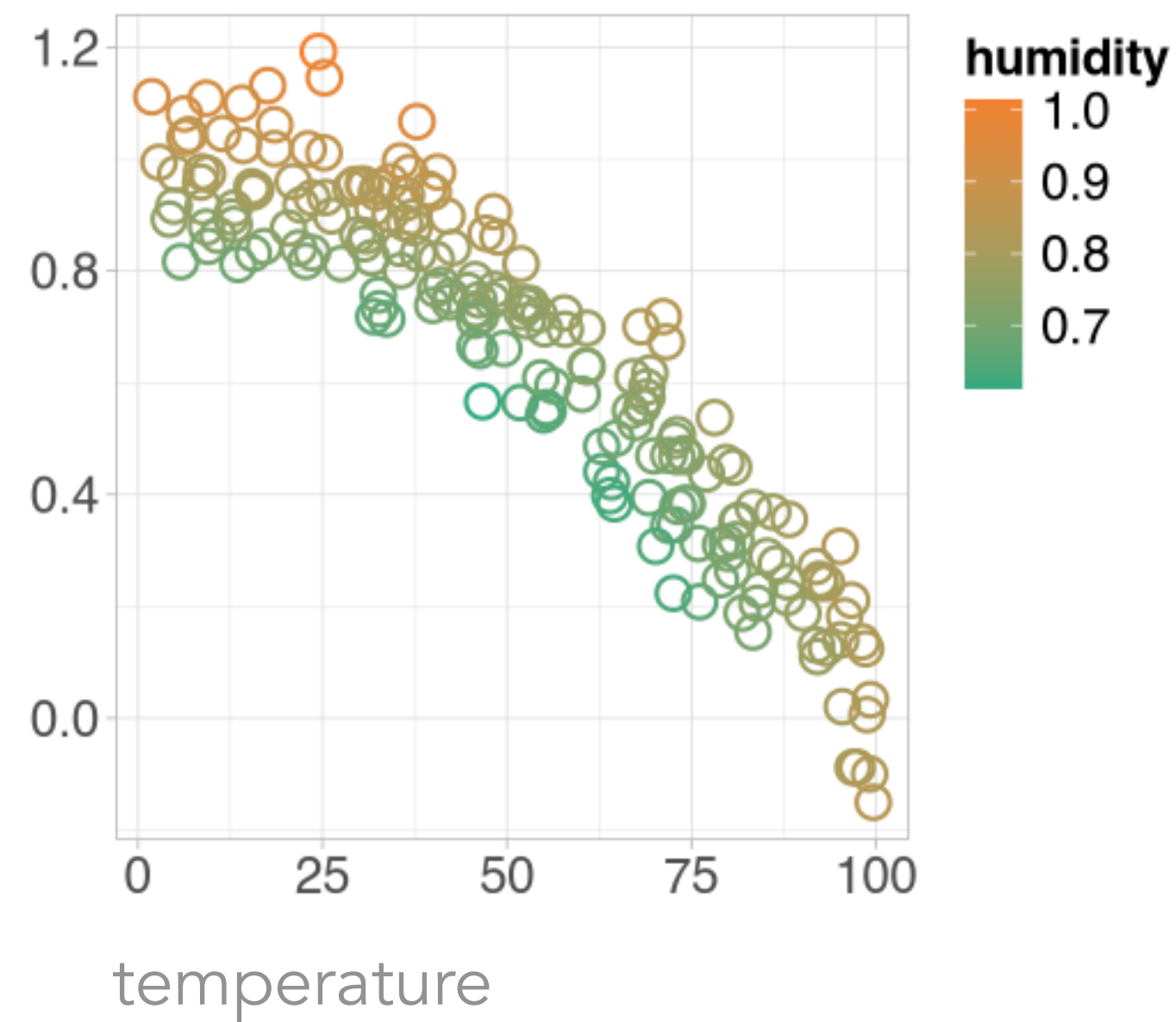
mark: point

$d_{\text{quantitative}} \rightarrow x$

$d_{\text{quantitative}} \rightarrow y$

$d_{\text{ordinal}} \rightarrow \text{color}$

Visual encoding

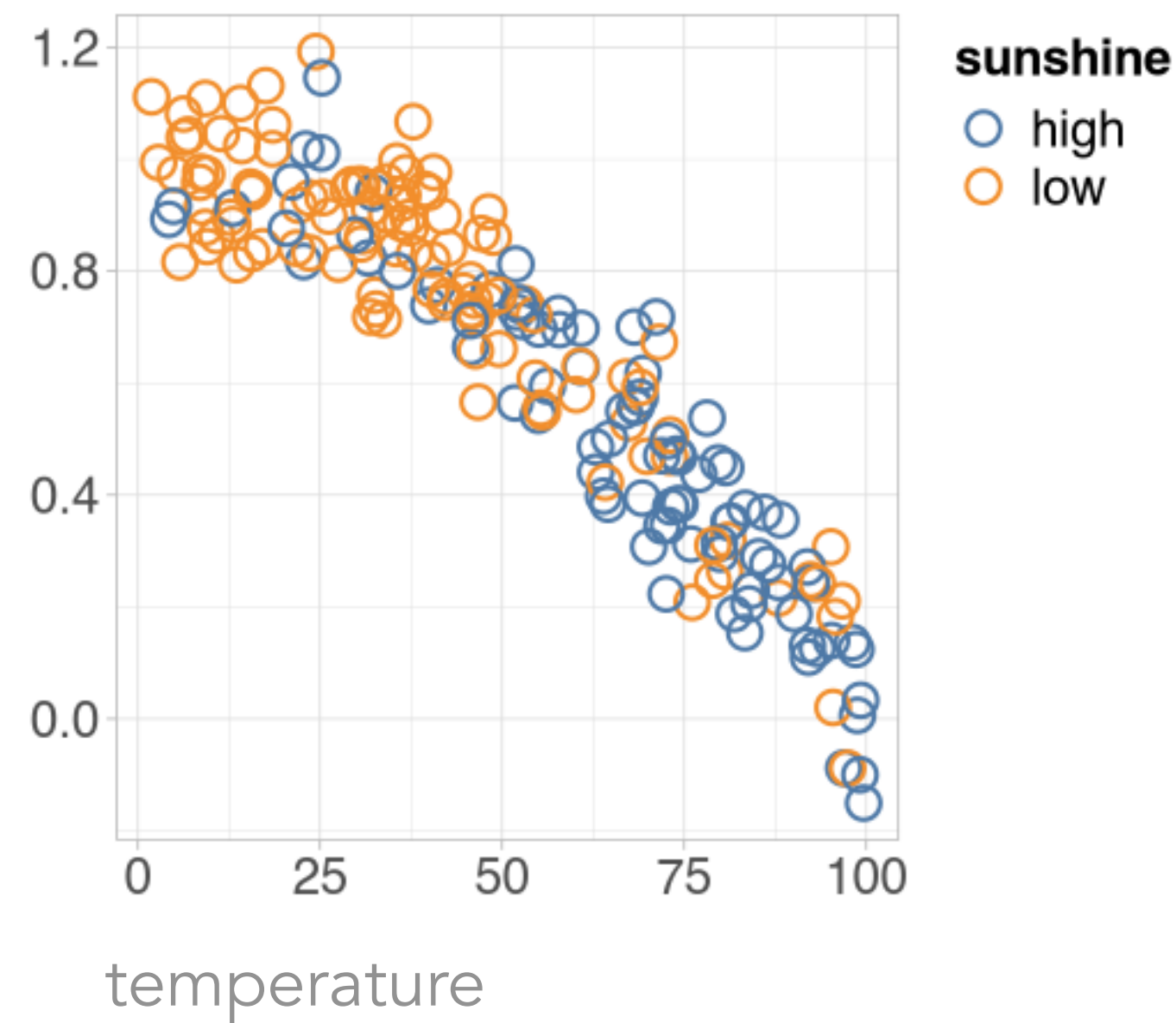


mark: point

$d_{\text{quantitative}} \rightarrow x$

$d_{\text{quantitative}} \rightarrow y$

$d_{\text{quantitative}} \rightarrow \text{color}$

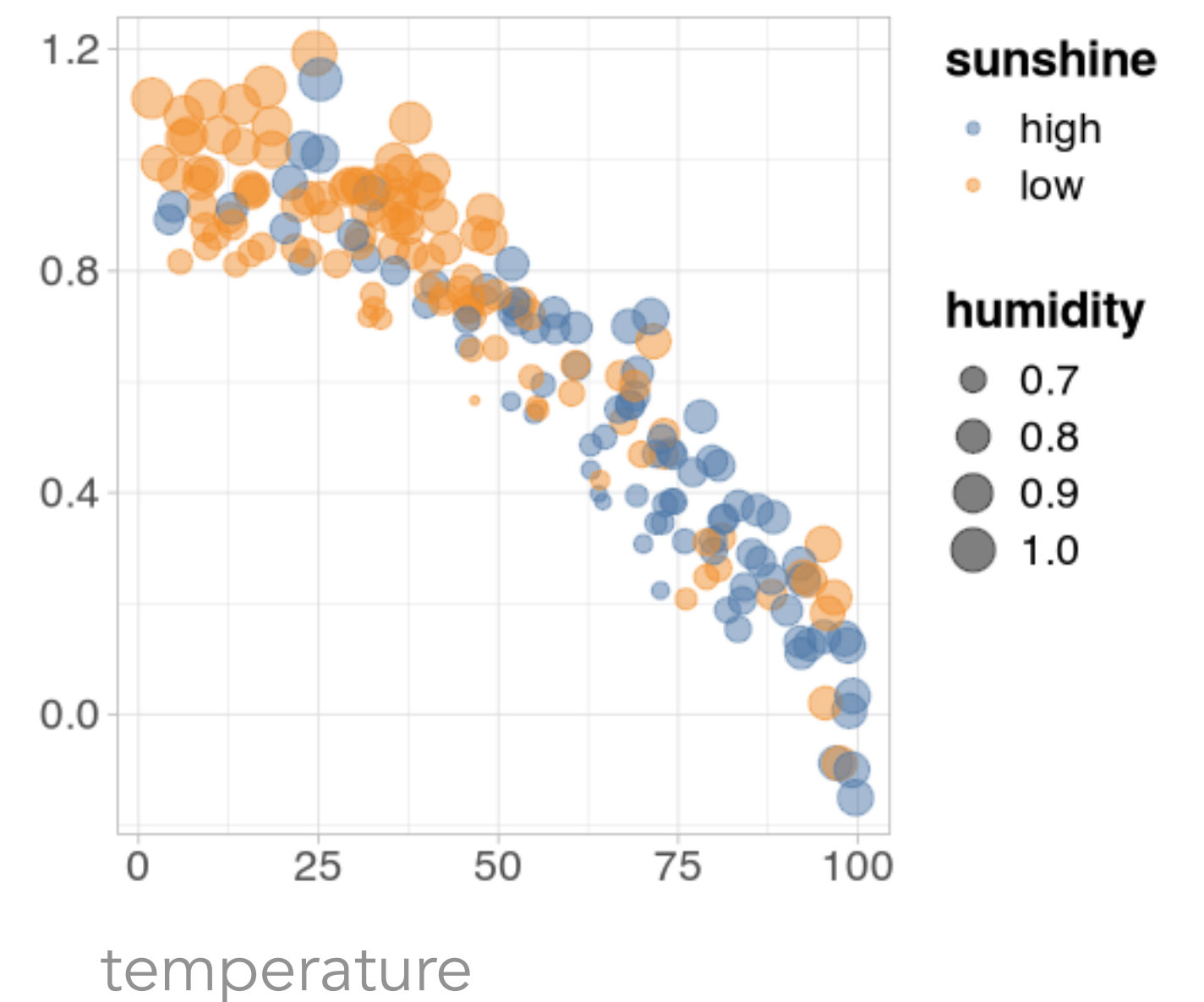


mark: point

$d_{\text{quantitative}} \rightarrow x$

$d_{\text{quantitative}} \rightarrow y$

$d_{\text{ordinal}} \rightarrow \text{color}$



mark: point

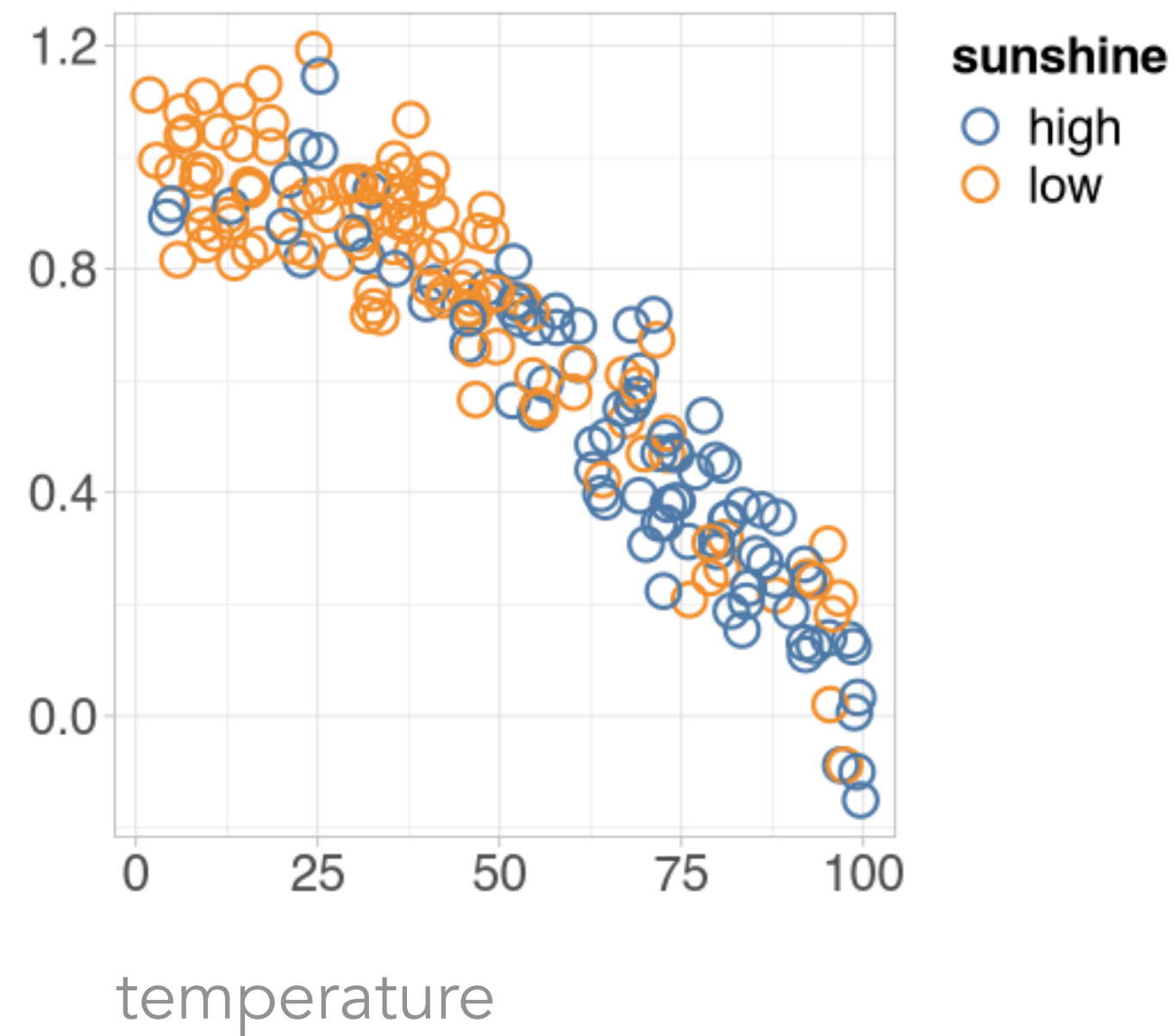
$d_{\text{quantitative}} \rightarrow x$

$d_{\text{quantitative}} \rightarrow y$

$d_{\text{ordinal}} \rightarrow \text{color}$

$d_{\text{quantitative}} \rightarrow \text{size}$

Visual encoding

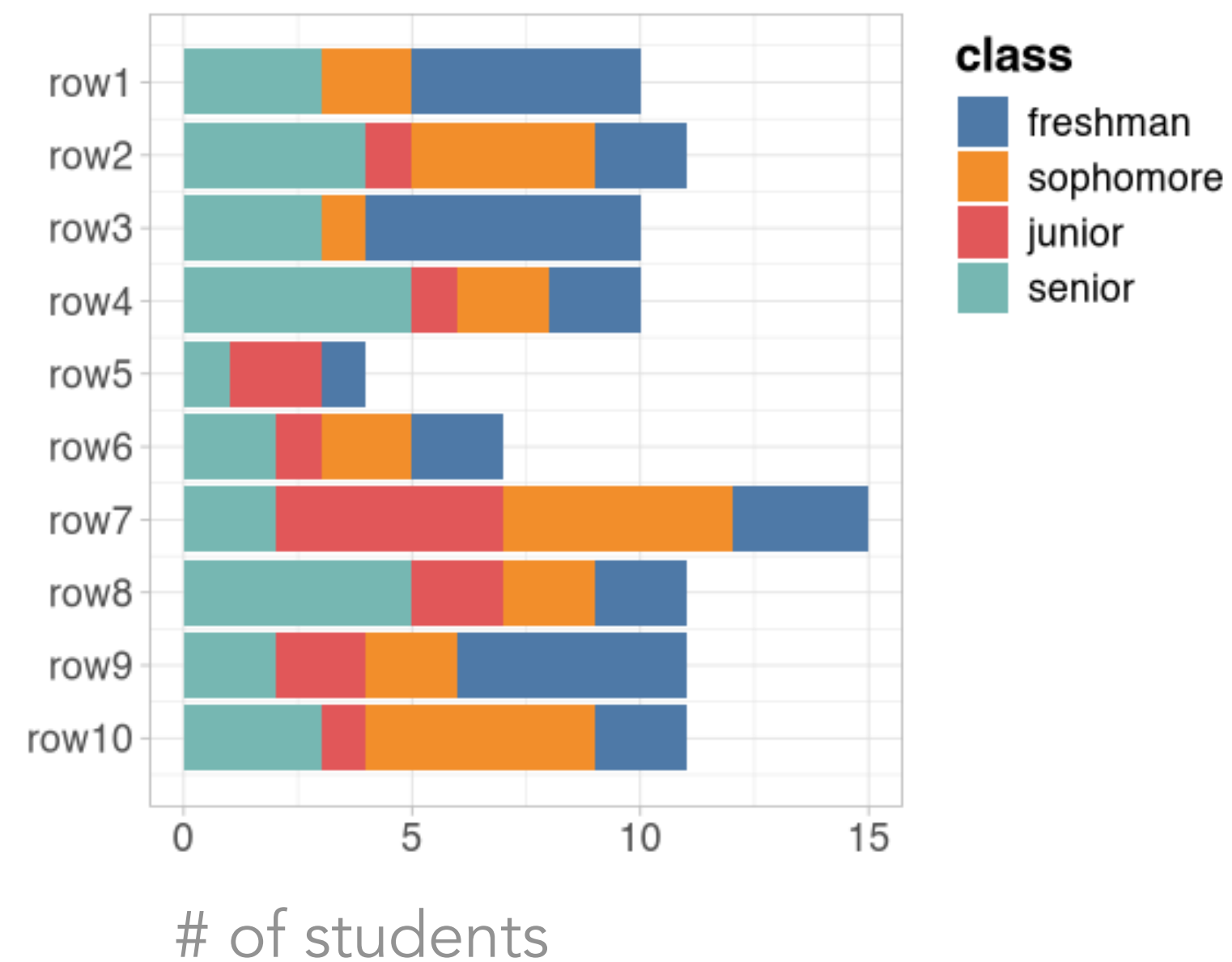


mark: point

$d_{\text{quantitative}} \rightarrow x$

$d_{\text{quantitative}} \rightarrow y$

$d_{\text{ordinal}} \rightarrow \text{color}$

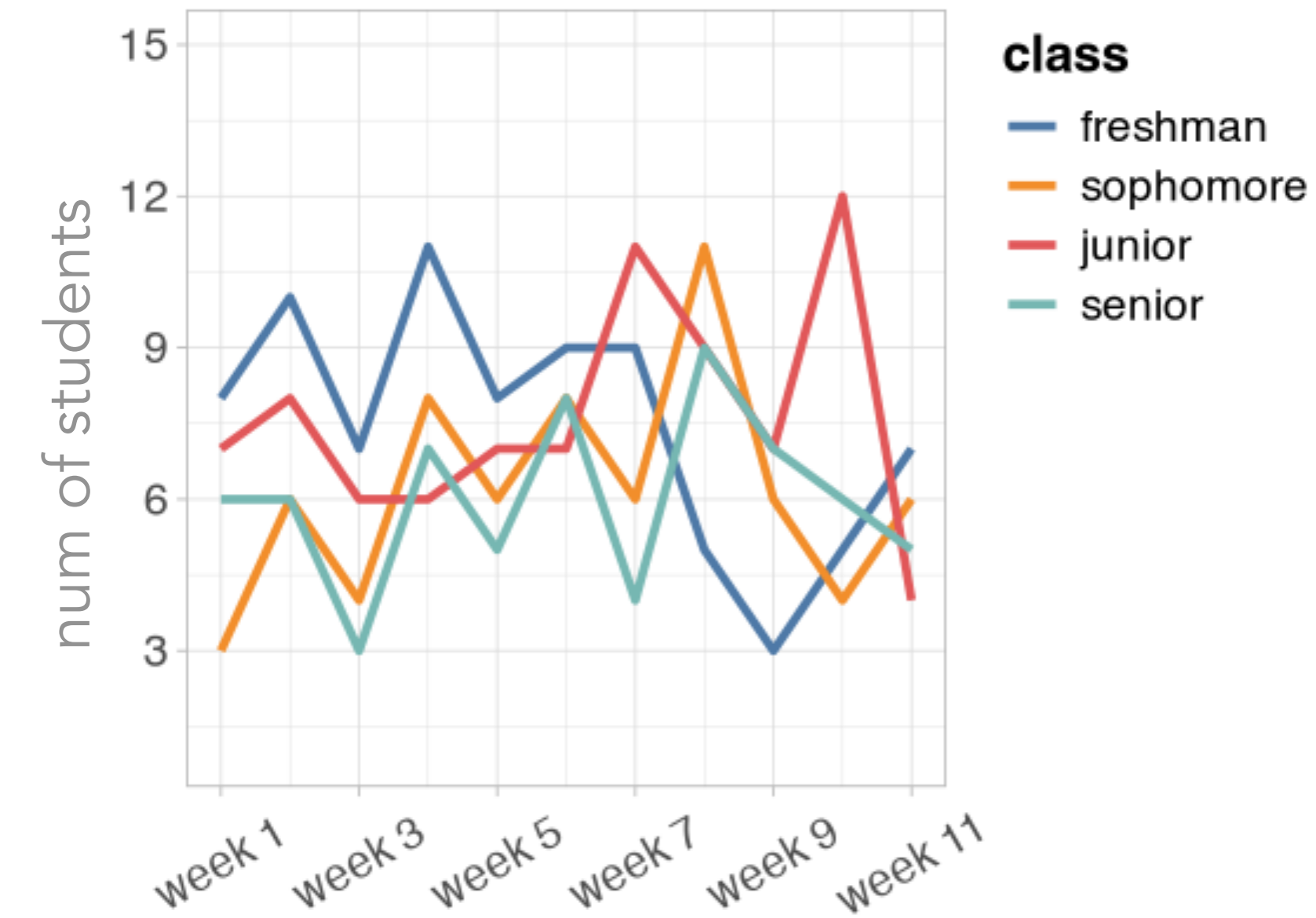


mark: bar

$d_{\text{quantitative}} \rightarrow x$

$d_{\text{ordinal}} \rightarrow y$

$d_{\text{ordinal}} \rightarrow \text{color}$



mark: line

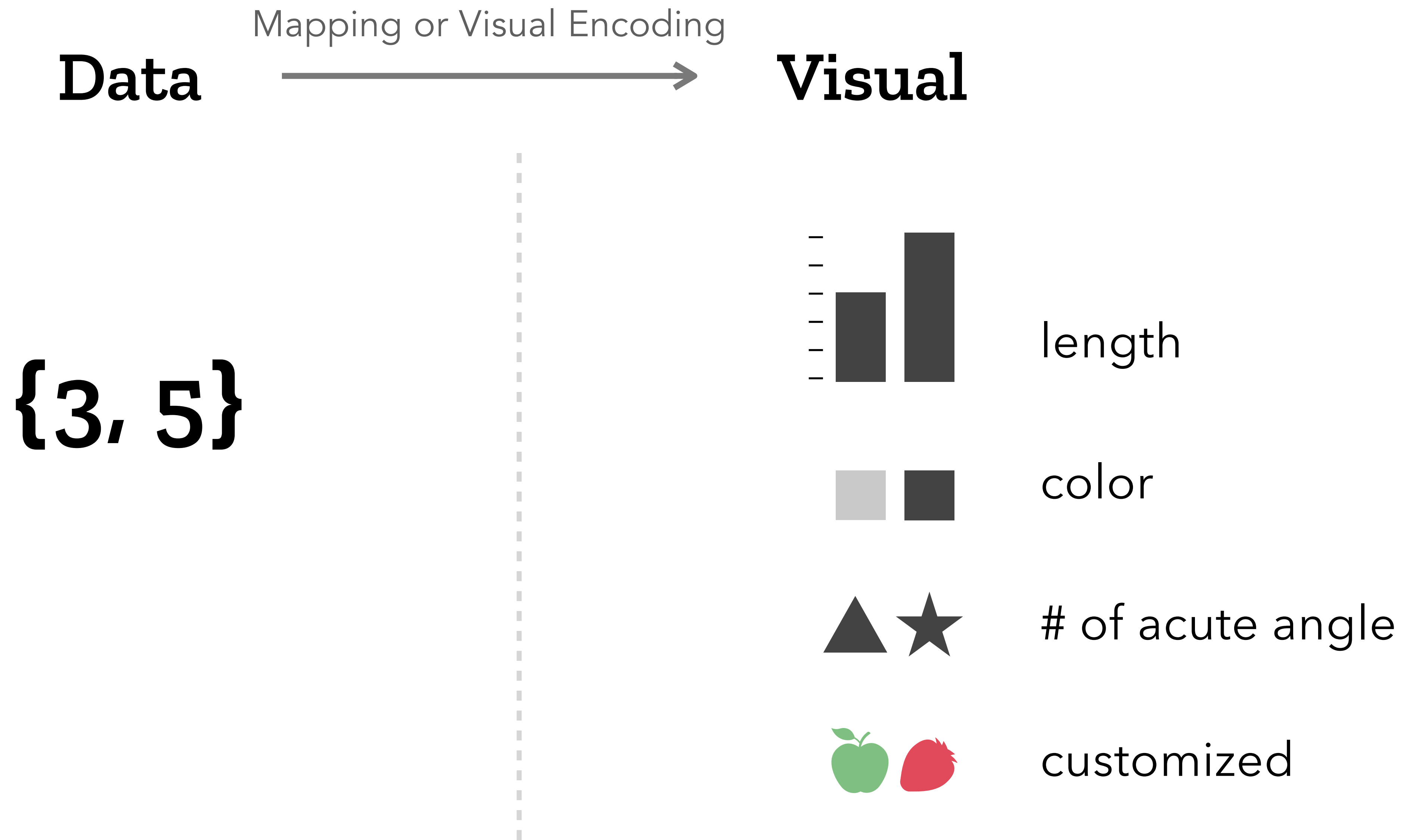
$d_{\text{temporal}} \rightarrow x$

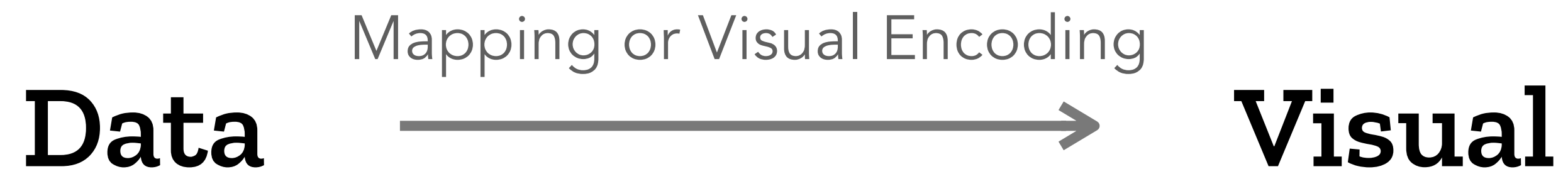
$d_{\text{quantitative}} \rightarrow y$

$d_{\text{ordinal}} \rightarrow \text{color}$

Question?

What makes a "good"
data visualization?





Expressiveness

A set of facts is **expressible** in a visual language if the sentences (i.e. the visualizations) in the language express *all the facts in the set of data, and only the facts in the data.*

Data

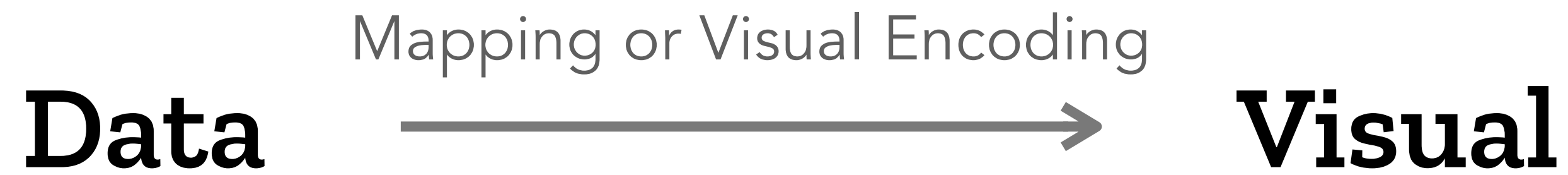
Mapping or Visual Encoding

Visual

Expressiveness



A multivariate dataset may be inexpressive in a single horizontal dot plot because multiple records are mapped to the same position.



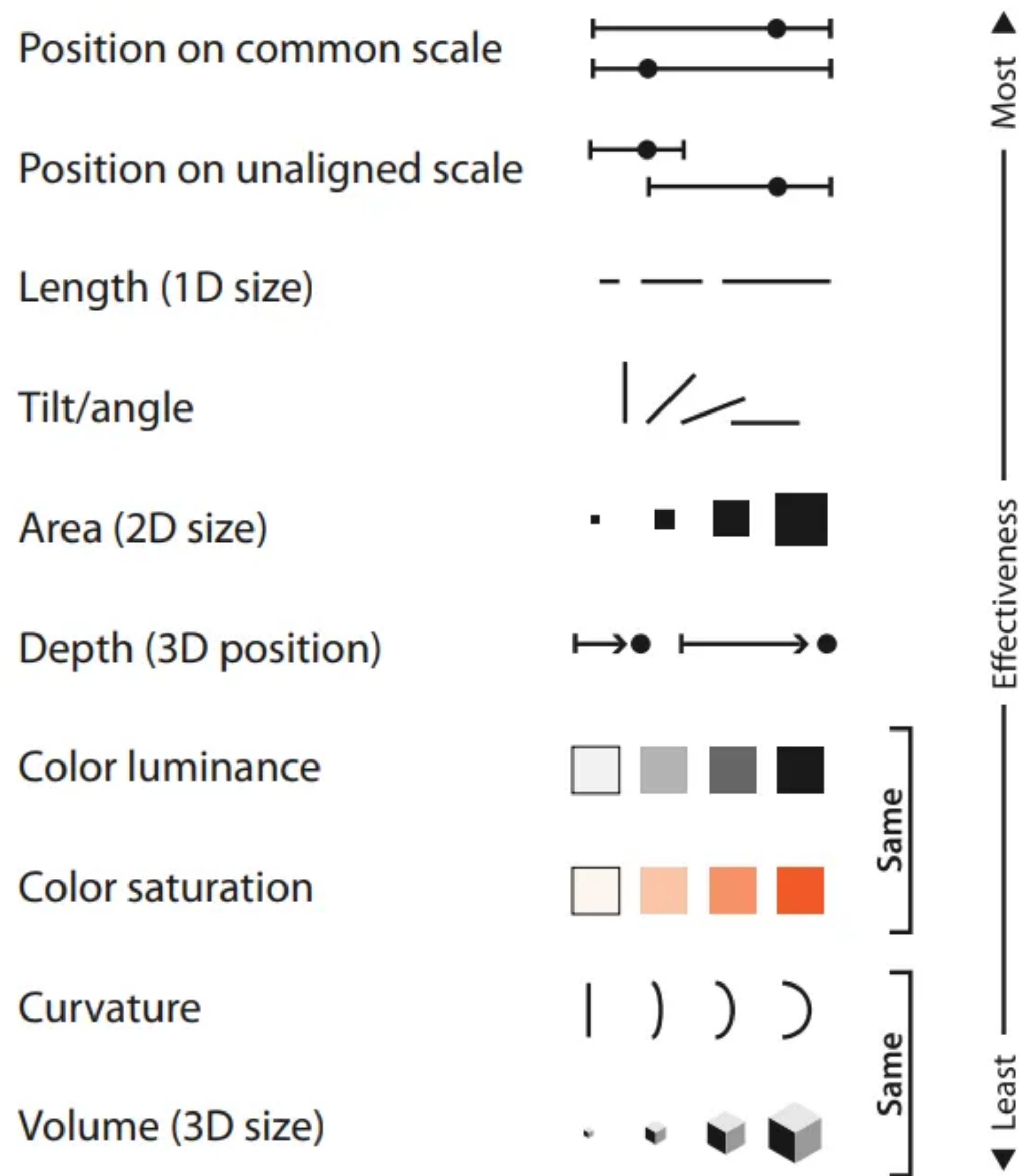
Expressiveness

A set of facts is **expressible** in a visual language if the sentences (i.e. the visualizations) in the language express **all the facts in the set of data, and only the facts in the data.**

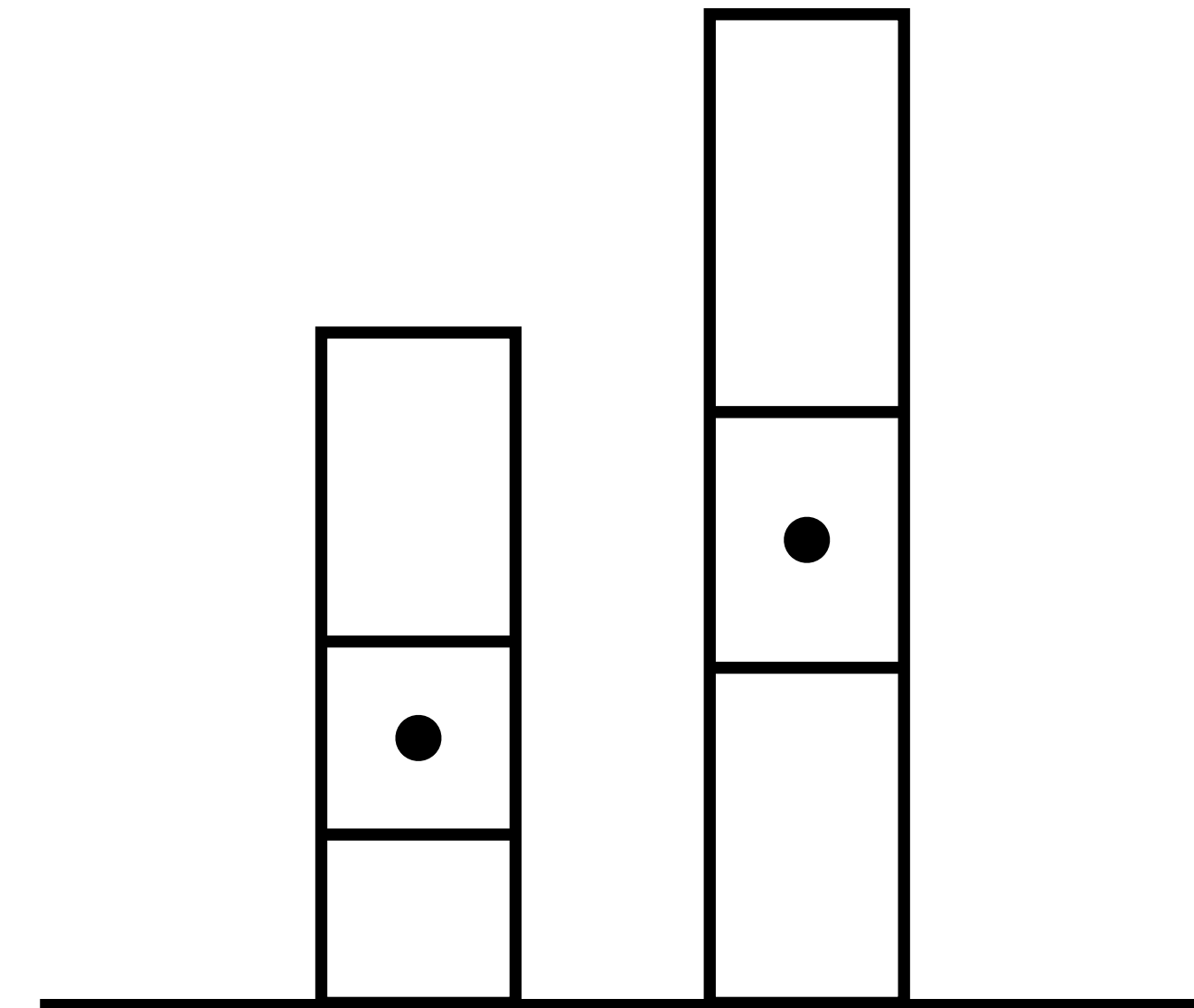
Effectiveness

A visualization is more **effective** than another if the information it conveys is **more readily perceived** than the information in the other visualization.

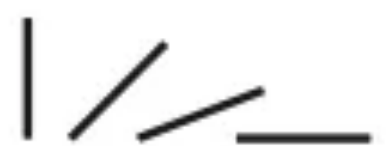
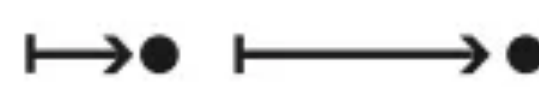
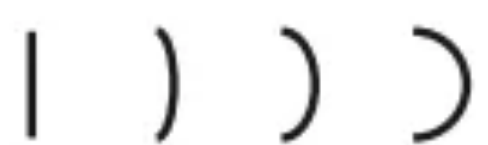
➔ Magnitude Channels: Ordered Attributes

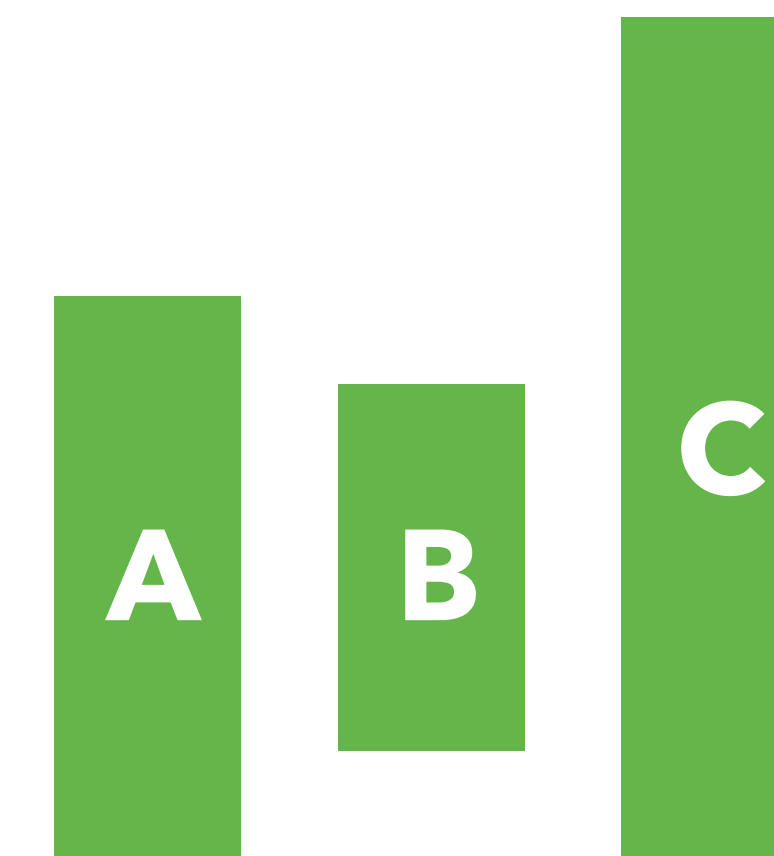
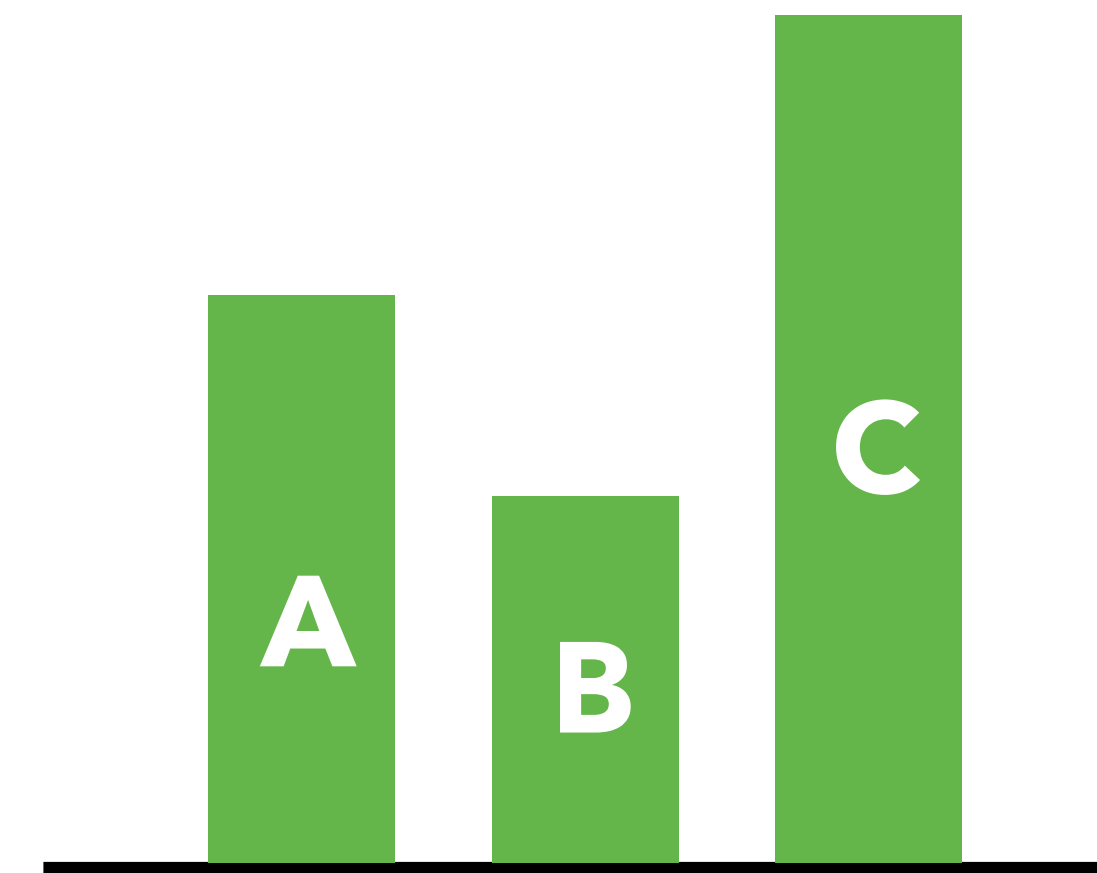


What's ratio of the taller bar to the shorter bar?



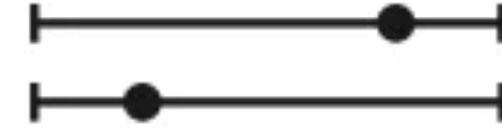
➔ Magnitude Channels: Ordered Attributes

Position on common scale		Most Effectiveness Least
Position on unaligned scale		
Length (1D size)		
Tilt/angle		
Area (2D size)		
Depth (3D position)		
Color luminance		
Color saturation		
Curvature		
Volume (3D size)		

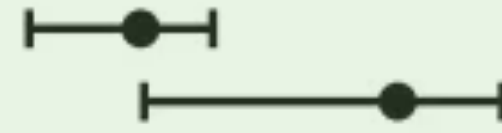


➔ Magnitude Channels: Ordered Attributes

Position on common scale



Position on unaligned scale



Length (1D size)



Tilt/angle



Area (2D size)



Depth (3D position)



Color luminance



Color saturation



Curvature



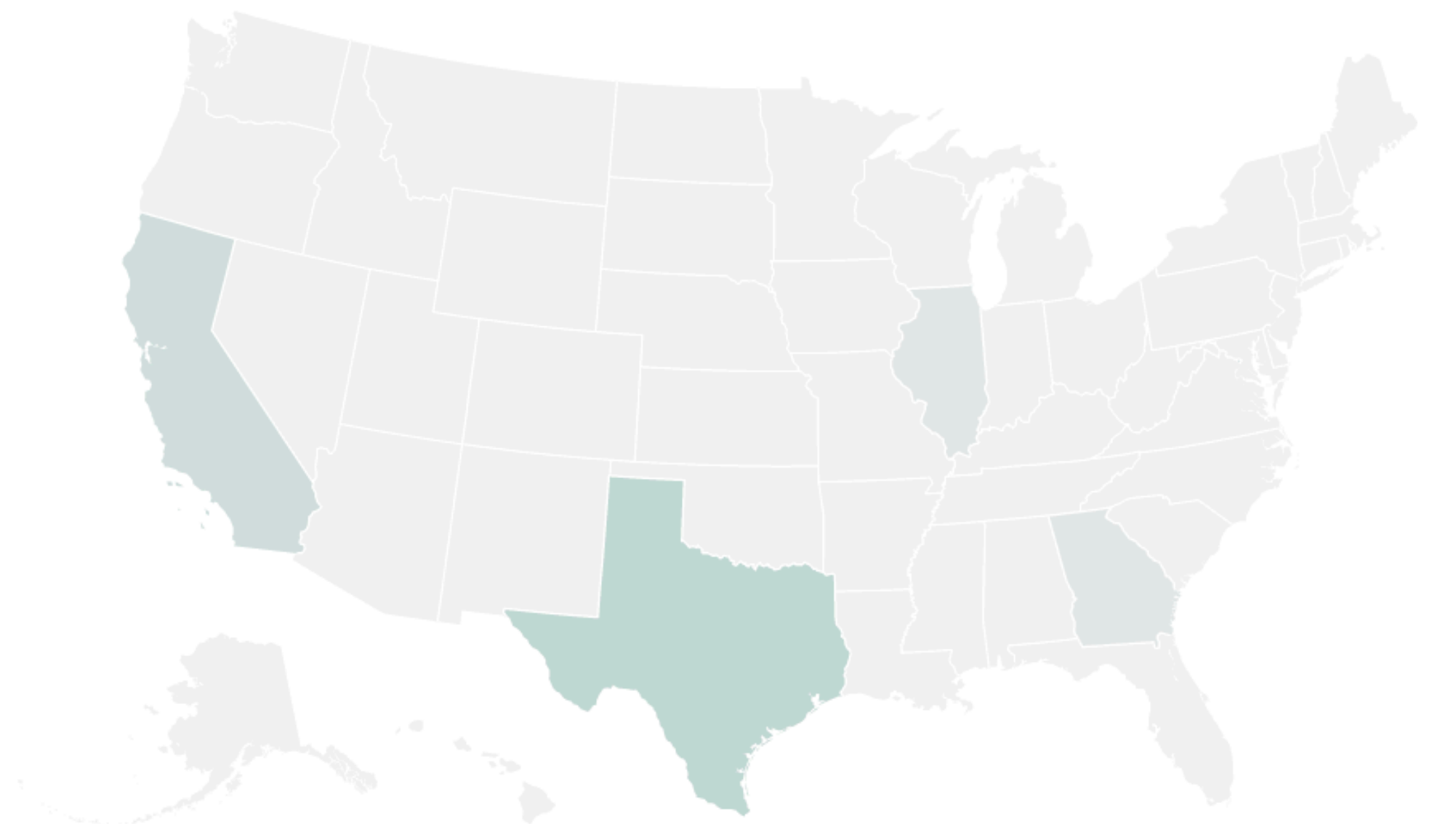
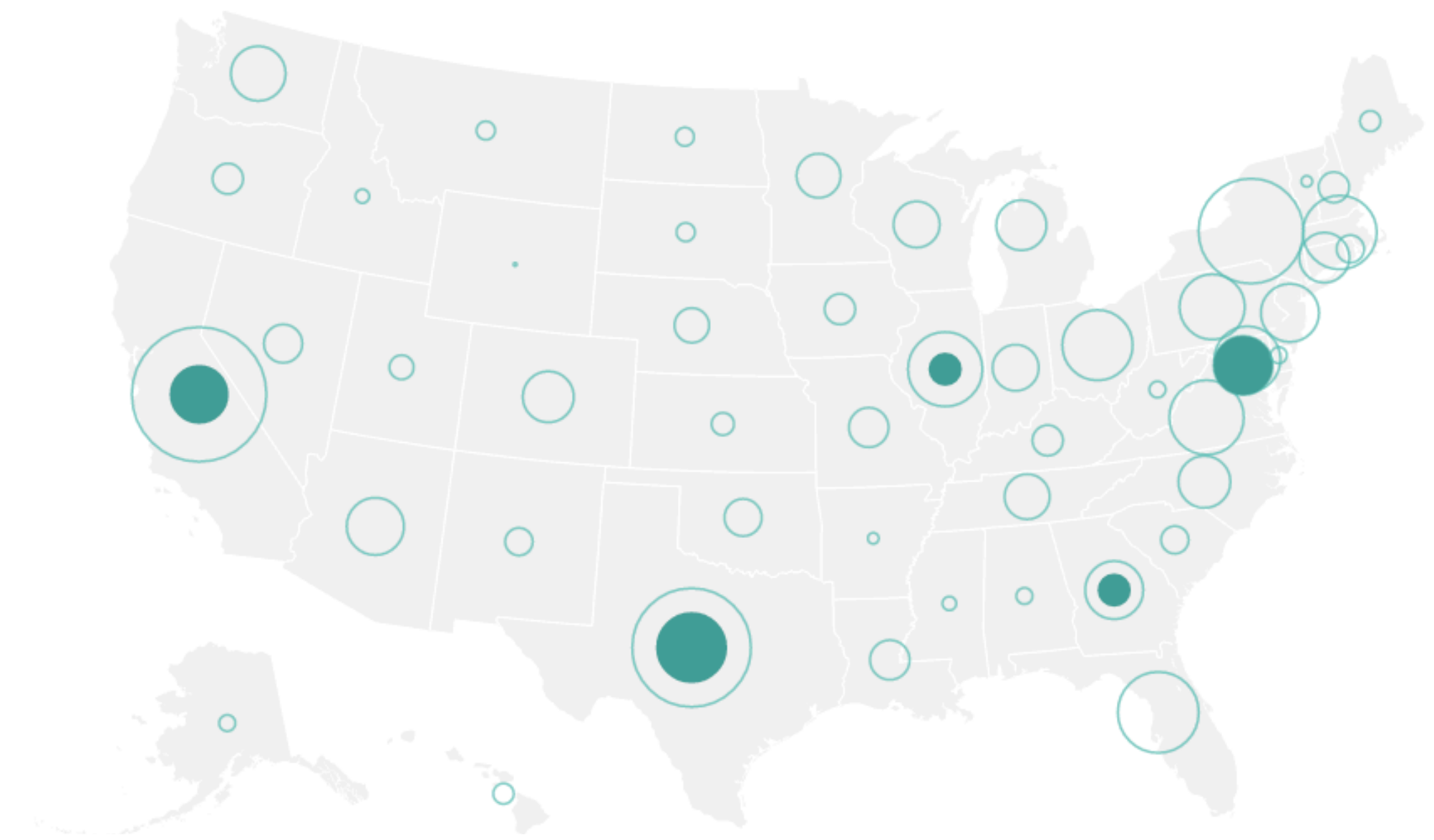
Volume (3D size)



Most

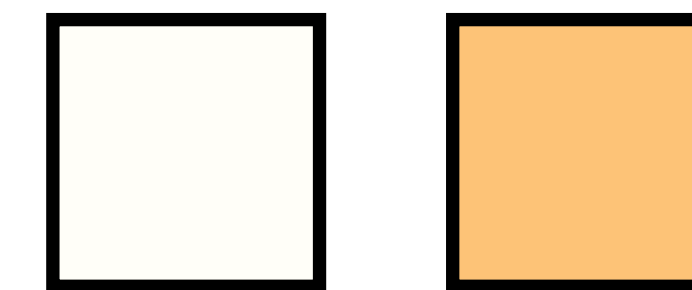
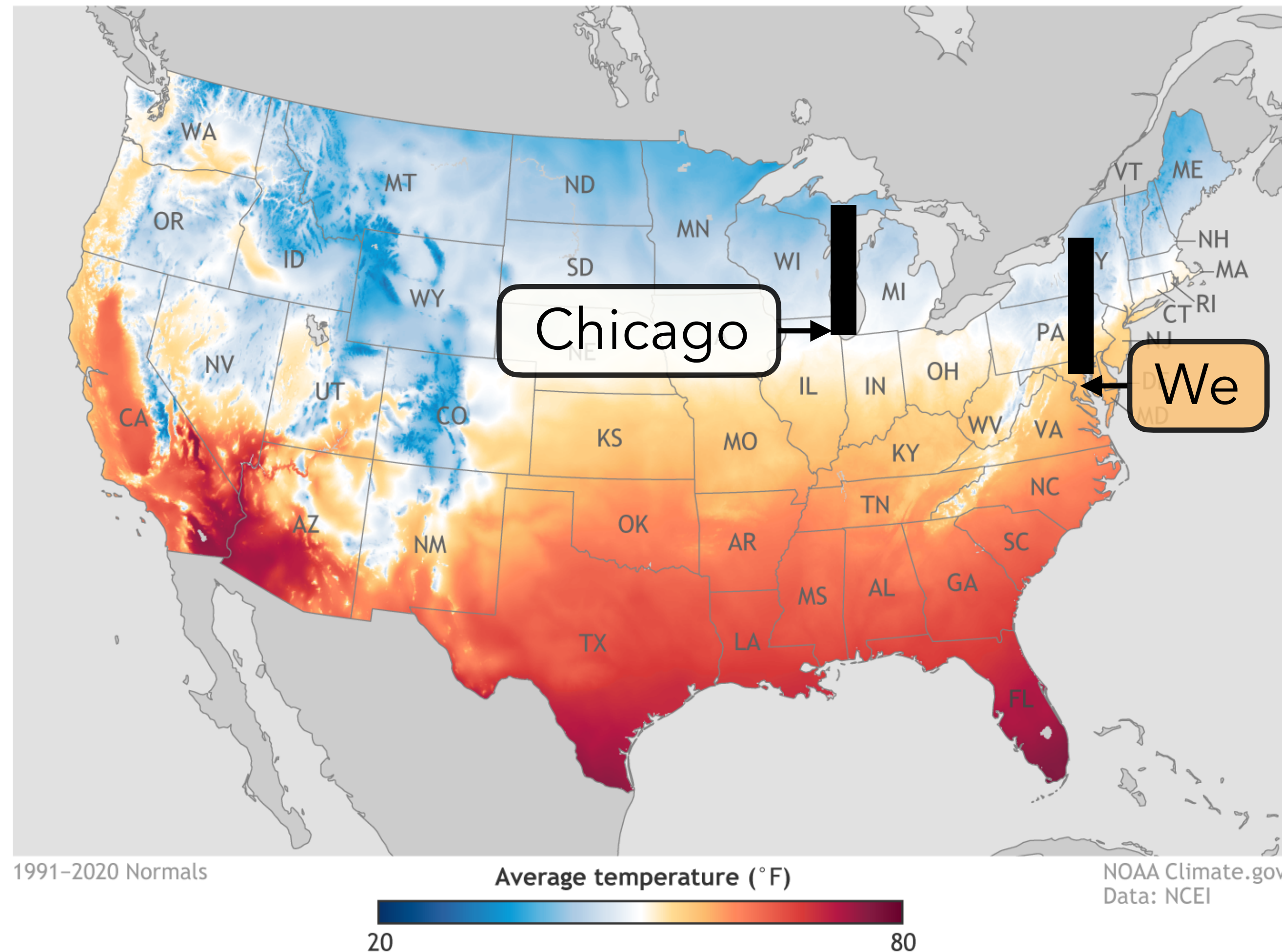
Effectiveness

Least



How many degrees warmer is College Park than Chicago?

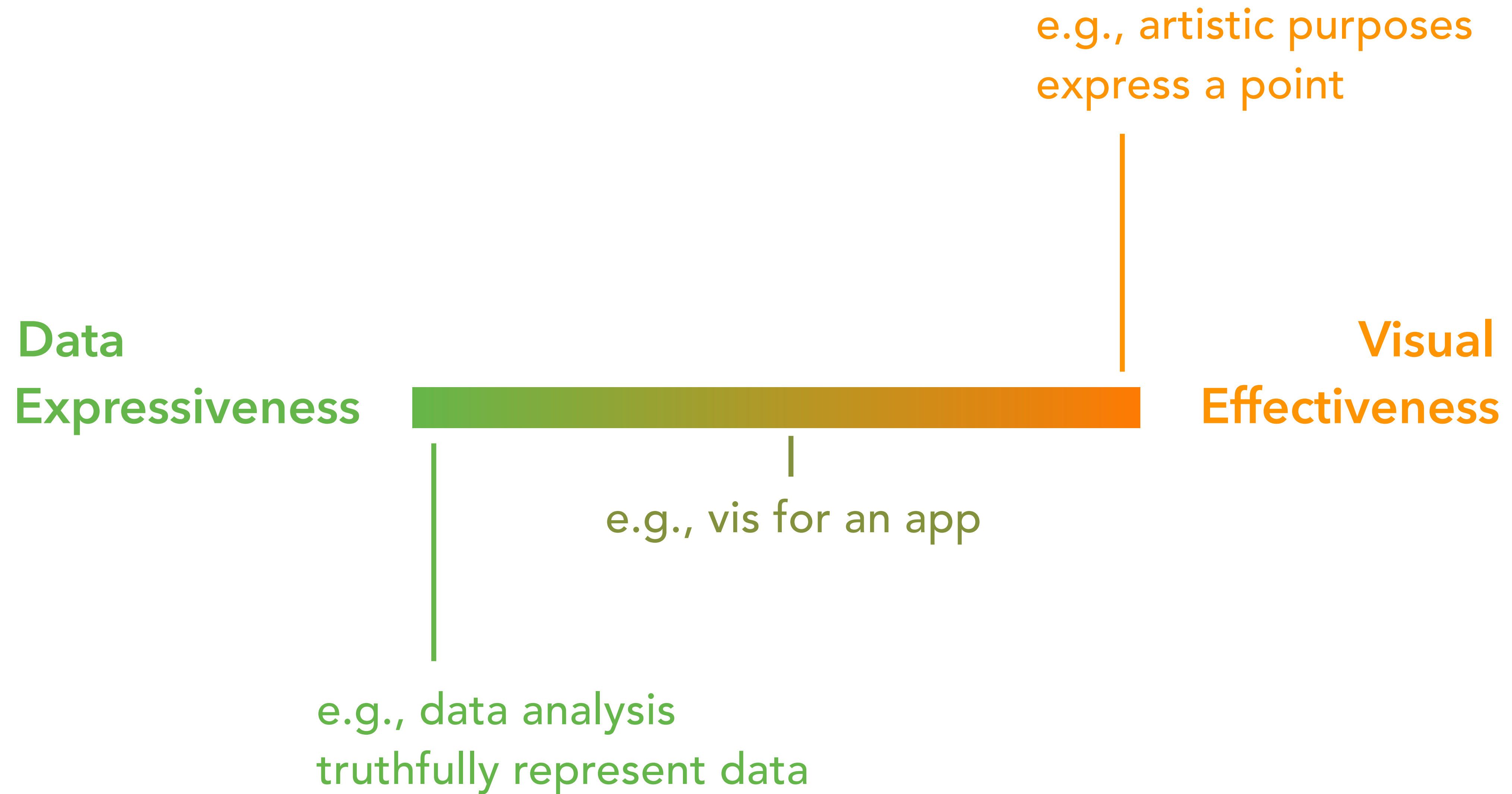
U.S. annual average temperature and precipitation (1991–2020)



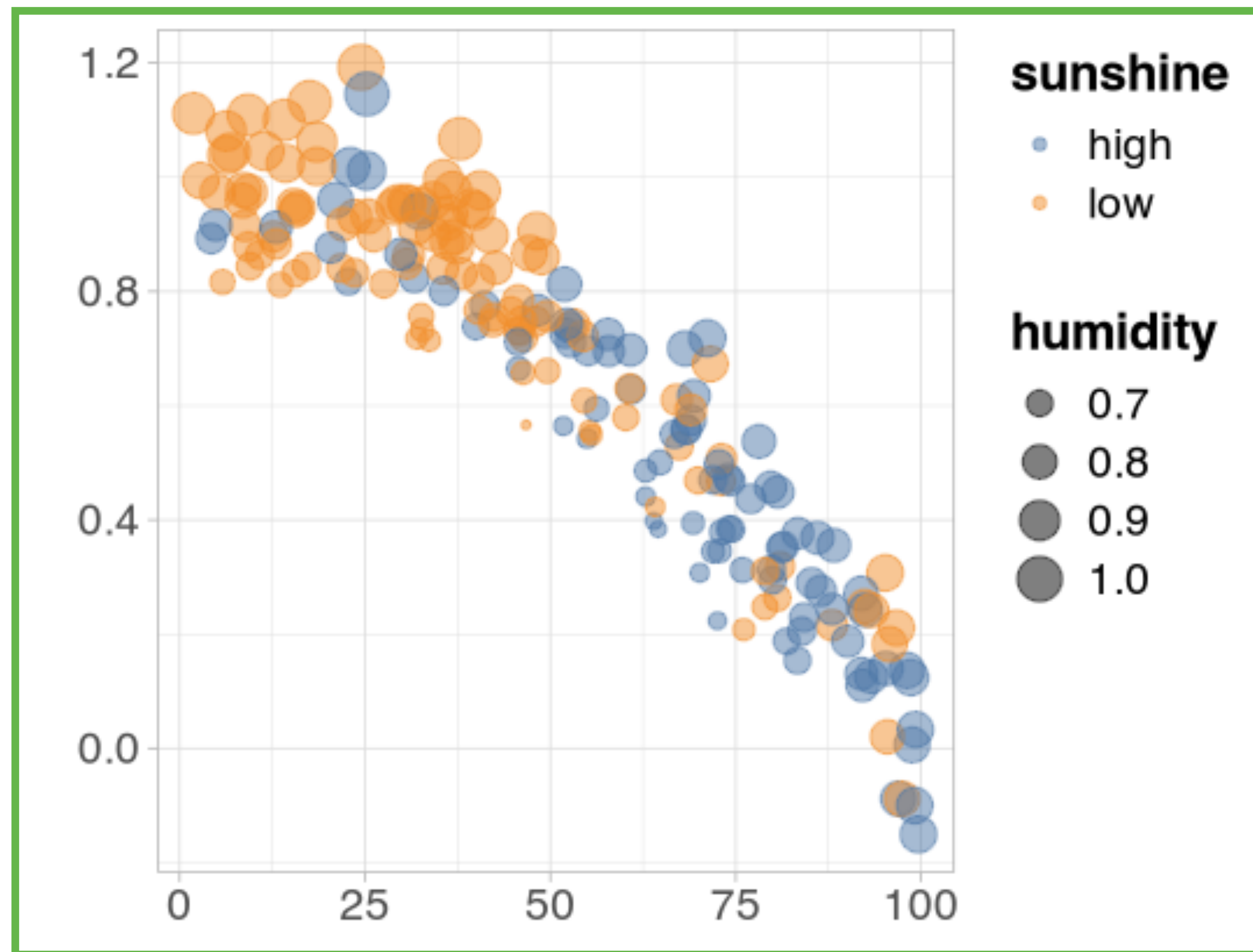
Not impossible

But difficult & prone-to-error for this task

Tradeoff, depending on your goals



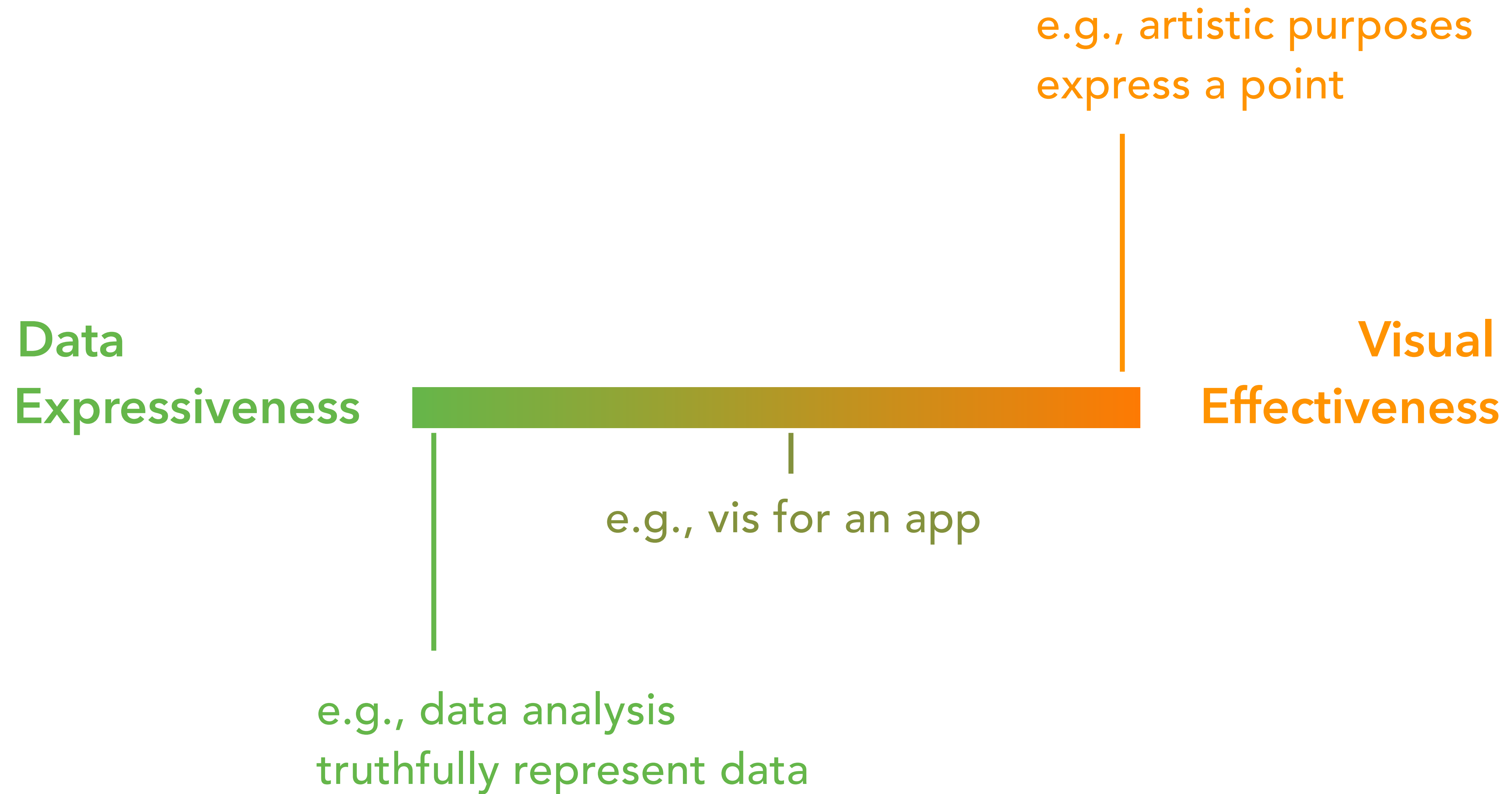
Tradeoff, depending on your goals



↖ e.g., data analysis
truthfully represent data

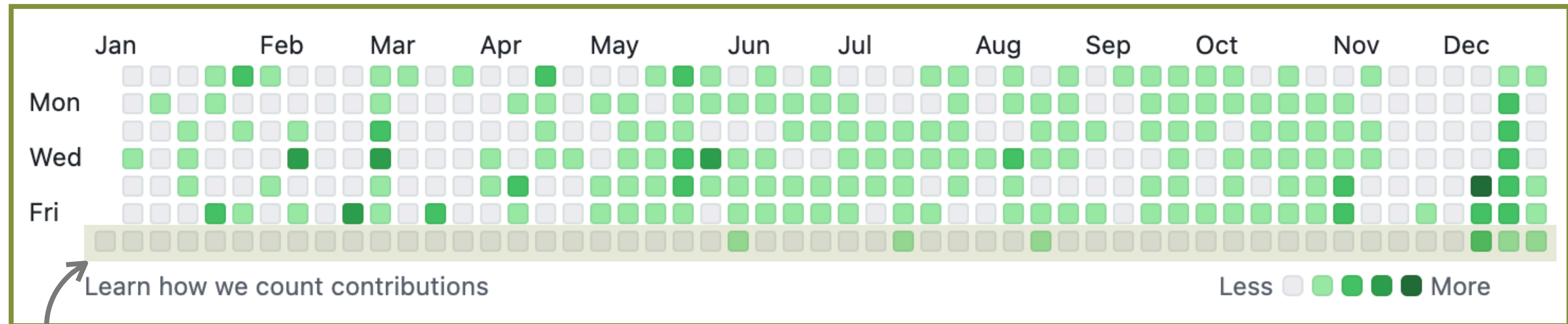
Visual
Effectiveness

Tradeoff, depending on your goals



Tradeoff, depending on your goals

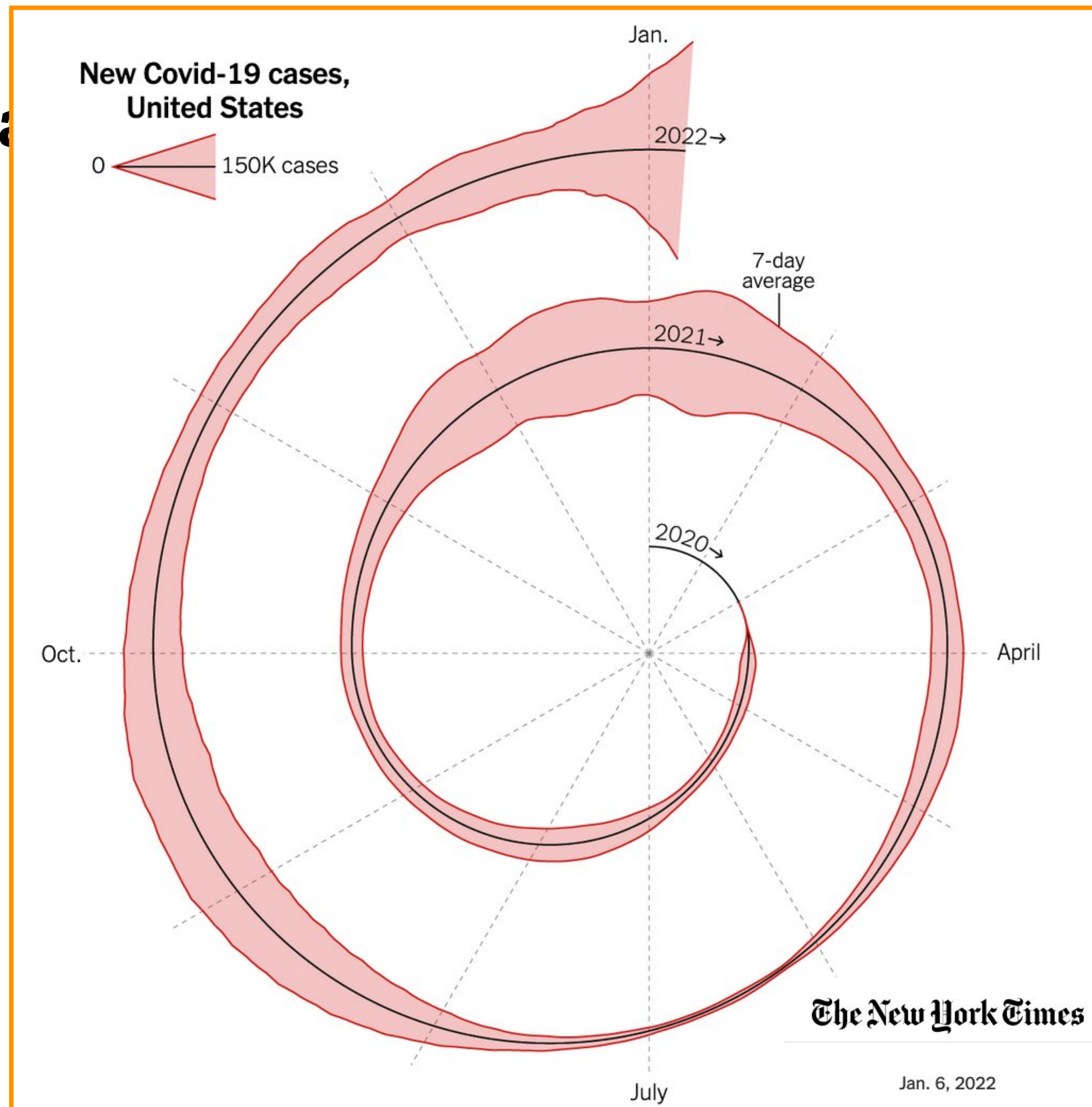
My Github contributions in 2022



I don't code much
on Saturdays

e.g., vis for an app
space constraints

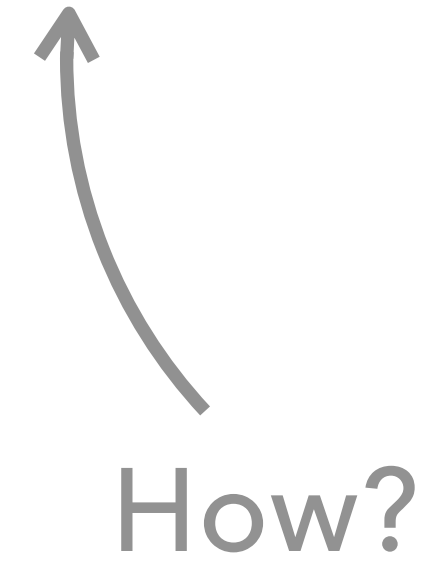
Tr



e.g., artistic purposes
express a point

Visual
Effectiveness

Tradeoff, depending on your goals



Data
Expressiveness



Visual
Effectiveness

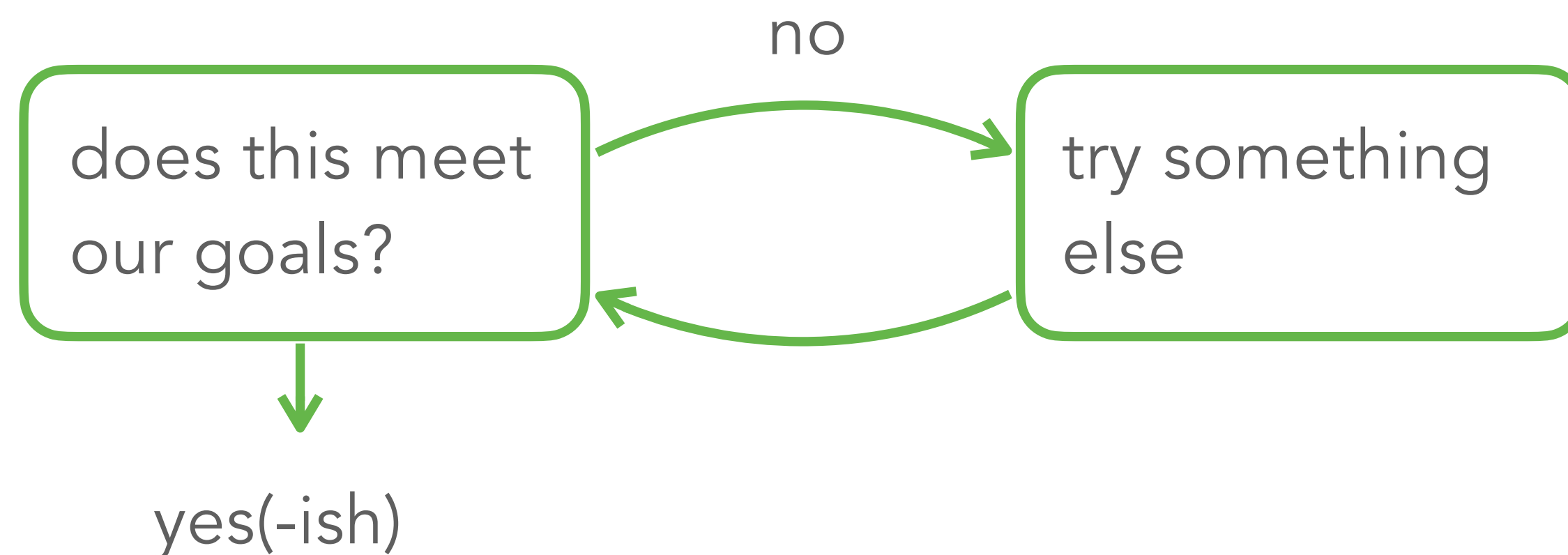
How to **meet** my goals?



If we know our goals...

How to meet our goals

Principles or empirical rules, existing idioms		We'll see a lot in the course
Understand viewer's perception		Next lecture
User study or controlled experiments		Typically more research work
Iterations, discussion with other designers		Work in groups



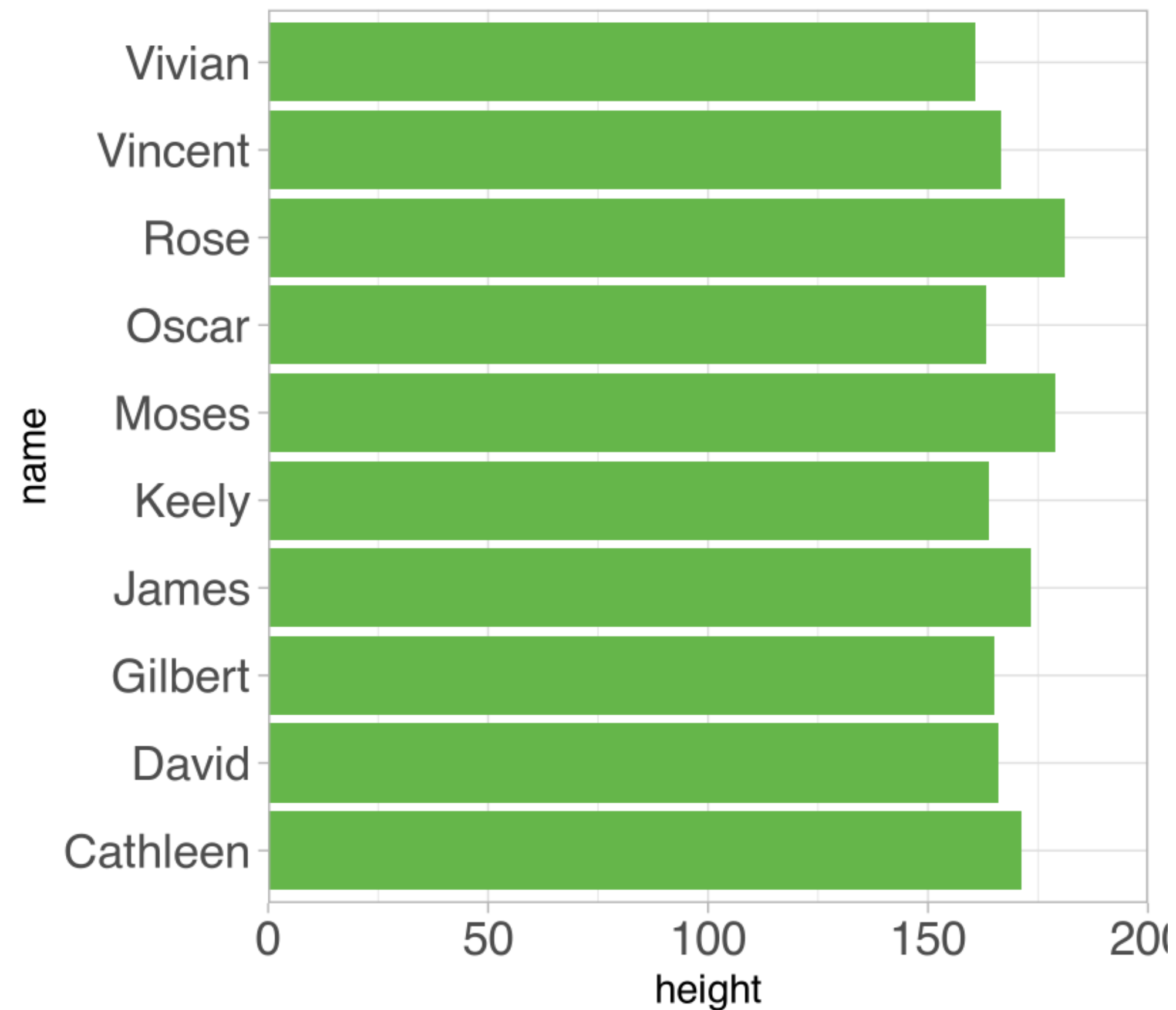
Principles or empirical rules

Using space (in)effectively ◀

Hiding/revealing data

(Mis)leading readers

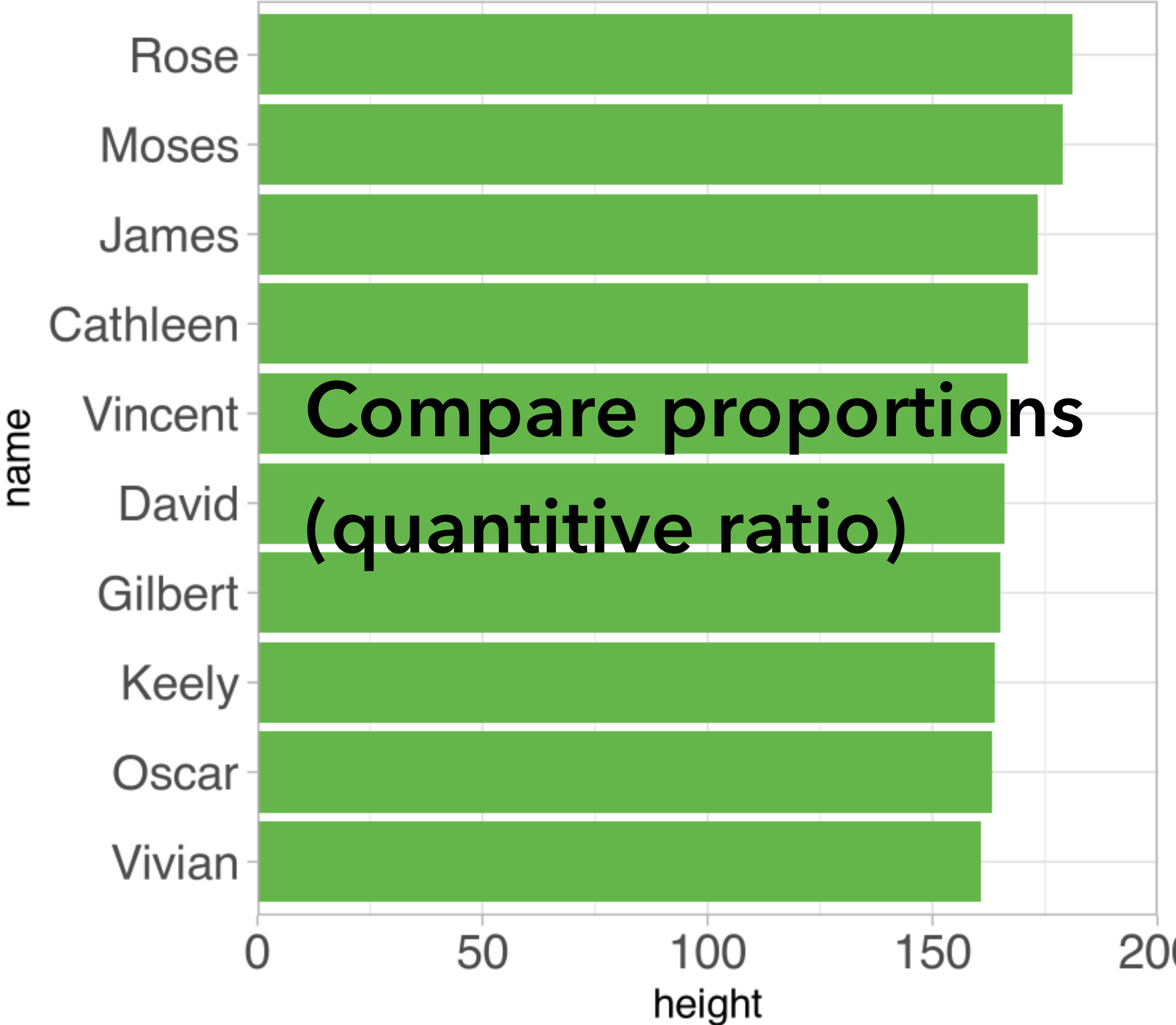
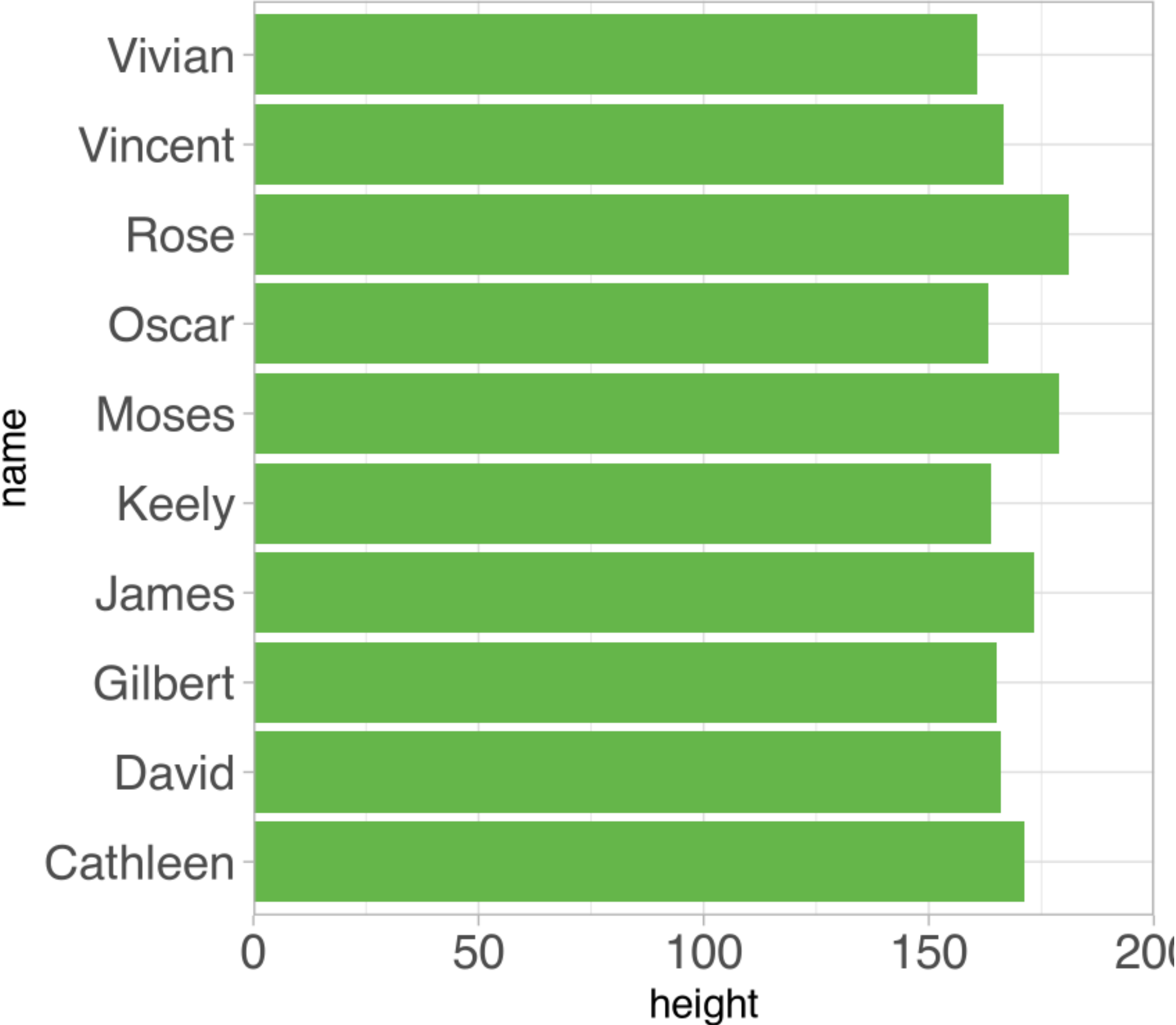
Using space (in)effectively



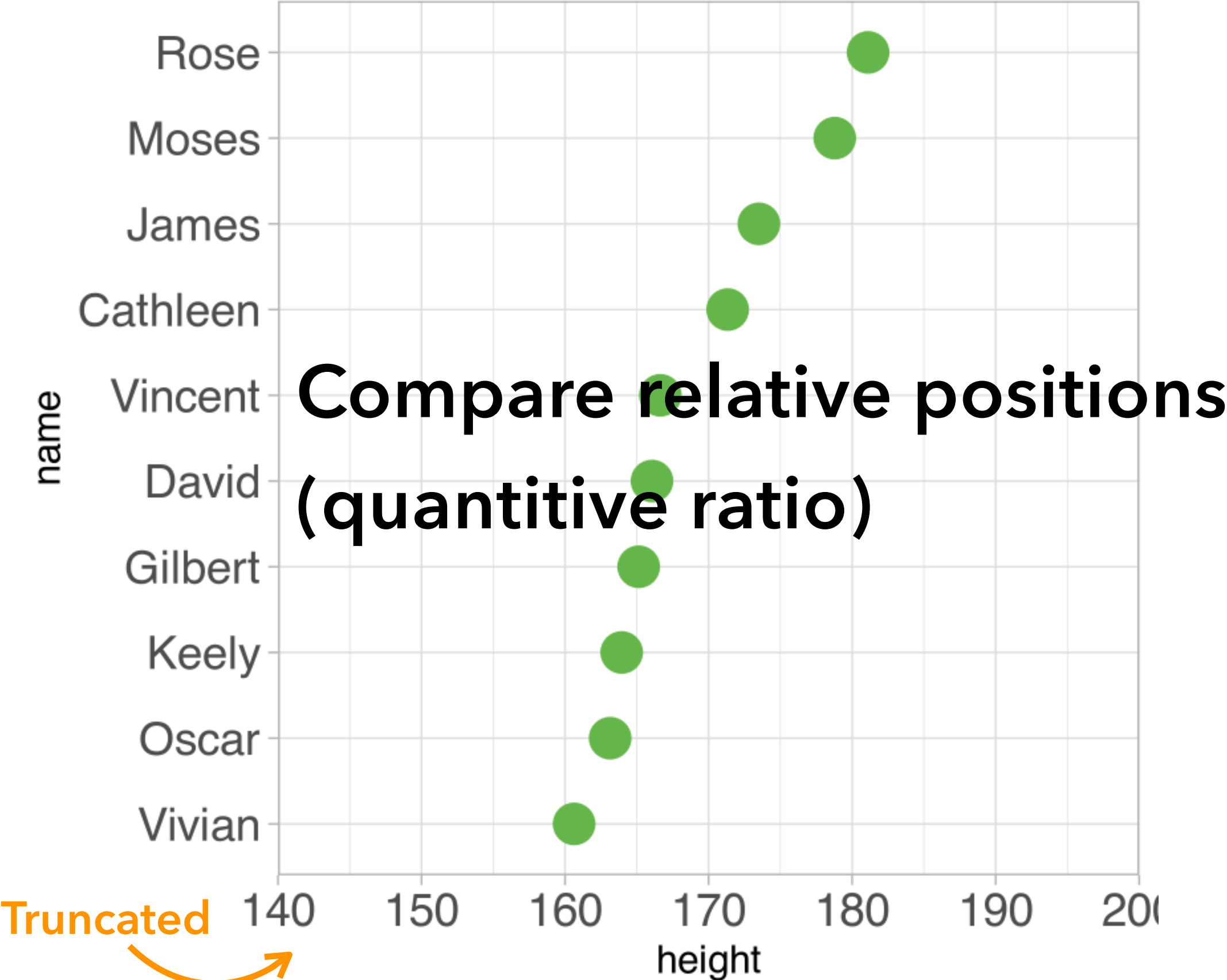
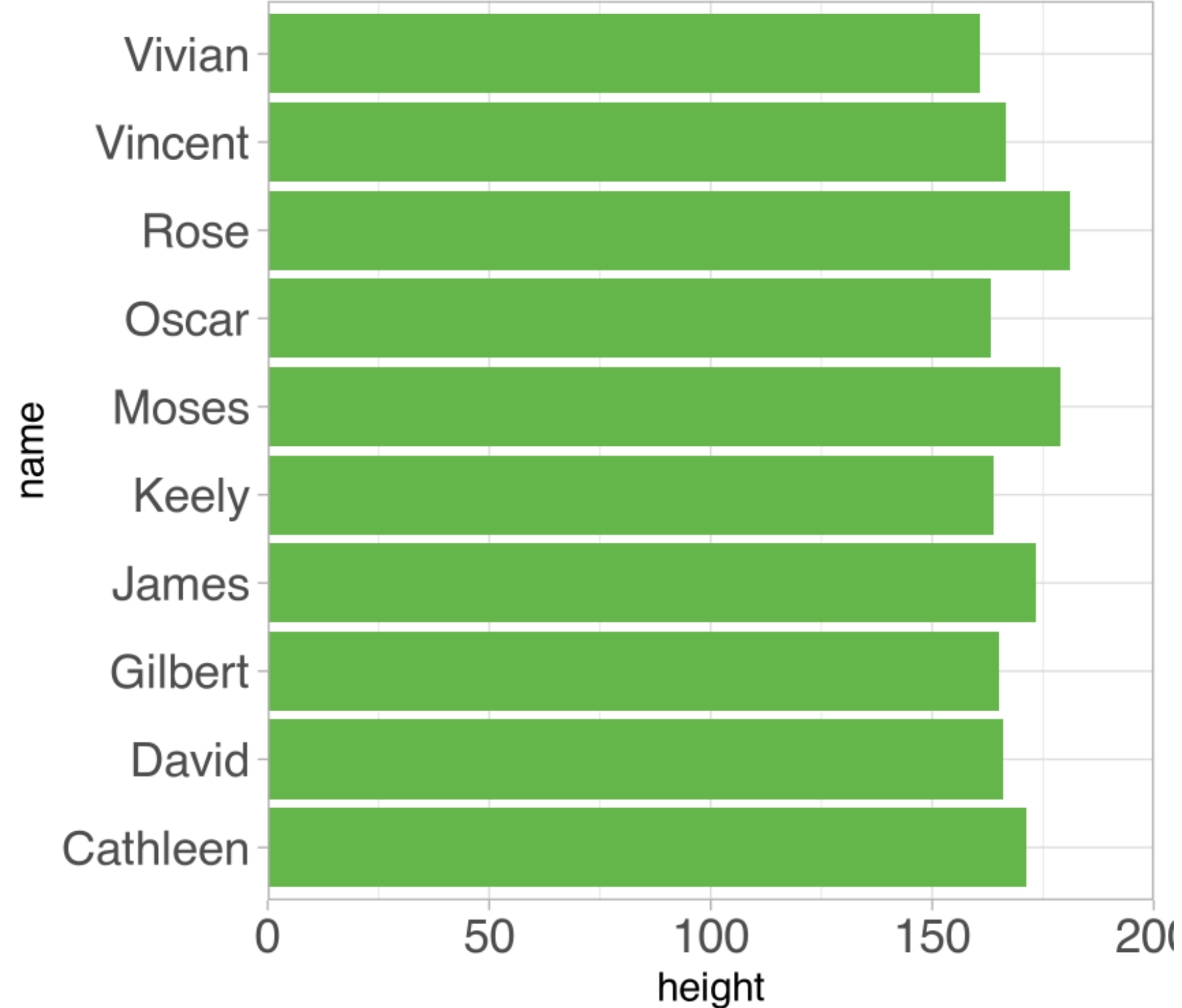
Why might this visualization be inexpressive?

What information is hard to find out through this visualization?

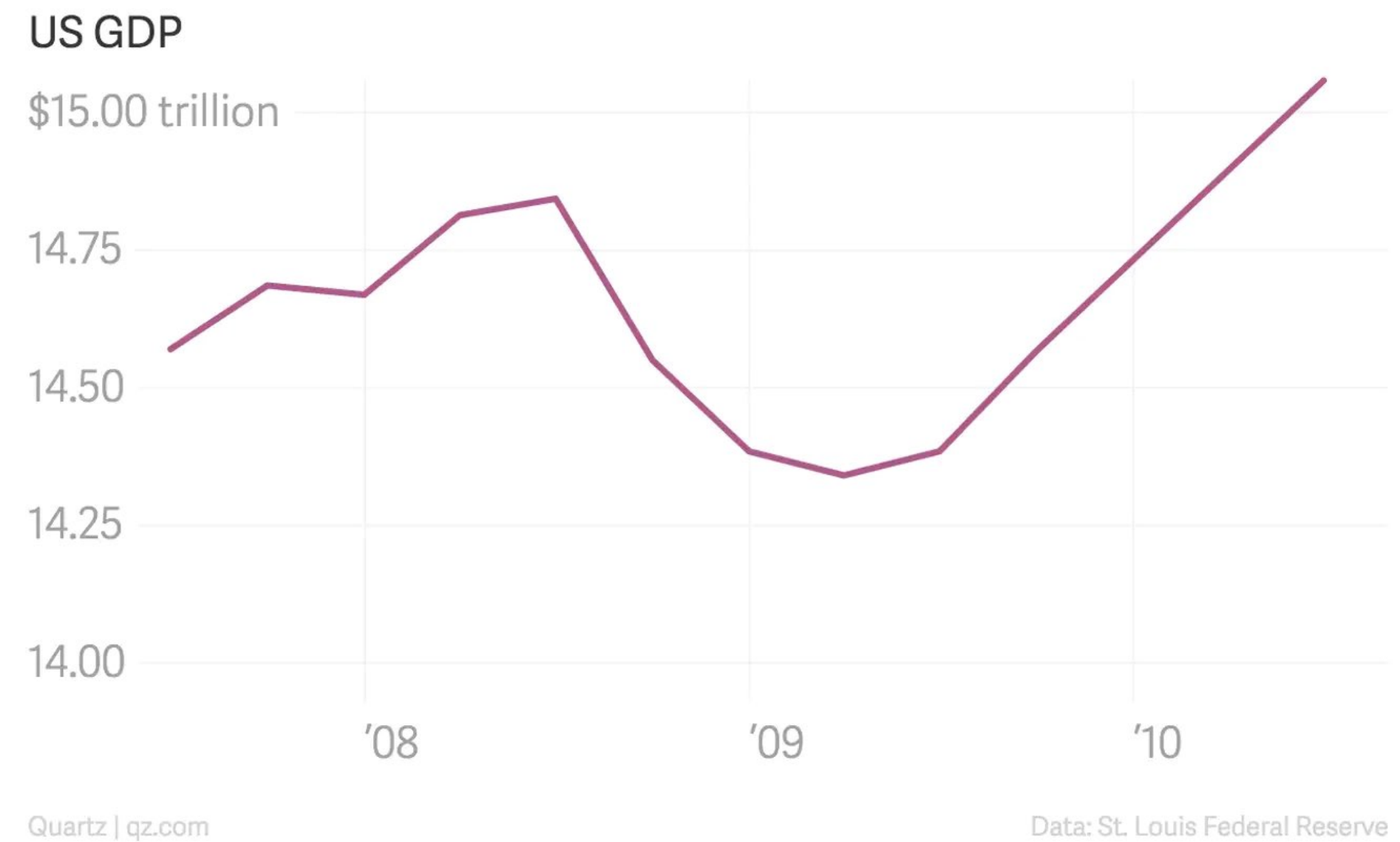
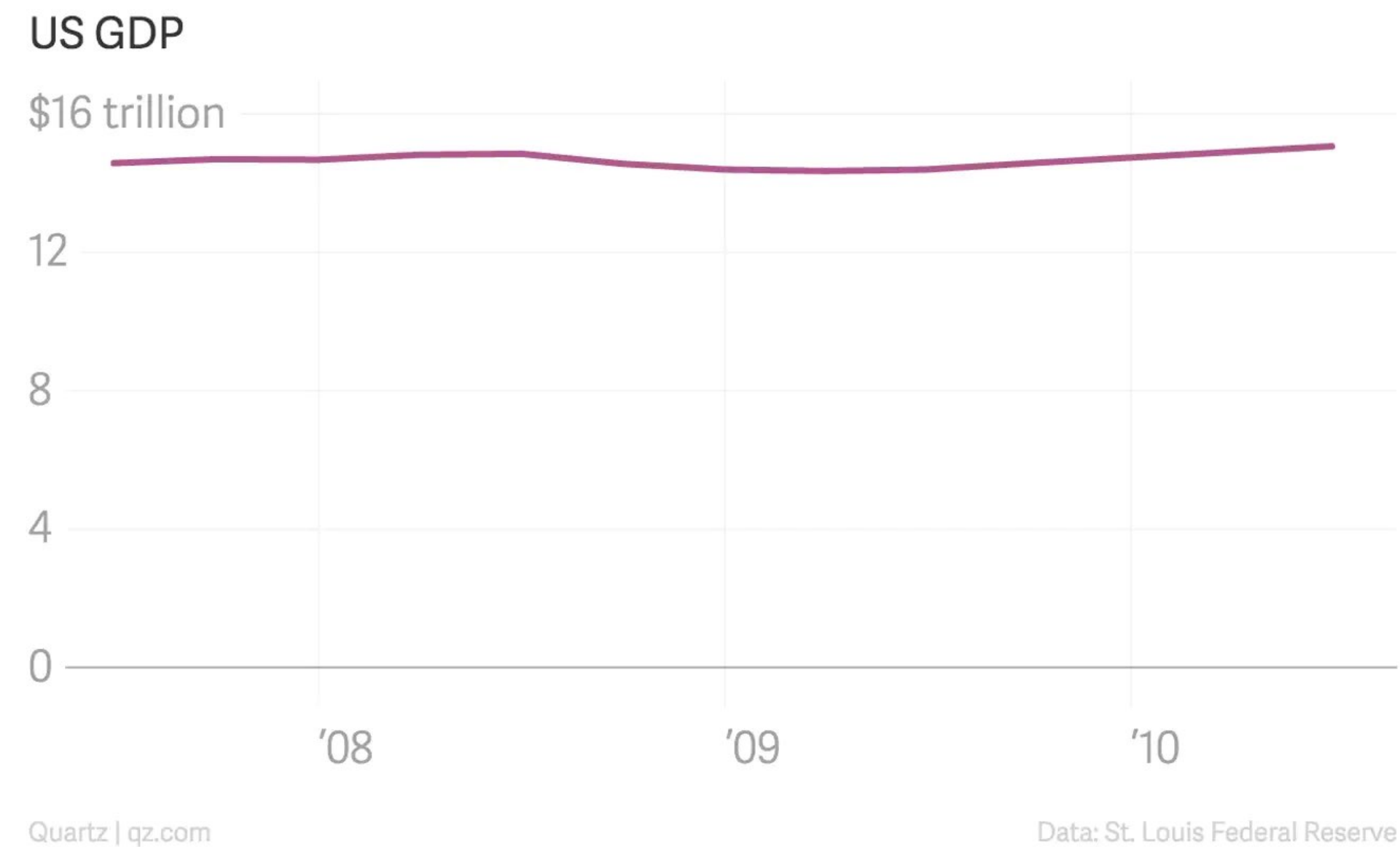
Using space (in)effectively



Using space (in)effectively



Truncate the y-axis: lie? effective?

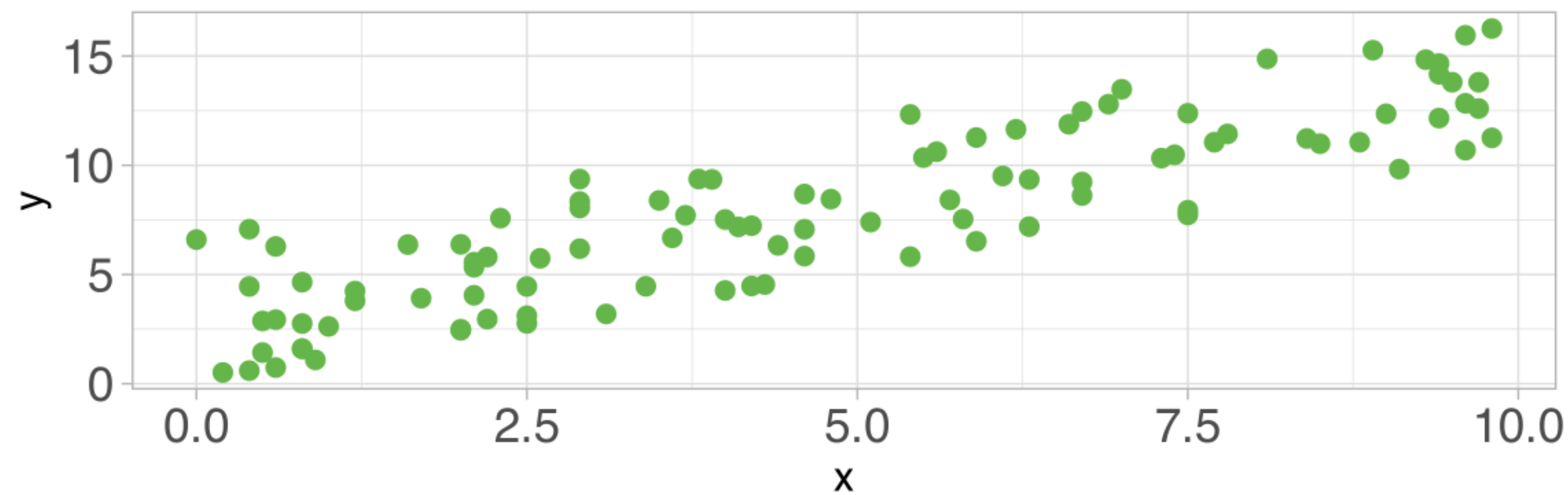


If the zero value doesn't make much sense.

To emphasize quantitative-interval (vs. quantitative-ratio) comparisons.

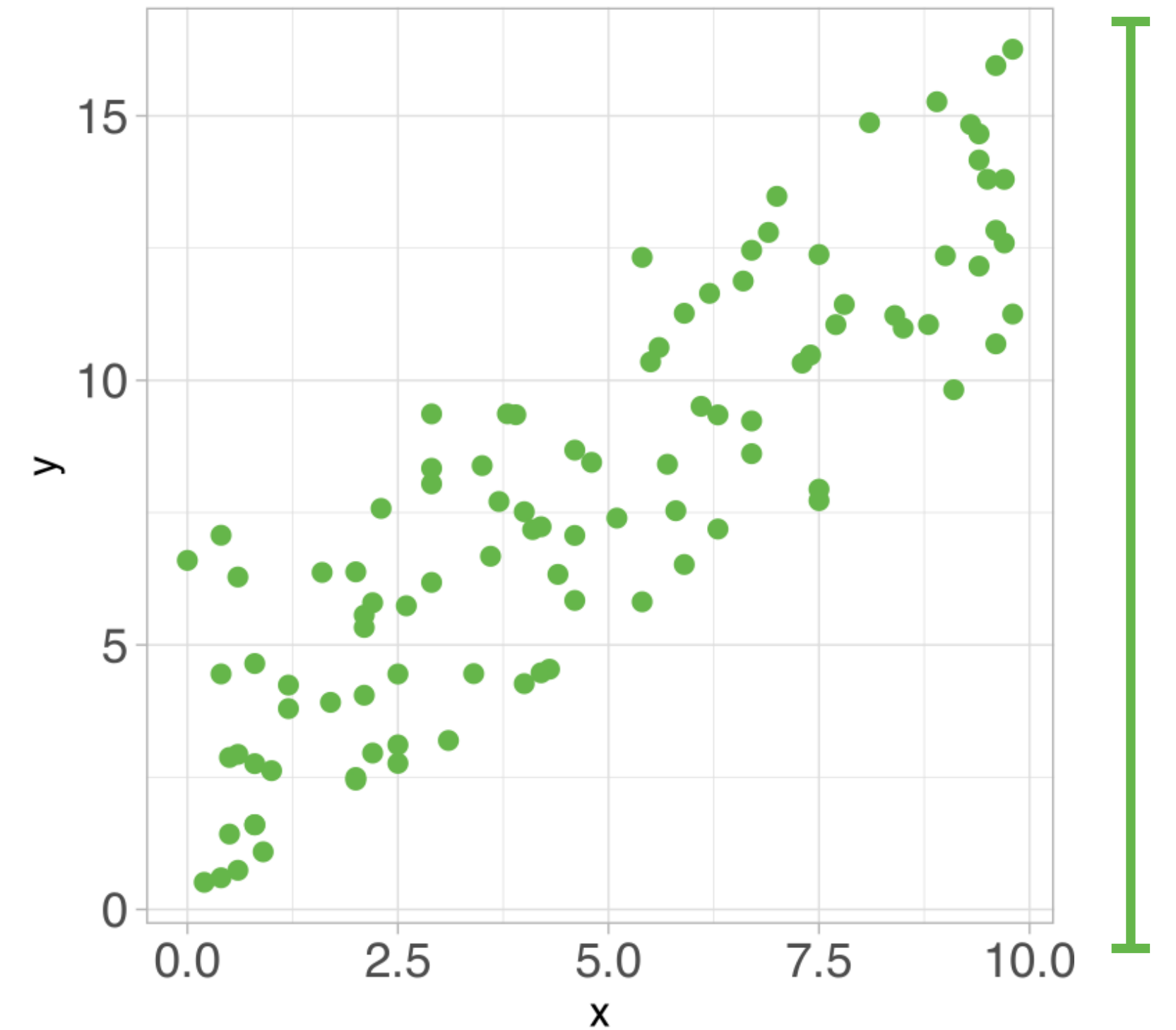
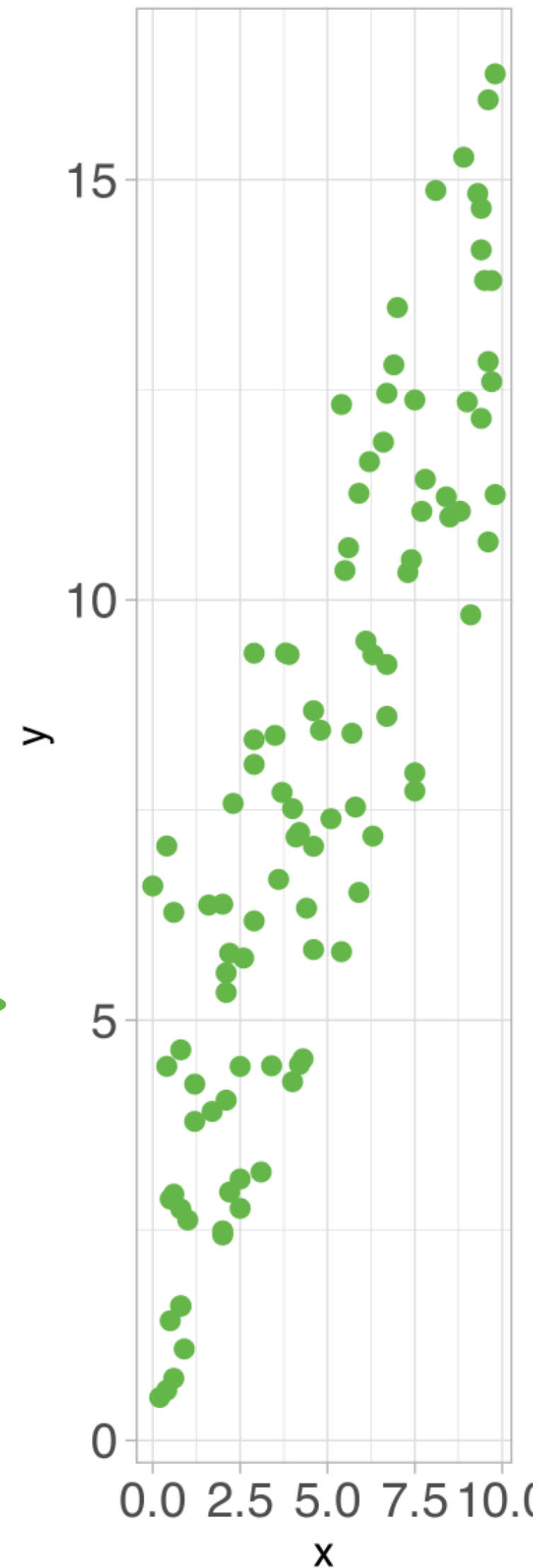
If it is the norm (e.g., stock charts).

Reasonable aspect ratio



Does one chart appear more correlated than the other?

They are the same data!

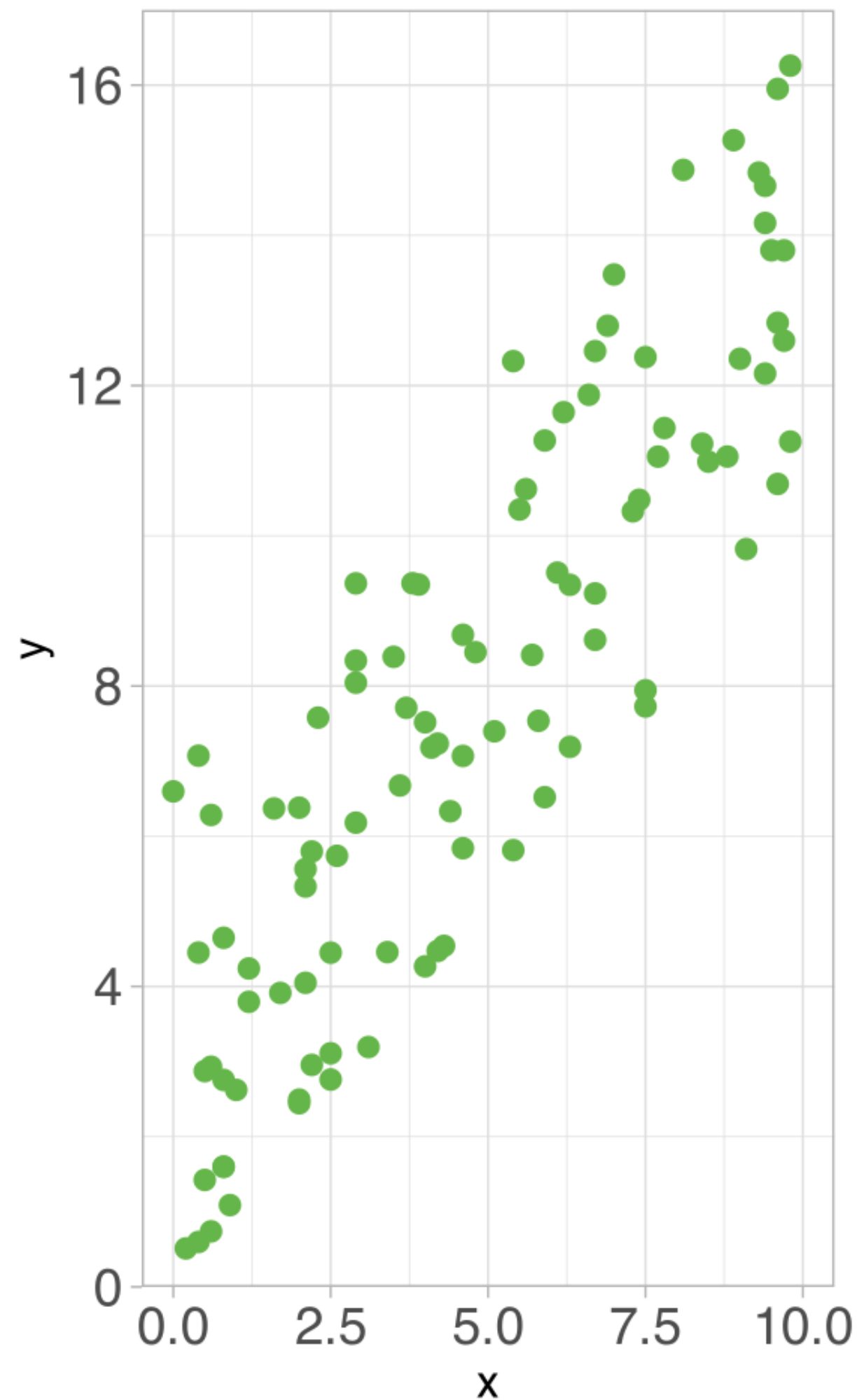


How about this chart?

$x_{\text{diff}} = 10$

$y_{\text{diff}} = \sim 20$

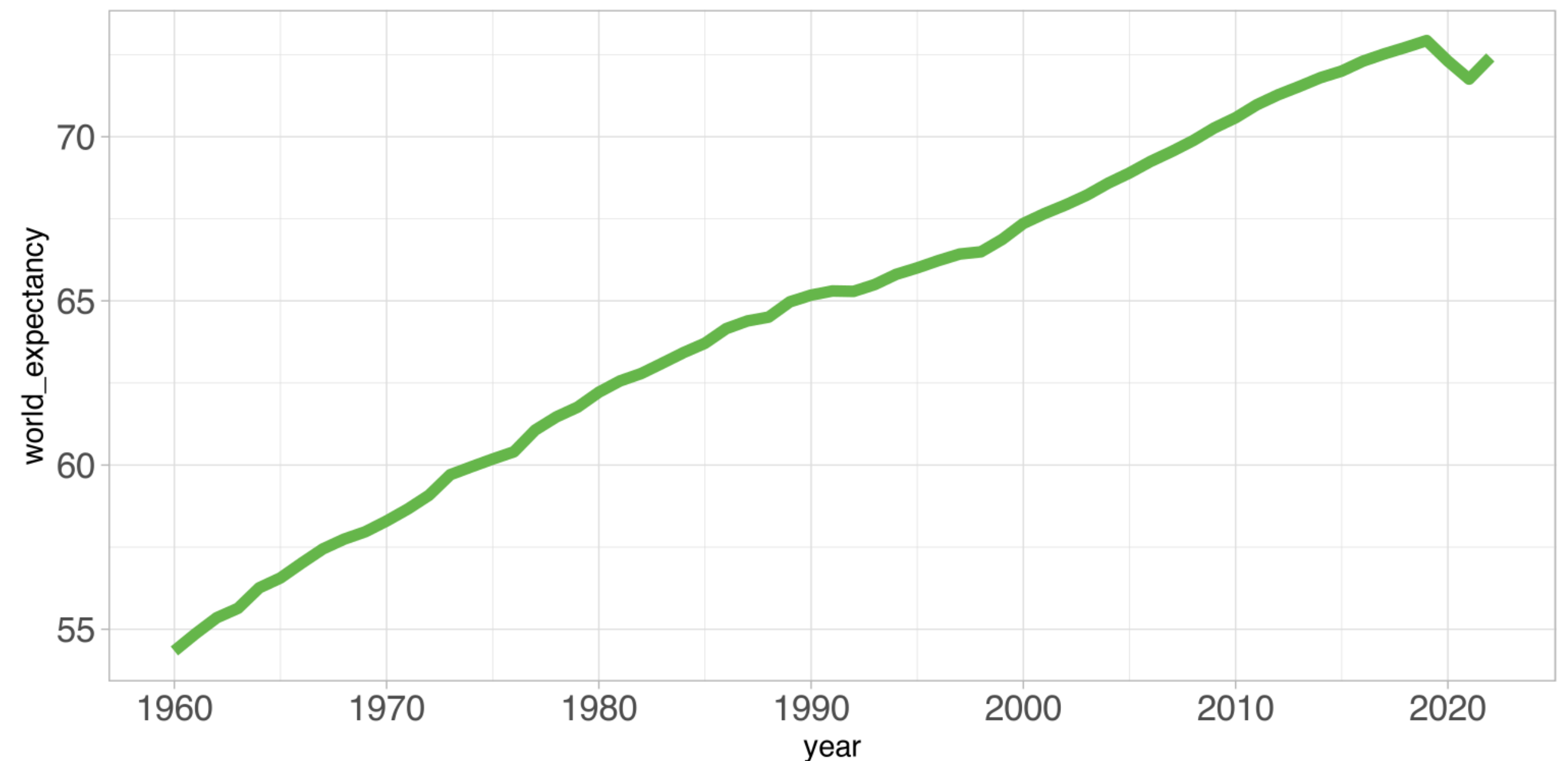
Reasonable aspect ratio



Use aspect ratio (**visual**) to reflect **data** ratio

$y_{\text{diff}} = \sim 17$, $x_{\text{diff}} = \sim 10$, so $y:x = \sim 1.7$

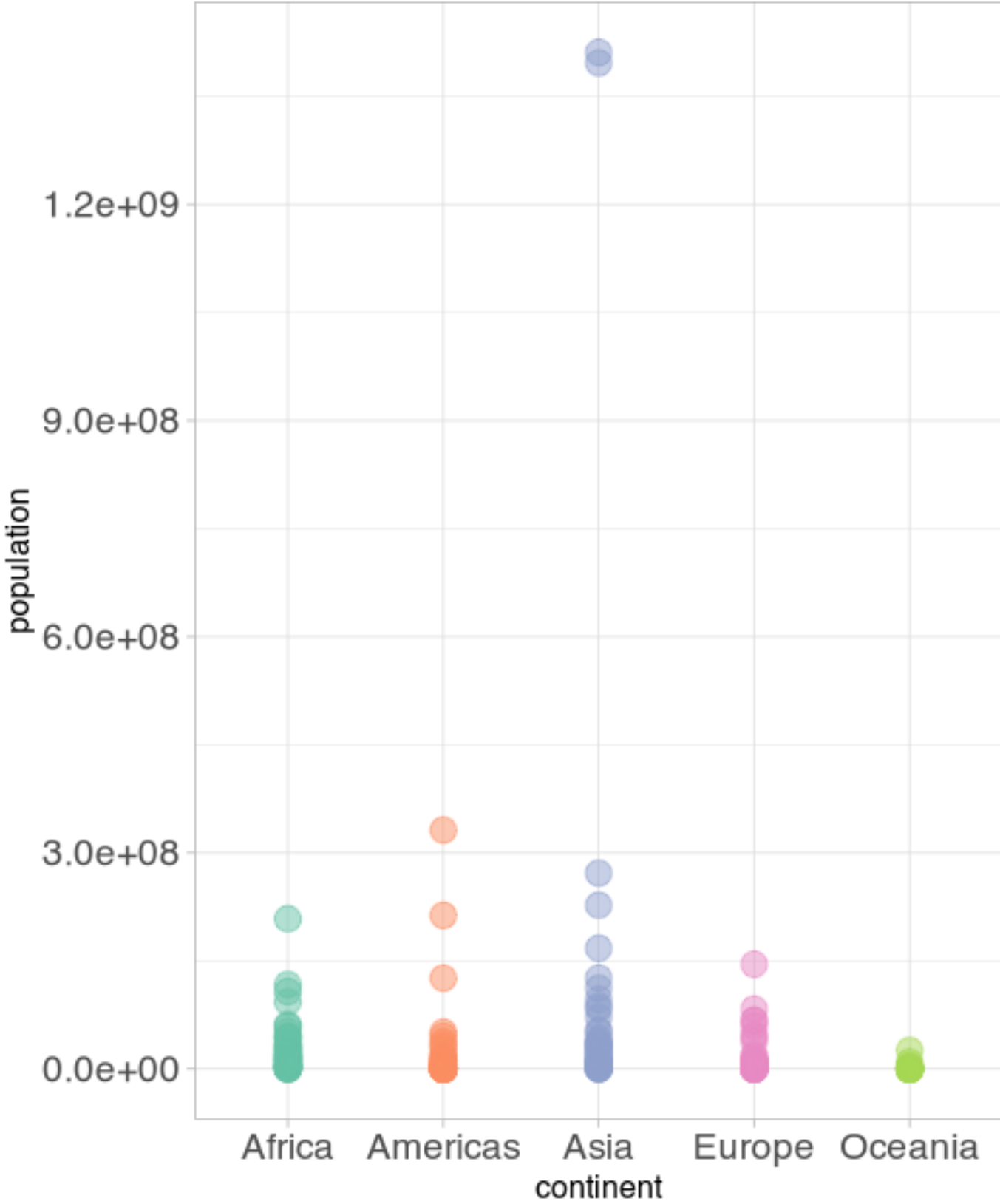
35% increase $\approx 1/3 \approx 1:3$ aspect ratio



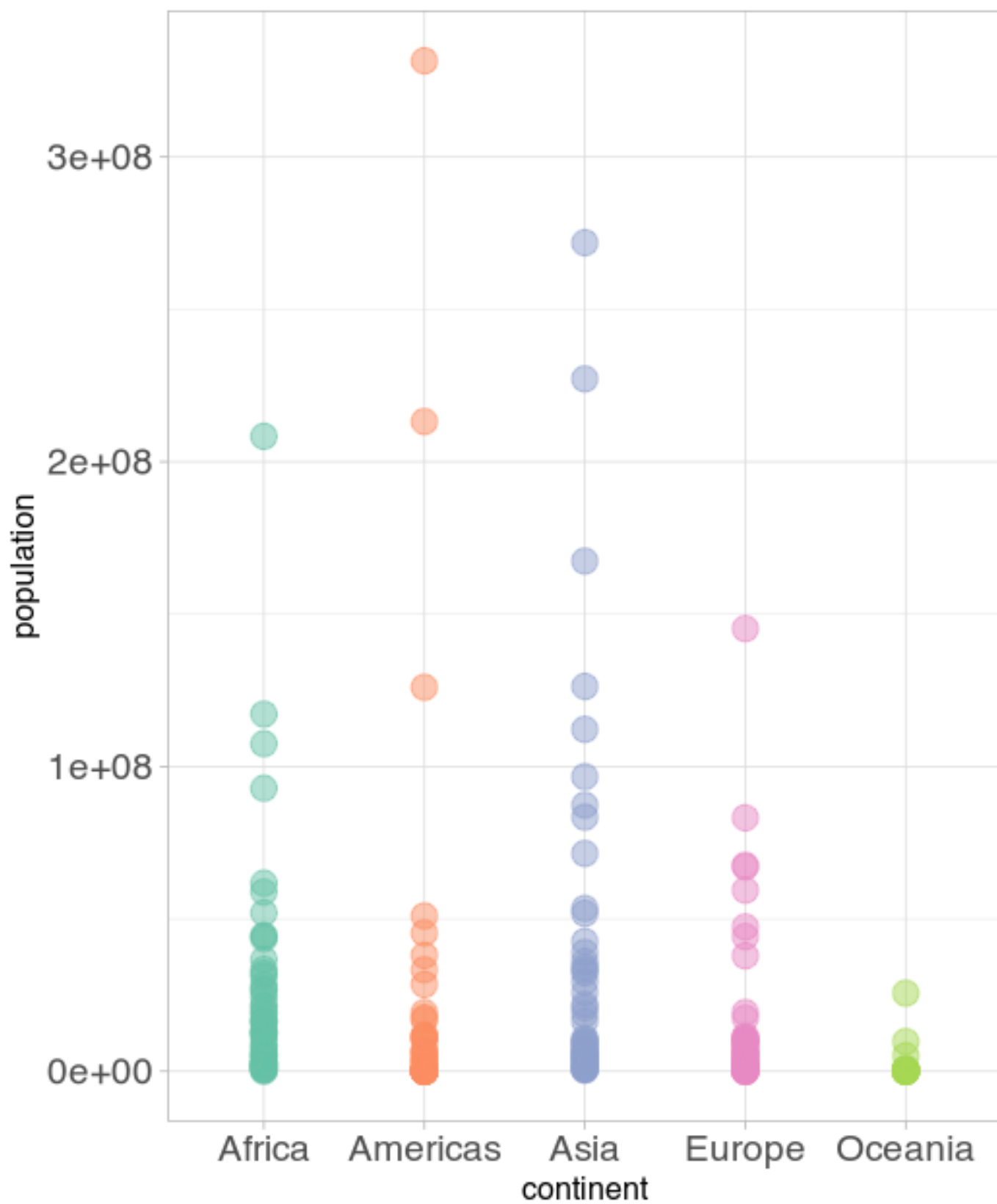
Scaling axes: outliers & skew

not easy to compare
but showing all data

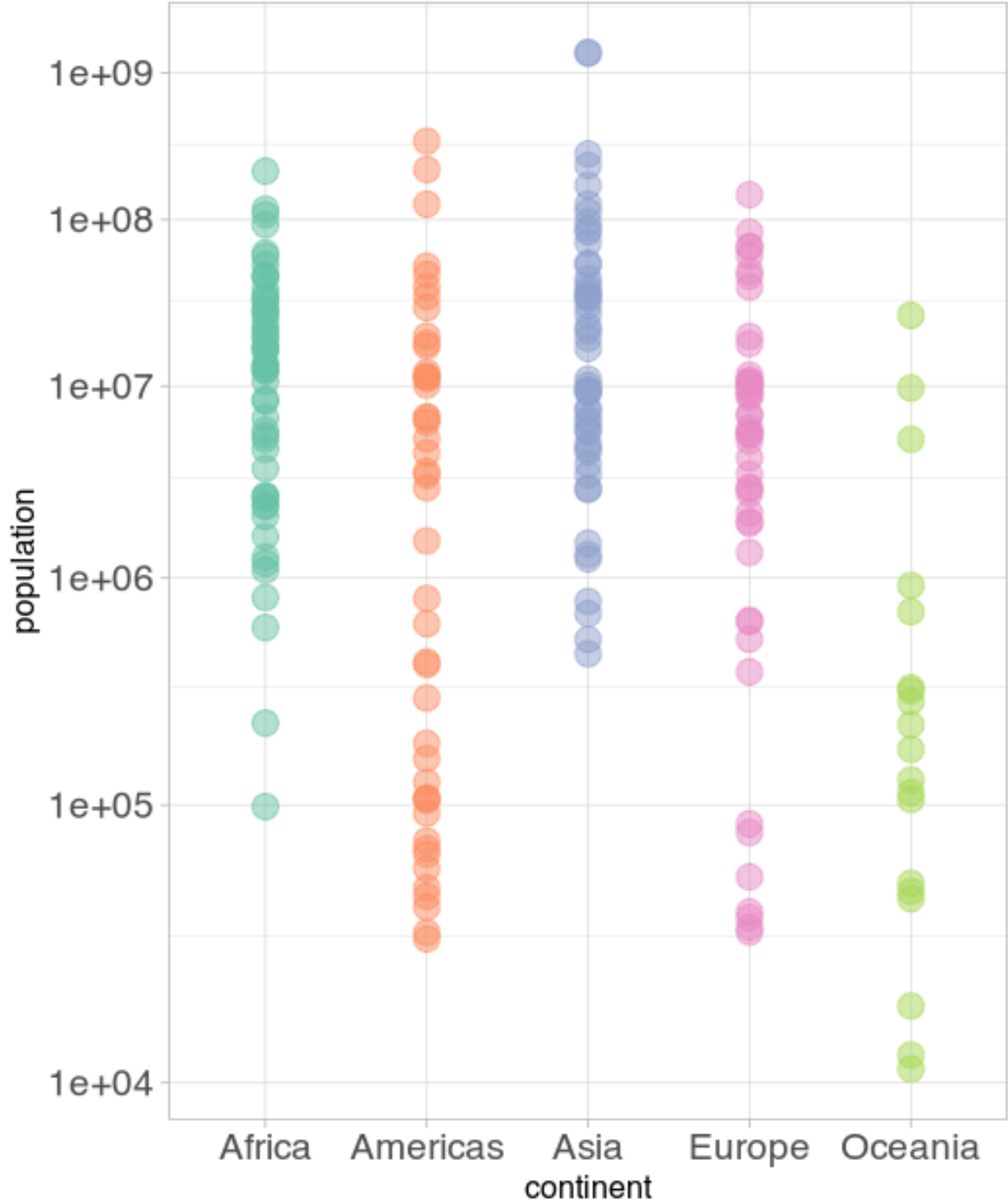
Not effective?



Clip them out?



Log scale



Principles or empirical rules

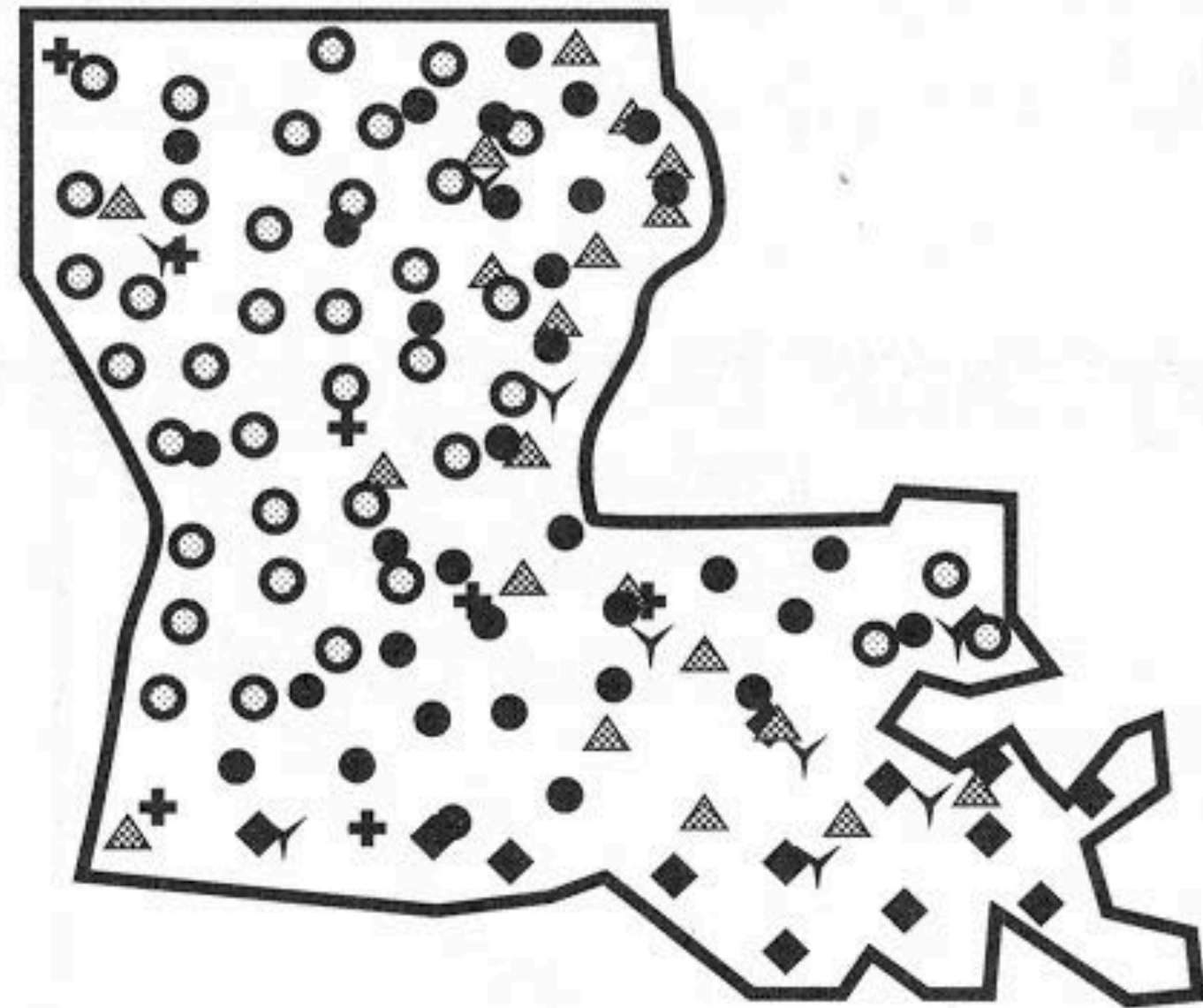
Using space (in)effectively

Hiding/revealing data ◀

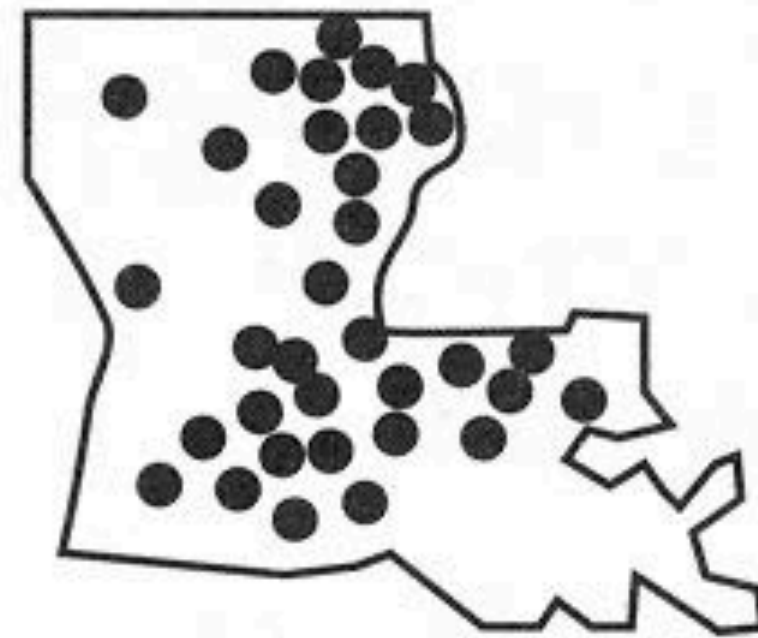
(Mis)leading readers

Small multiples

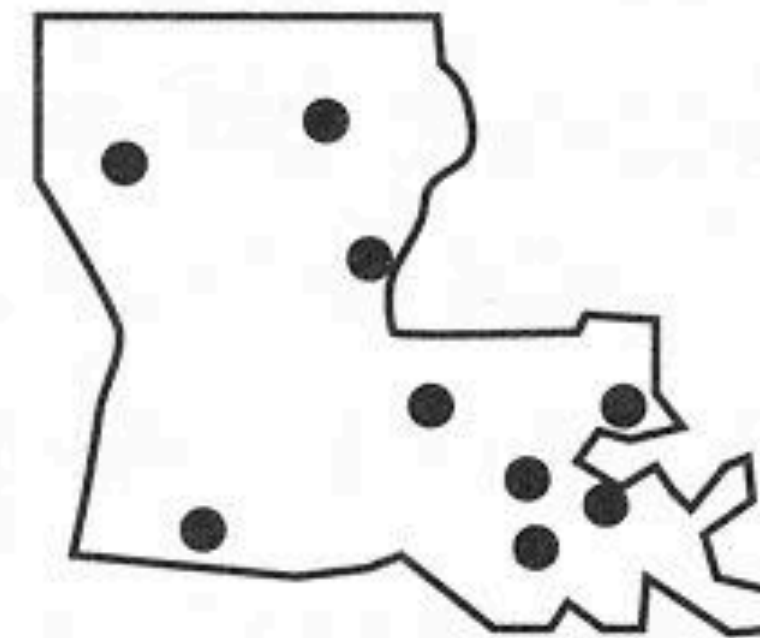
Different types mapped
in Louisiana



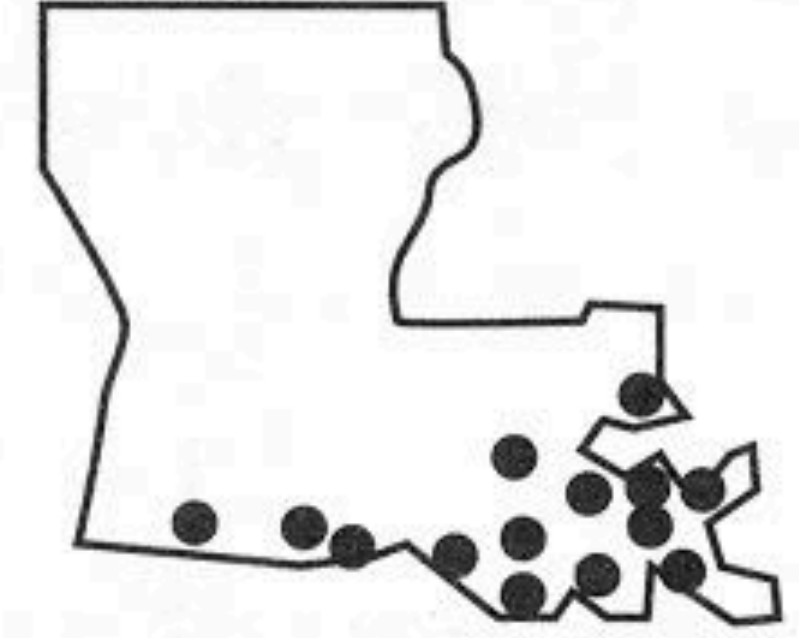
alfisol



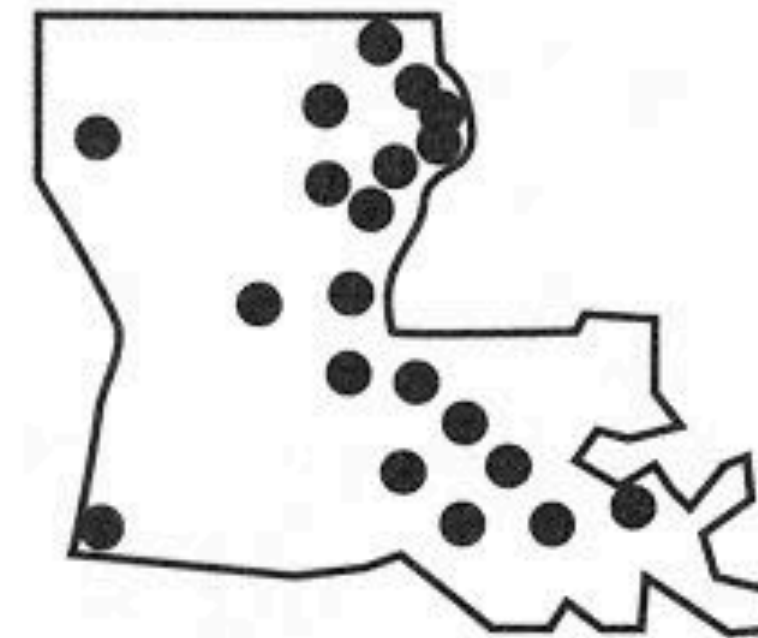
entisol



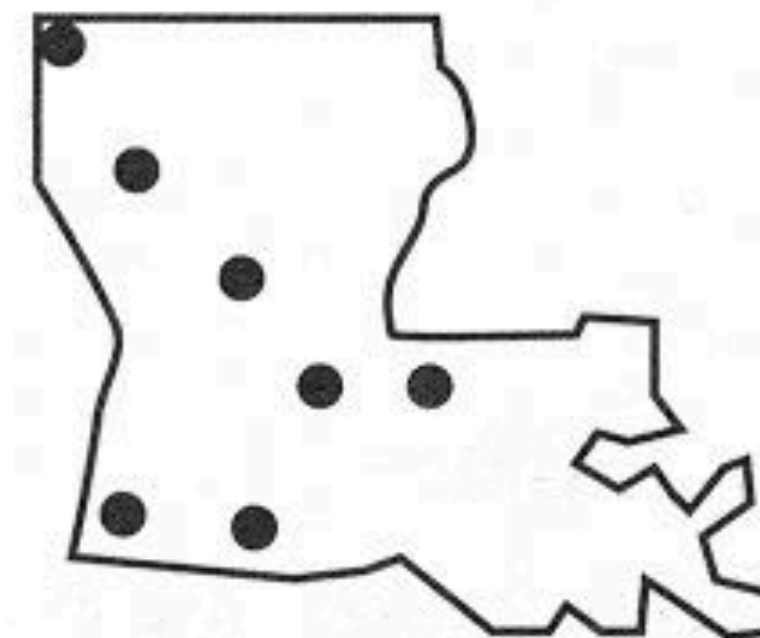
histosol



inceptisol



mollisol



ultisol

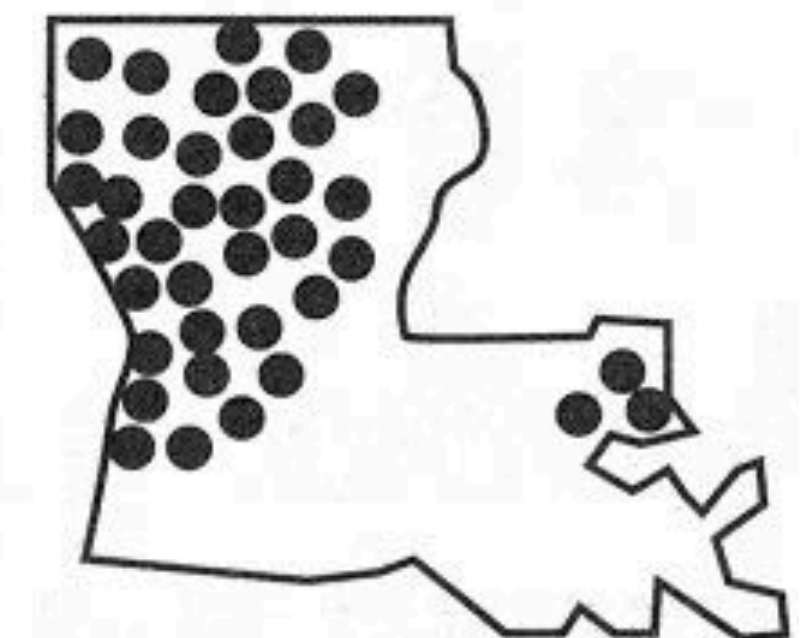
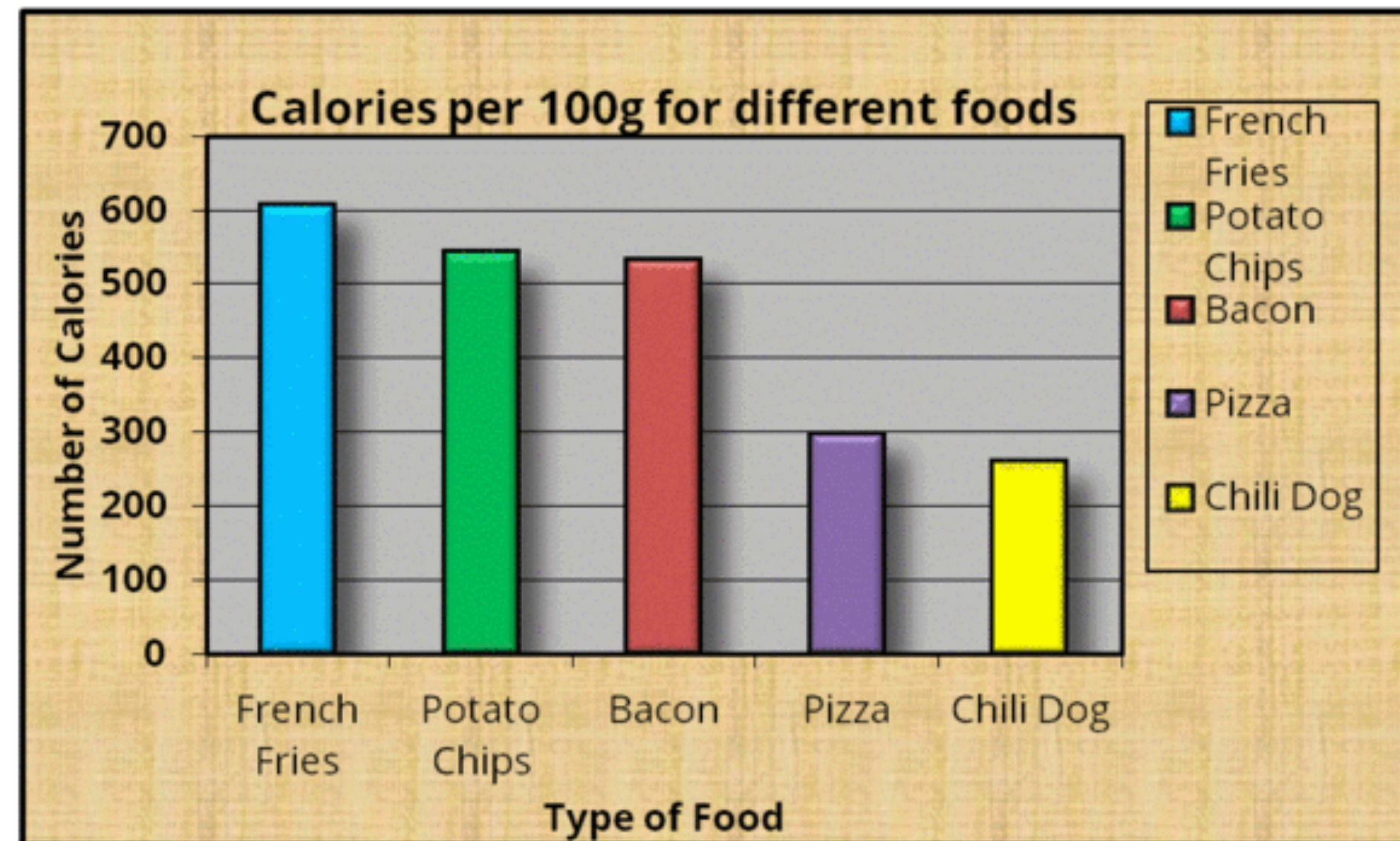


Chart junk



remove
unnecessary
elements

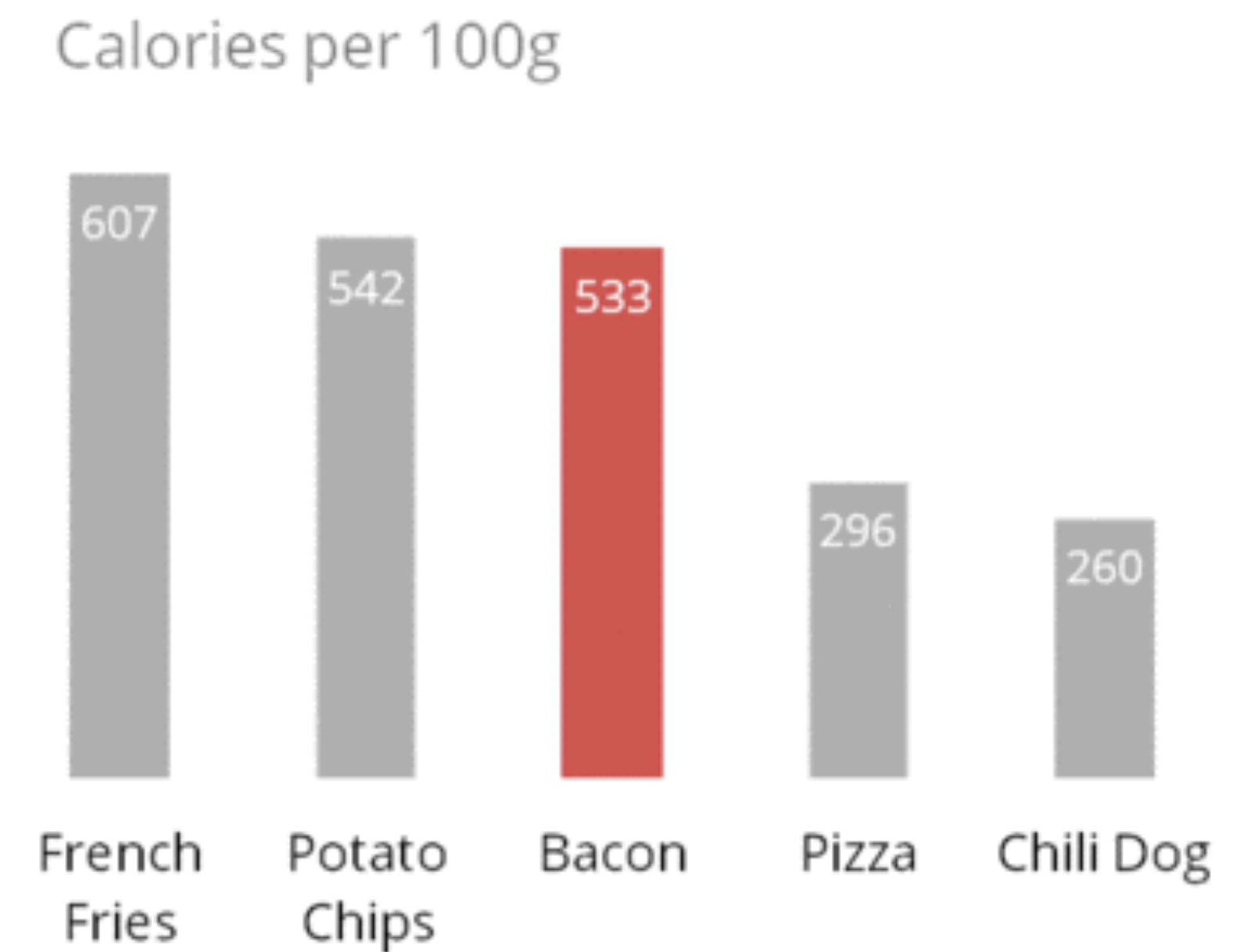
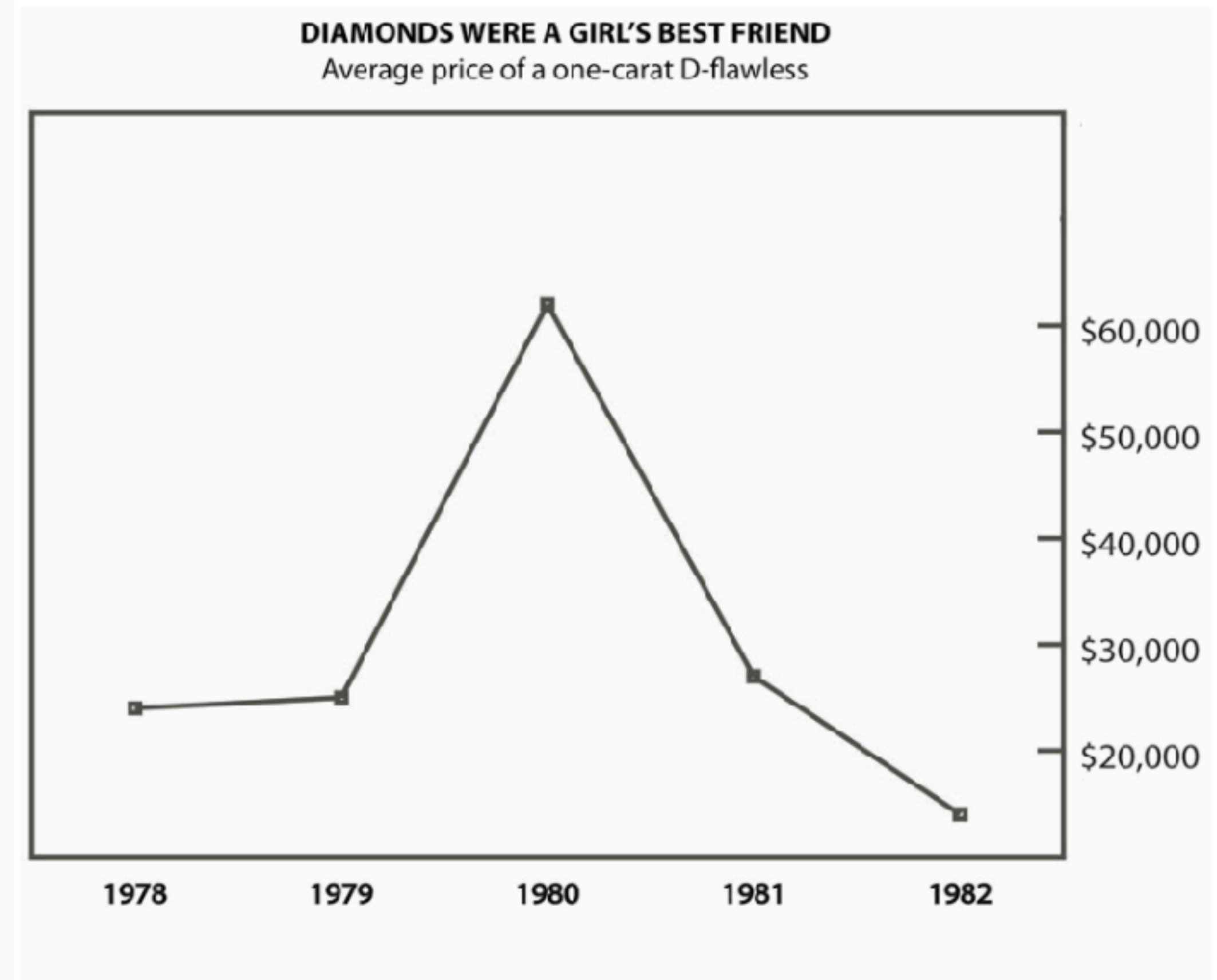
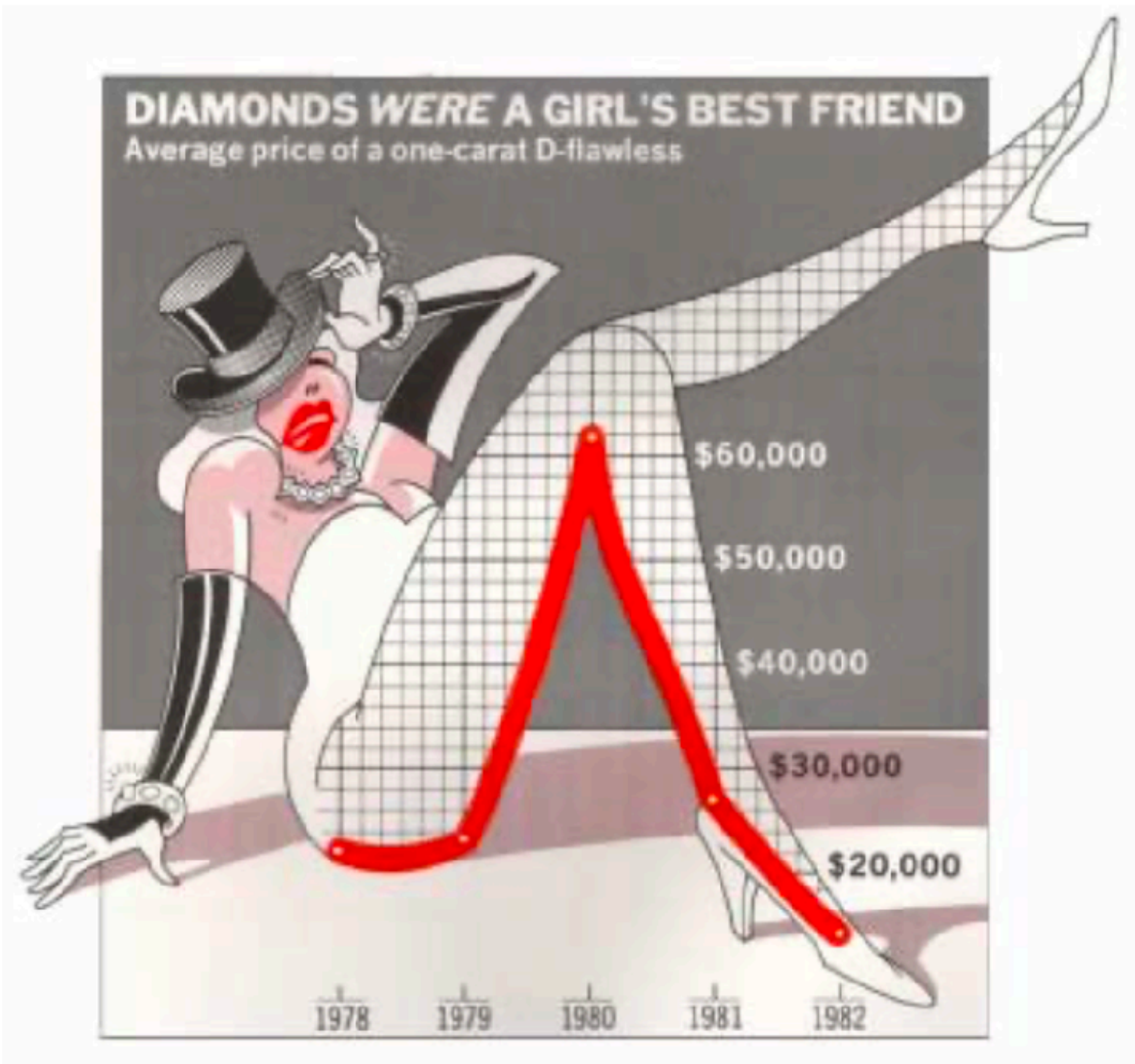


Chart junk



Principles or empirical rules

Using space (in)effectively

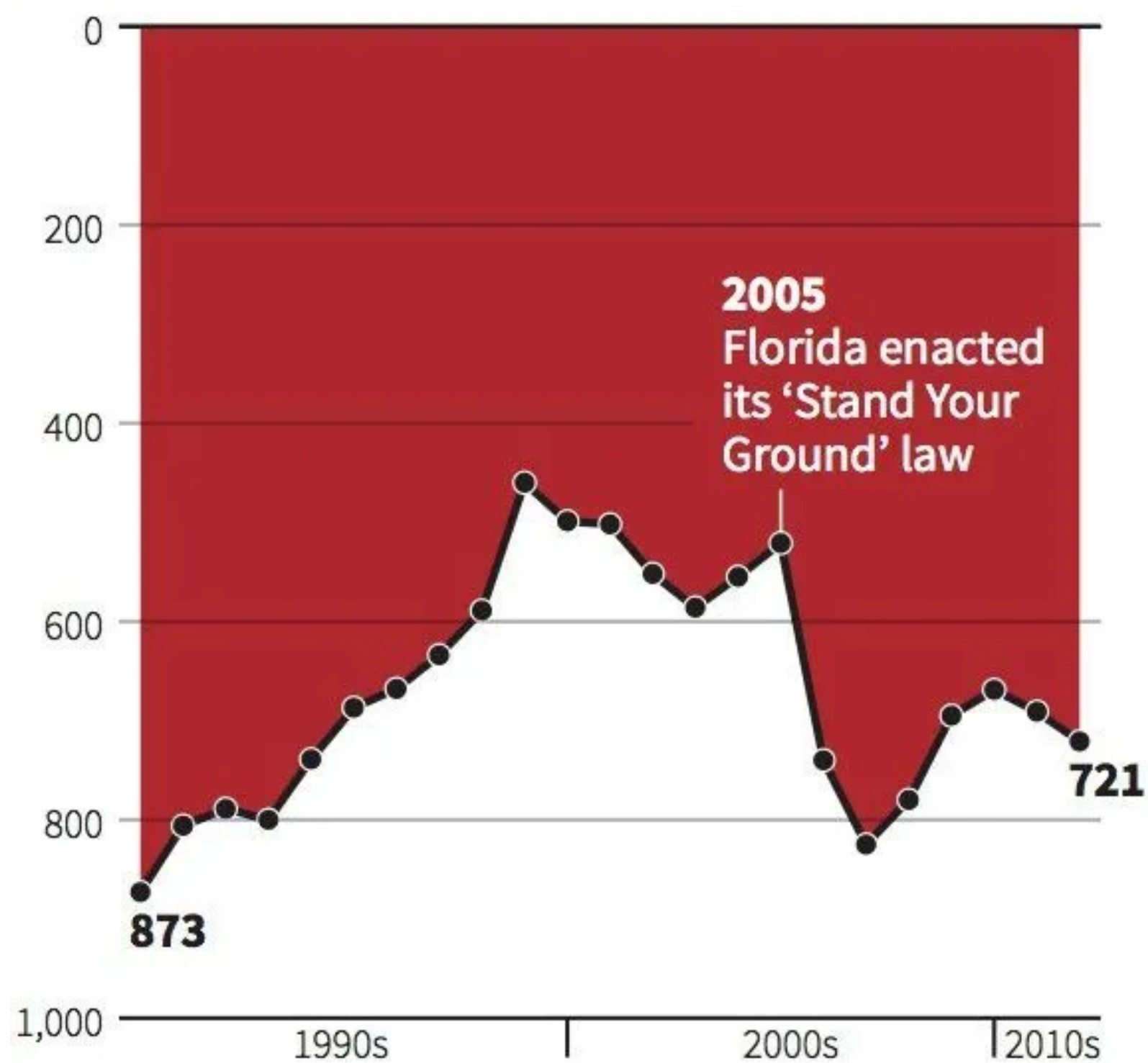
Hiding/revealing data

(Mis)leading readers ◀

Inverted y-axis

Gun deaths in Florida

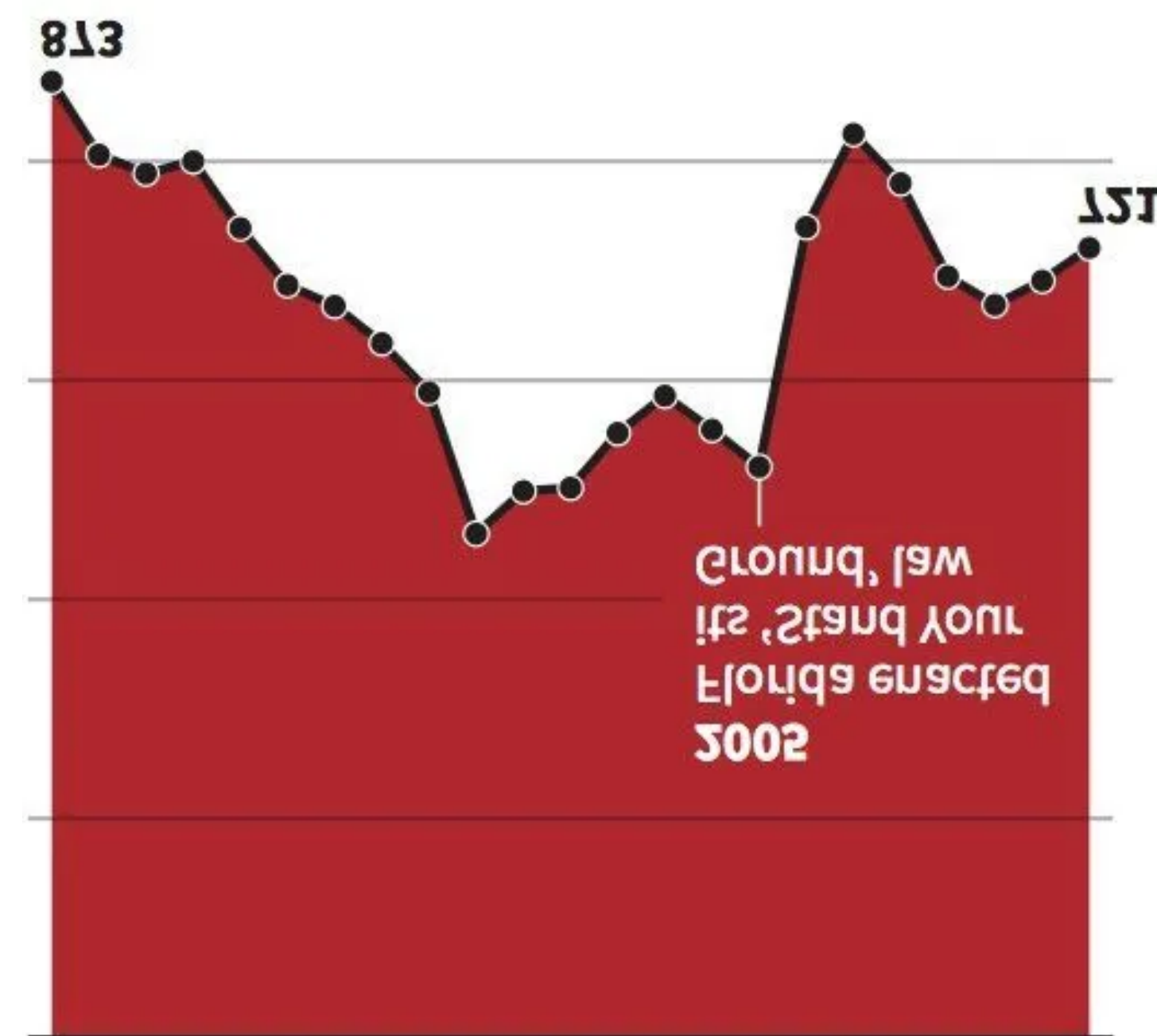
Number of murders committed using firearms



Source: Florida Department of Law Enforcement

C. Chan 16/02/2014

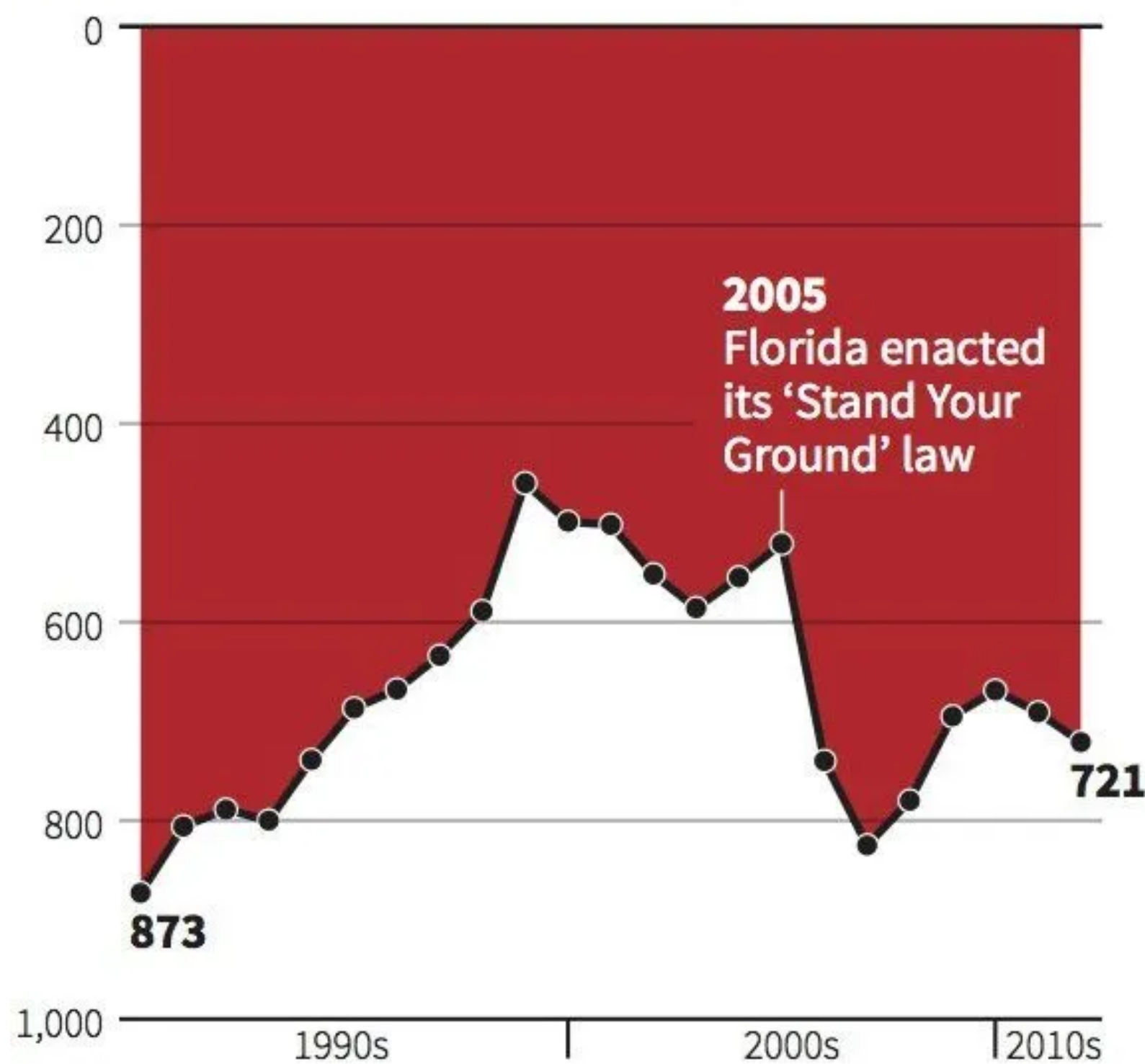
REUTERS



Inverted y-axis

Gun deaths in Florida

Number of murders committed using firearms



Source: Florida Department of Law Enforcement

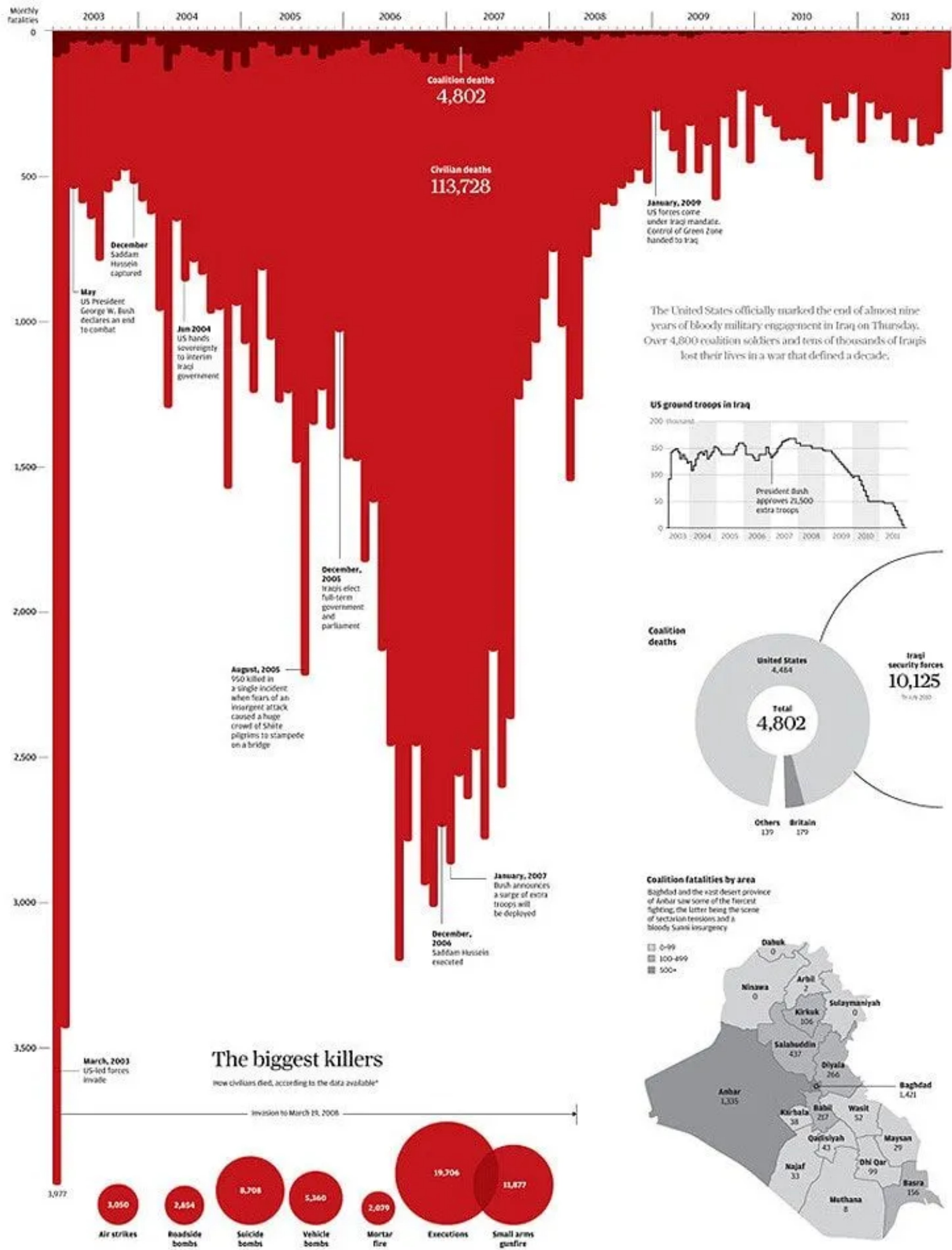
C. Chan 16/02/2014



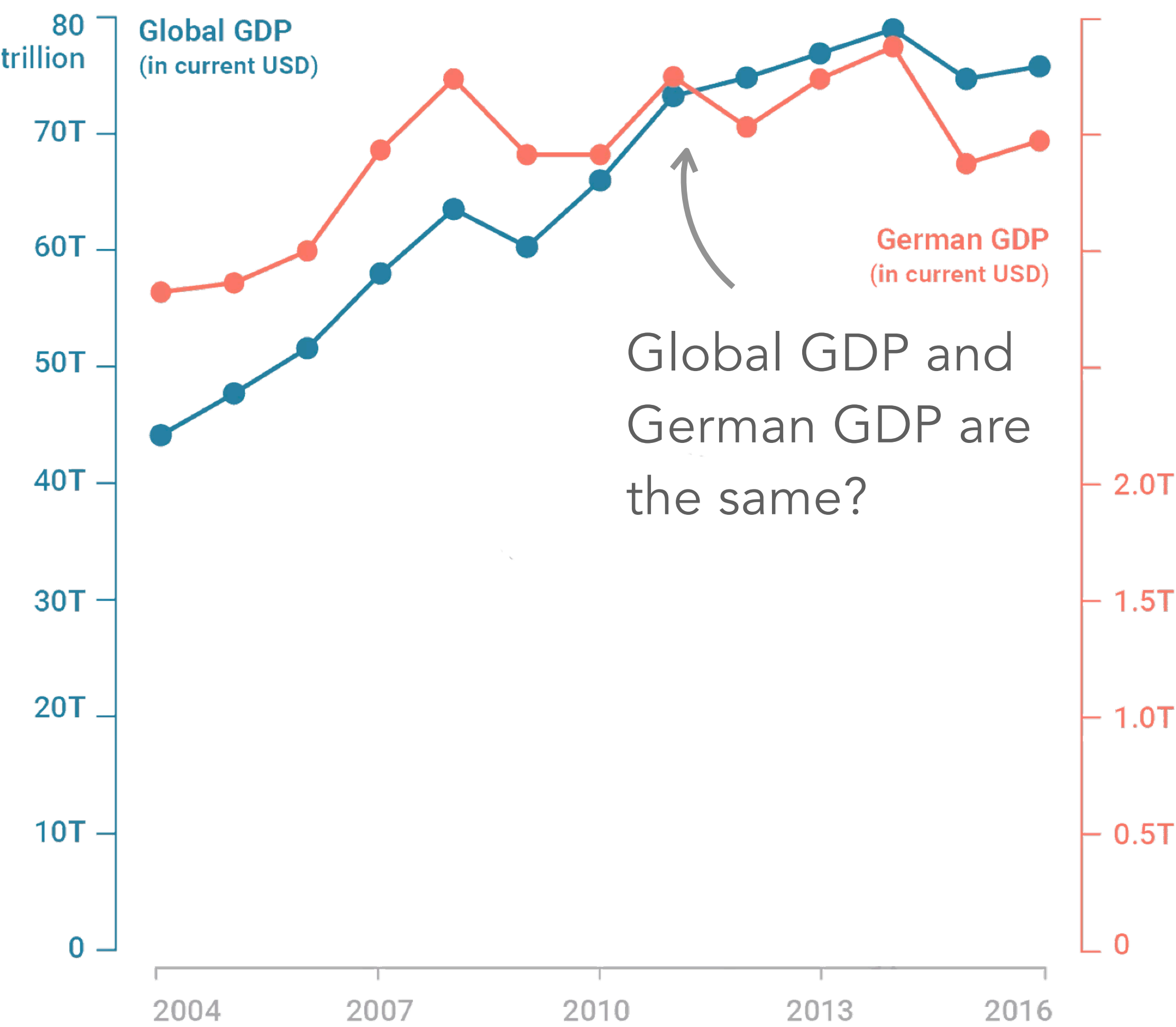
but...



Iraq's bloody toll



Dual axis charts

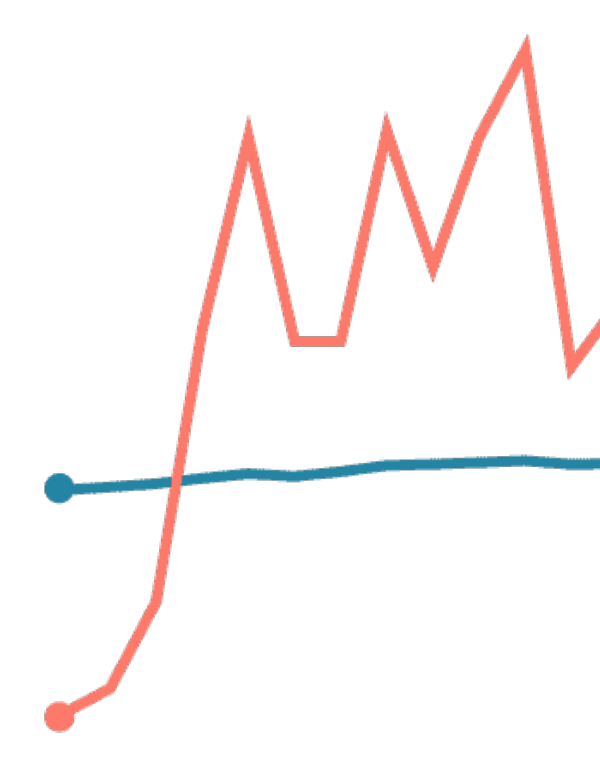


Dual axis charts

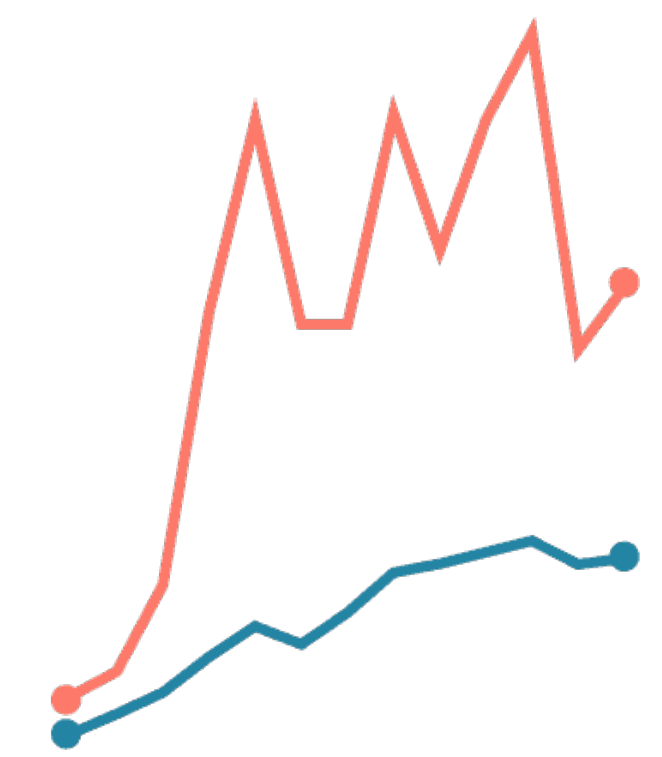
We can make all kinds of statements with our two data sets if we just tweak the scales a little bit....



Orange steady,
Blue massively increasing.



Blue steady,
Orange increasing.



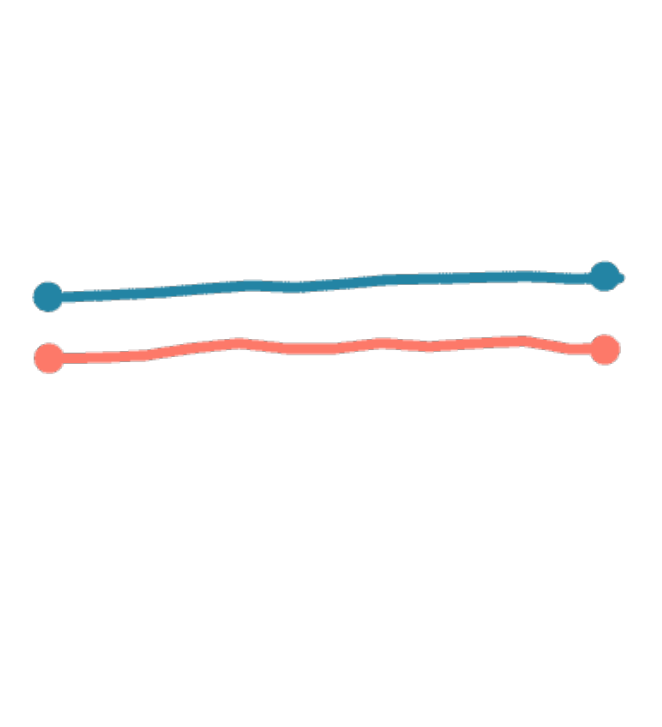
Both started at the same
level, but Orange increased
far more than Blue.



Both started at the same
level, but Blue increased far
more than Orange.

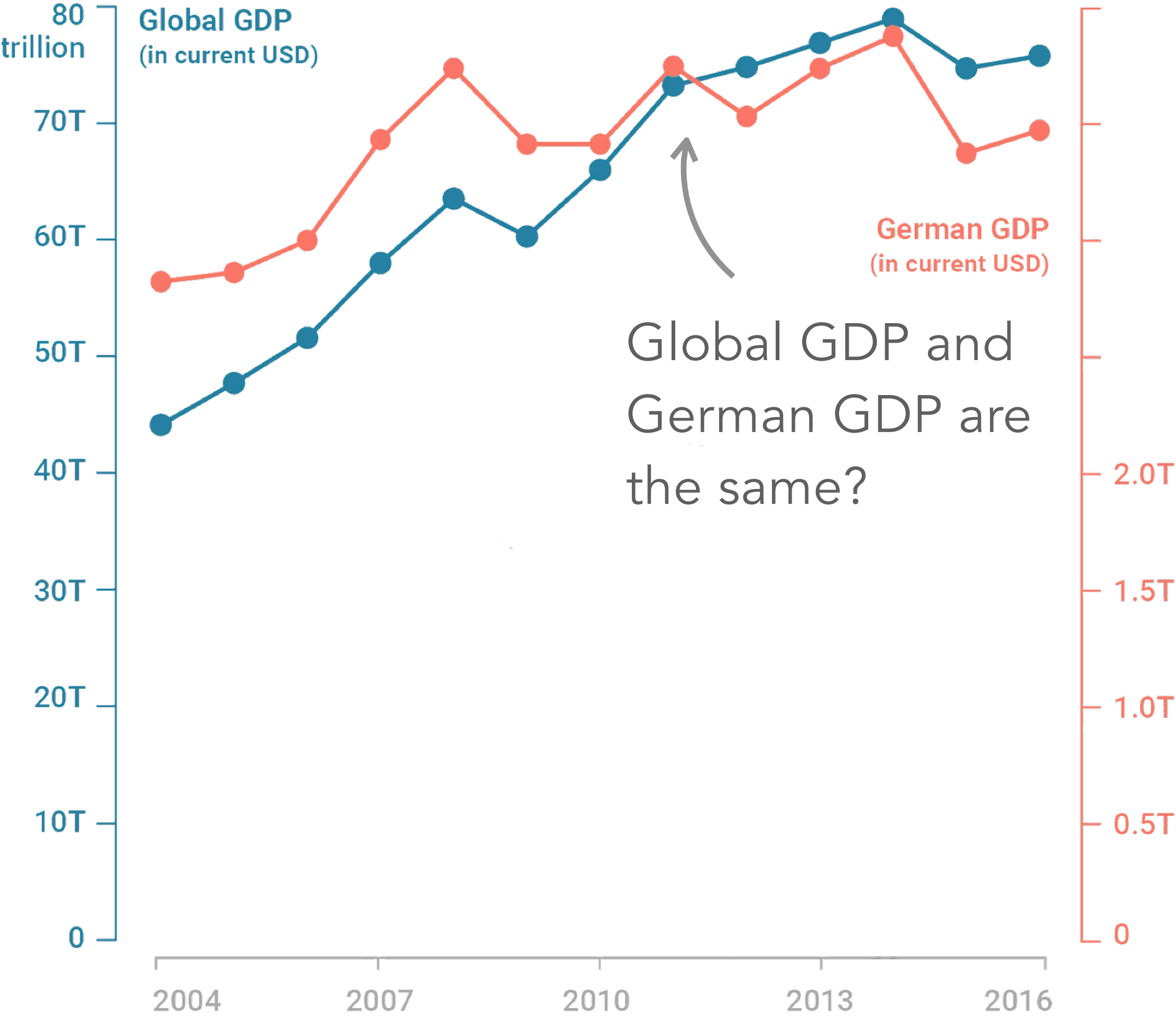


Both started with the
same increase, then Blue
raced to the top.

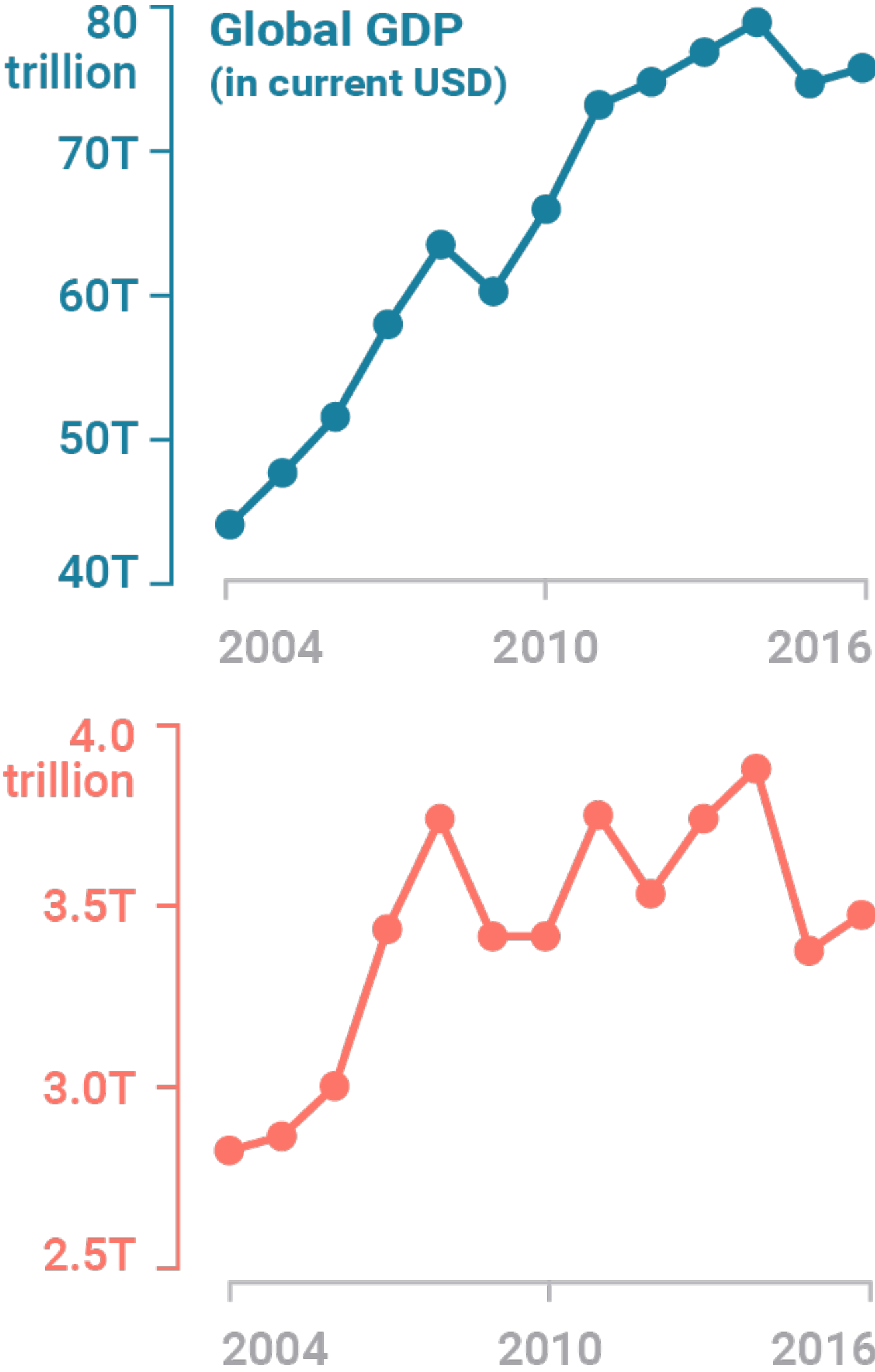


Both steady.

Dual axis charts



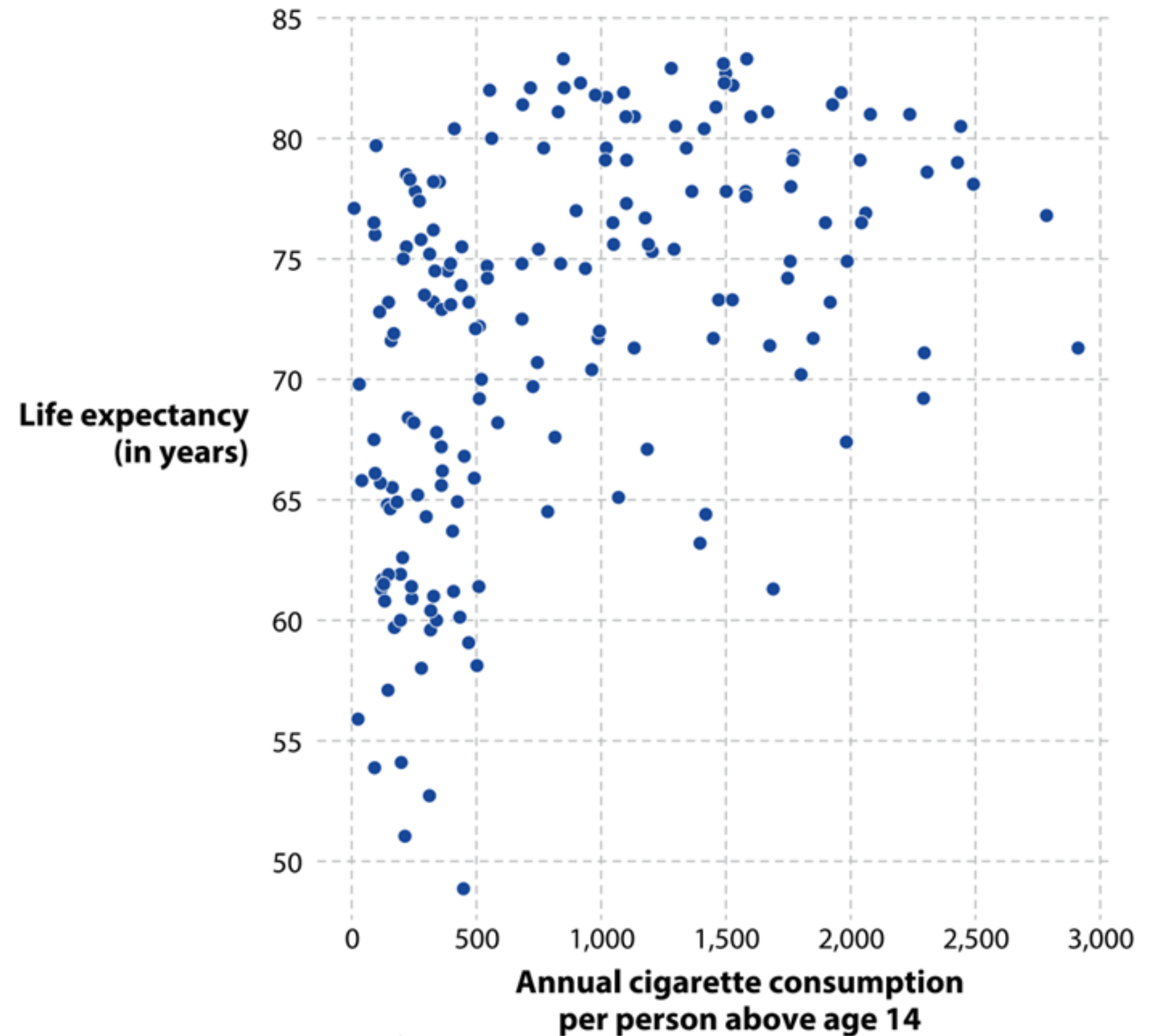
Don't ask
one chart to
do too much



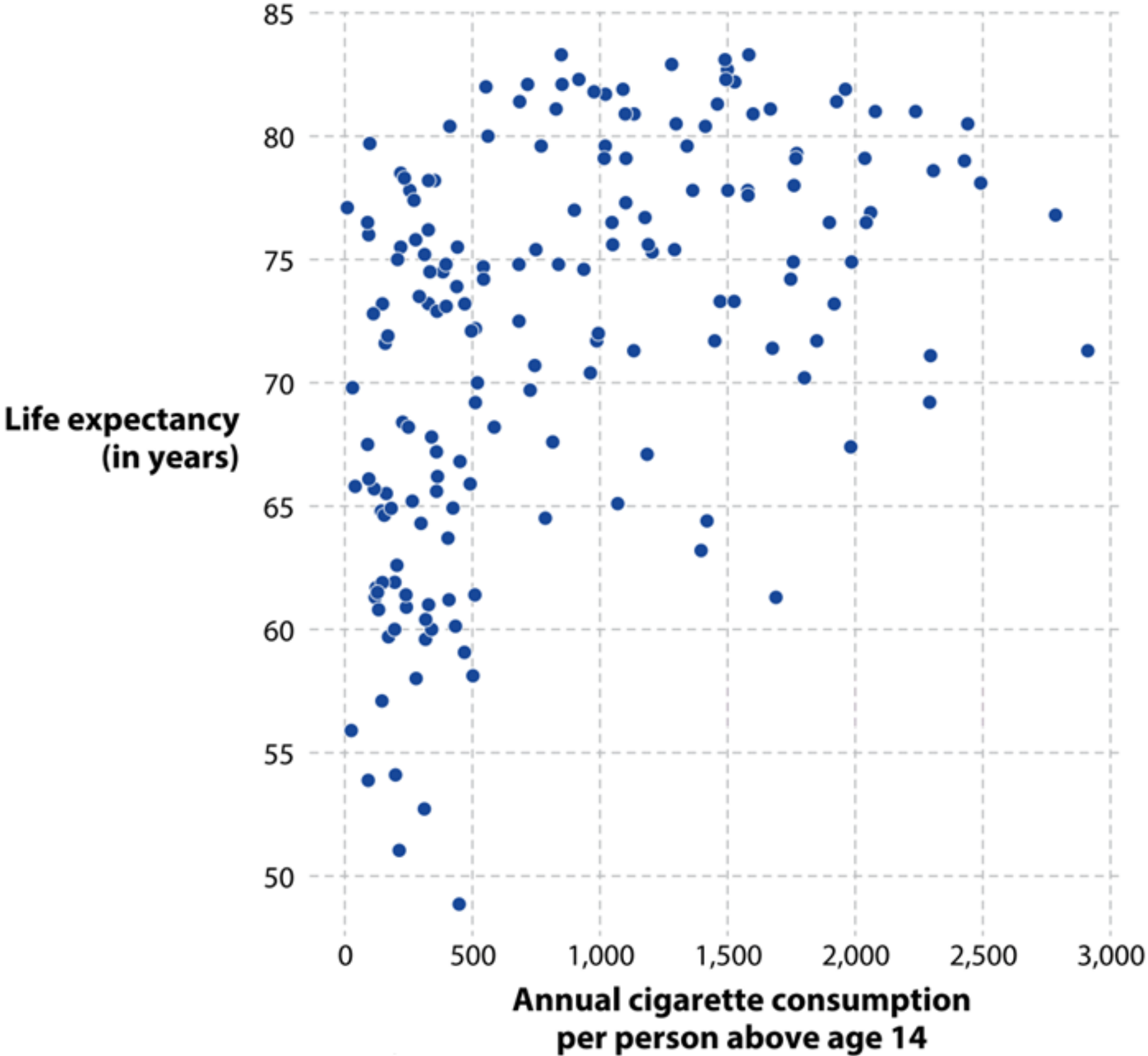
Correlation vs. causality

The more cigarettes we consume, the more we live...?

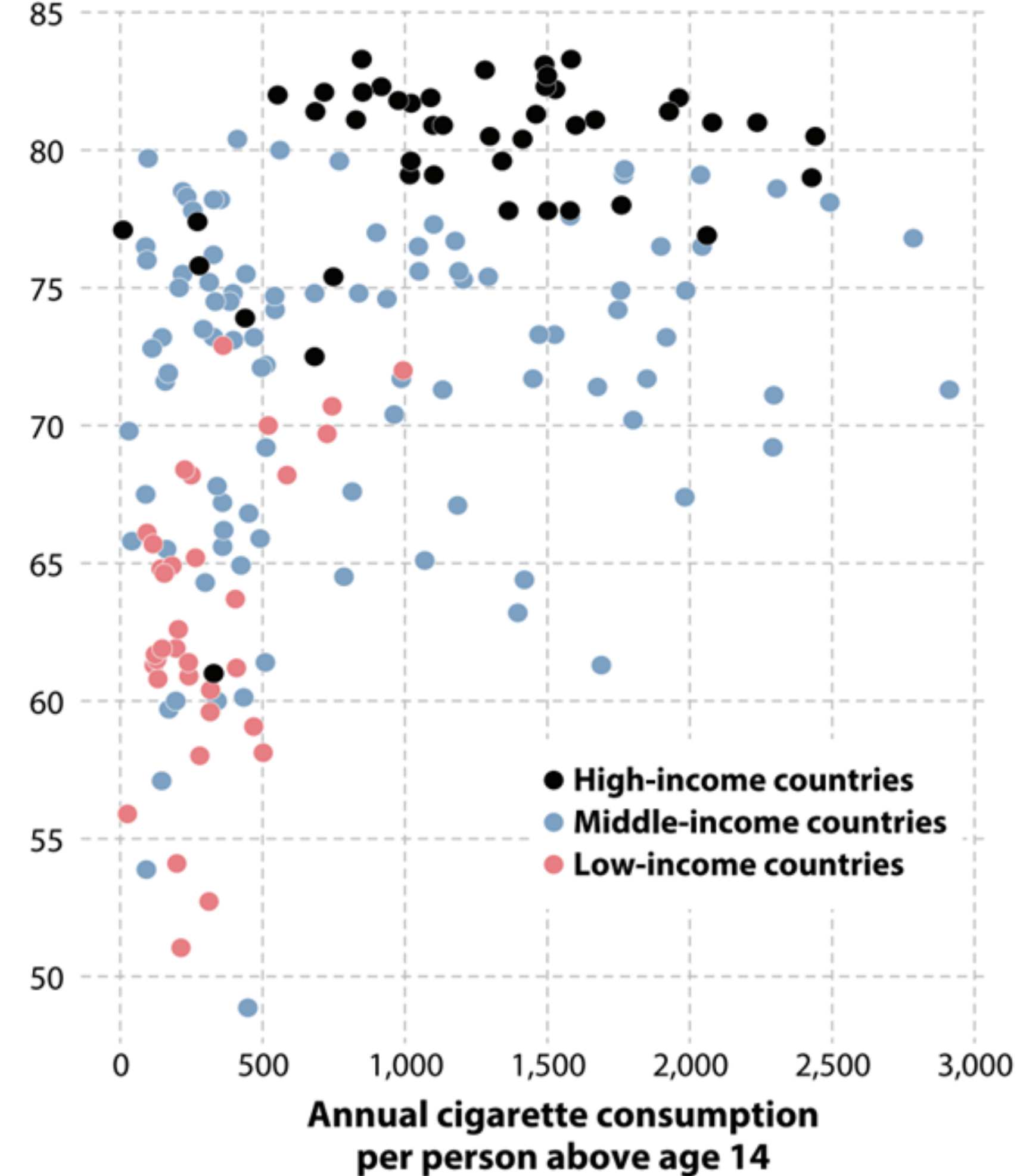
There is a positive relationship between cigarette consumption and life expectancy.



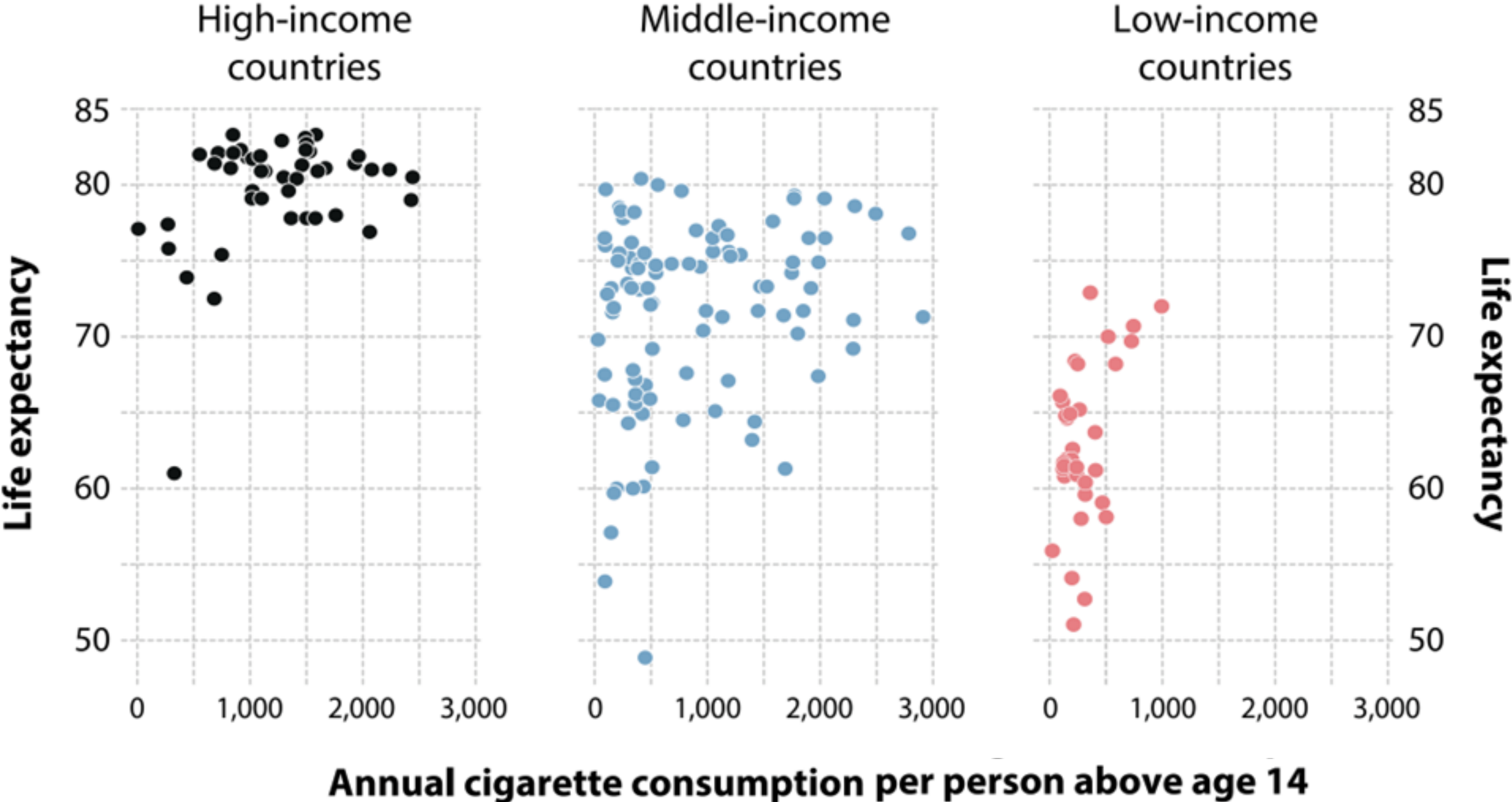
Correlation vs. causality



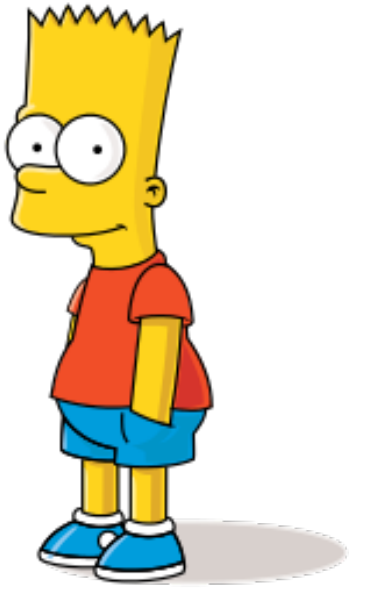
by income
→



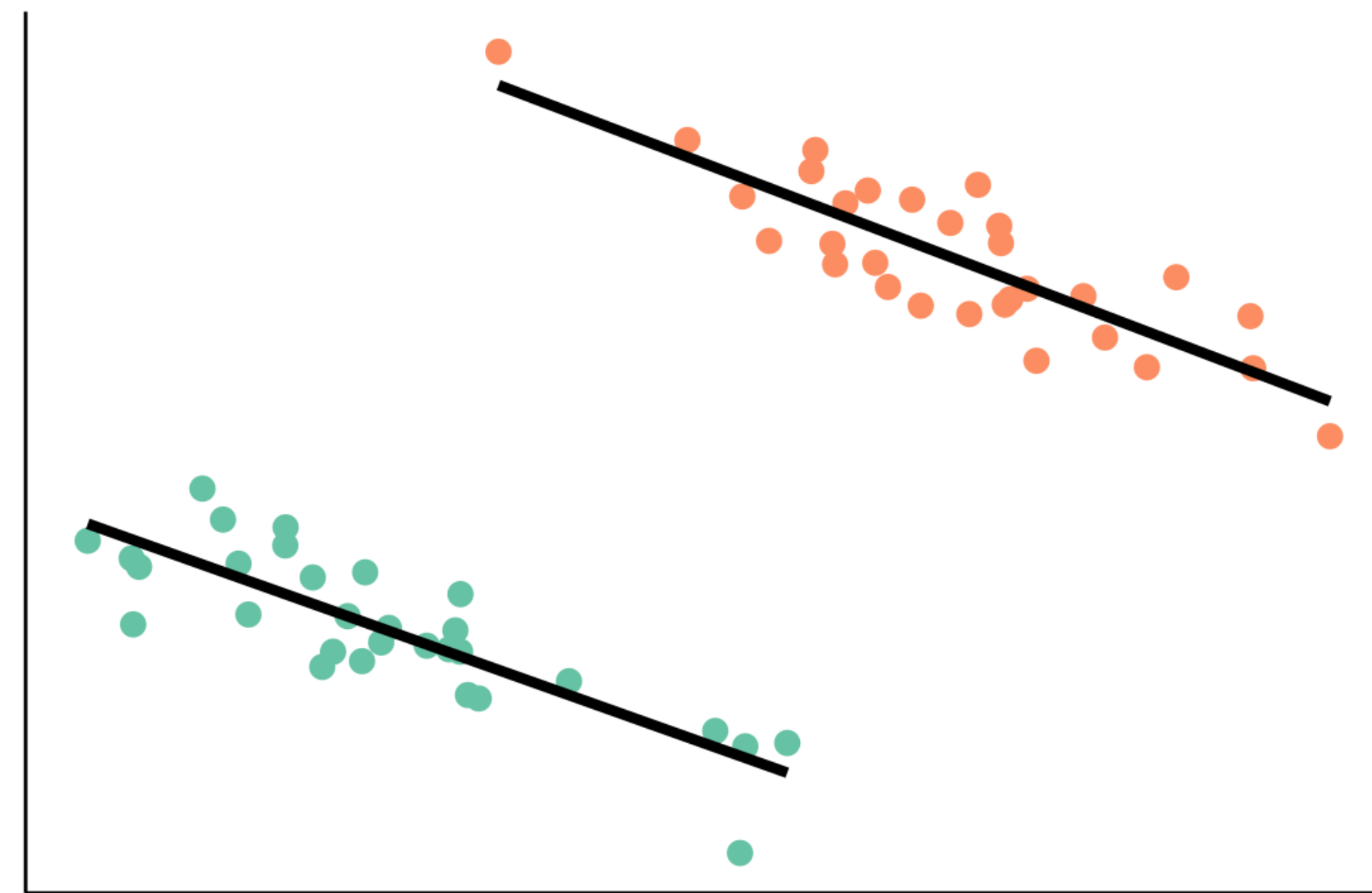
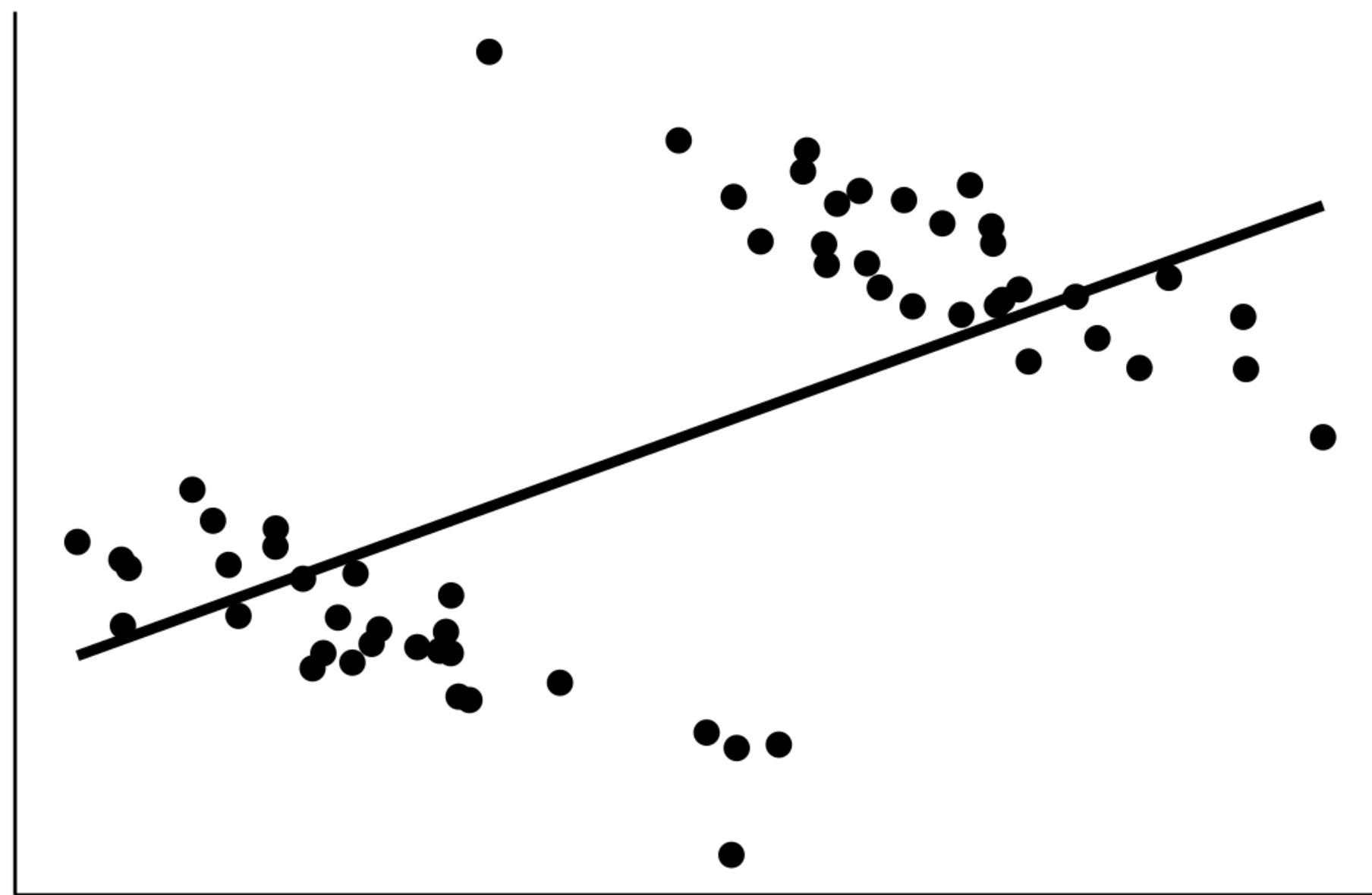
Correlation vs. causality



Simpson's Paradox



A trend or result that is present when data is put into groups that reverses or disappears when the data is combined.



Recap

Data: attributes, dataset types

Visual: visual channels, graphical marks

Data → **Visual**: expressiveness, effectiveness, tradeoffs

Goals & Lies

Group Activity

Group Activity

What are the variable types?

Nominal, Ordinal, Quantitative (interval),
Quantitative (ratio)

country	year	population Population, total	continent
Afghanistan	2020	38972230	Asia
Albania	2020	2837849	Europe
Algeria	2020	43451666	Africa
American Samoa	2020	46189	Oceania
Andorra	2020	77700	Europe
Angola	2020	33428486	Africa
Antigua and Barbuda	2020	92664	Americas
Argentina	2020	45376763	Americas
Armenia	2020	2805608	Asia
Aruba	2020	106585	Americas
Australia	2020	25649248	Oceania
Austria	2020	8916864	Europe
Azerbaijan	2020	10093121	Asia
Bahamas, The	2020	406471	Americas
Bahrain	2020	1477469	Asia
Bangladesh	2020	167420951	Asia
Barbados	2020	280693	Americas
Belarus	2020	9379952	Europe
Belgium	2020	11538604	Europe

Group Activity

Variable types?

Channels/encodings?

Marks?

Author's intention?

Effective (for the intention)?



Group Activity

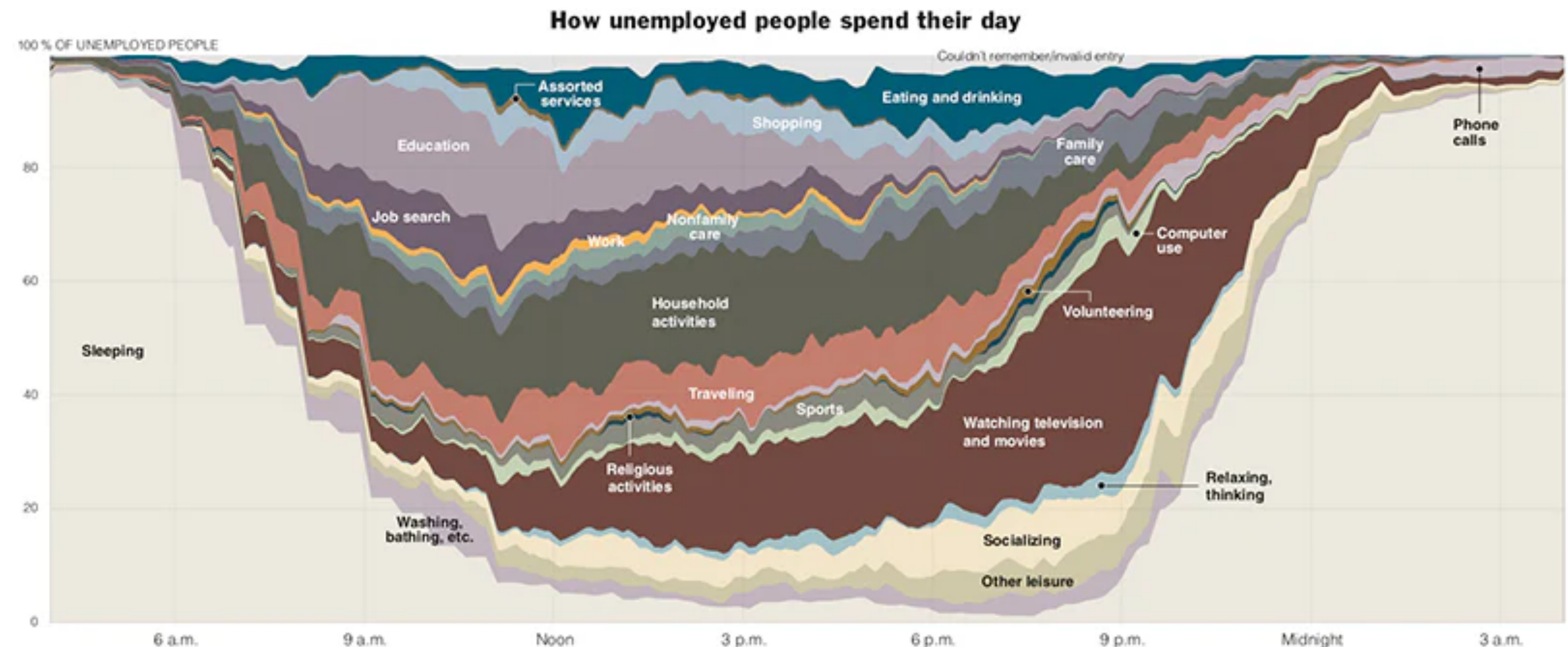
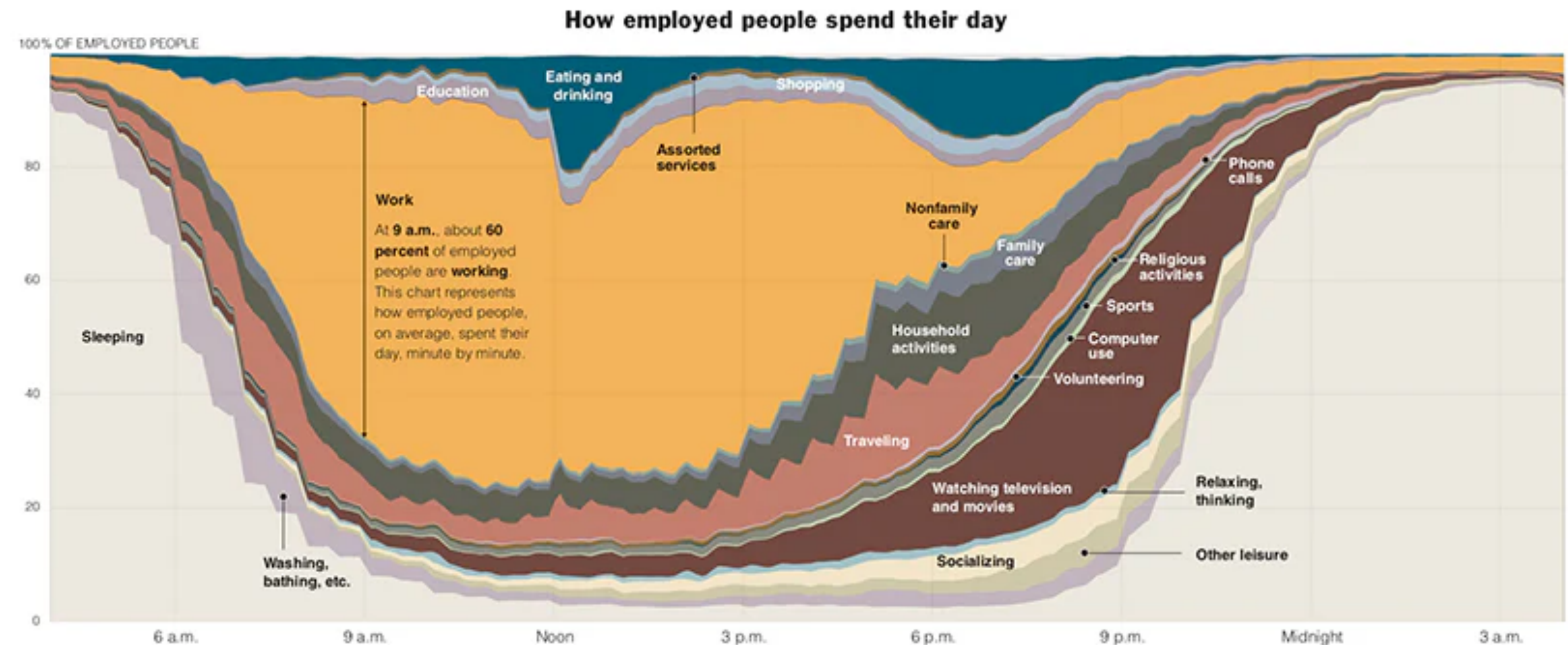
Variable types?

Channels/encodings?

Marks?

Author's intention?

Effective?



Wed Lab

Lab 2: Building a Basic Bar Chart

